01_MME_JAS_2025

April 29, 2025

#####

#

AGHRYMET RCC-WAS

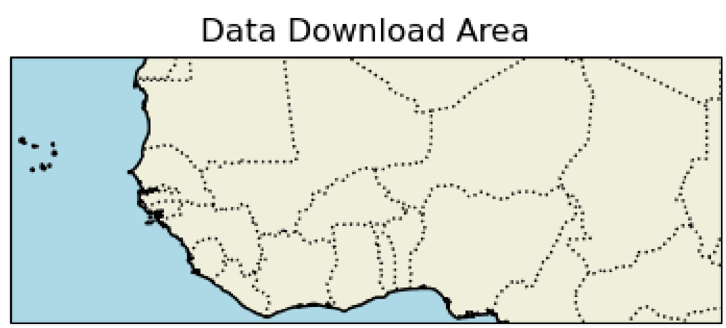
0.1 Library

Set forecast working directory

Set Climatological Years

0.2 Download Observation and Process

Load Downloader



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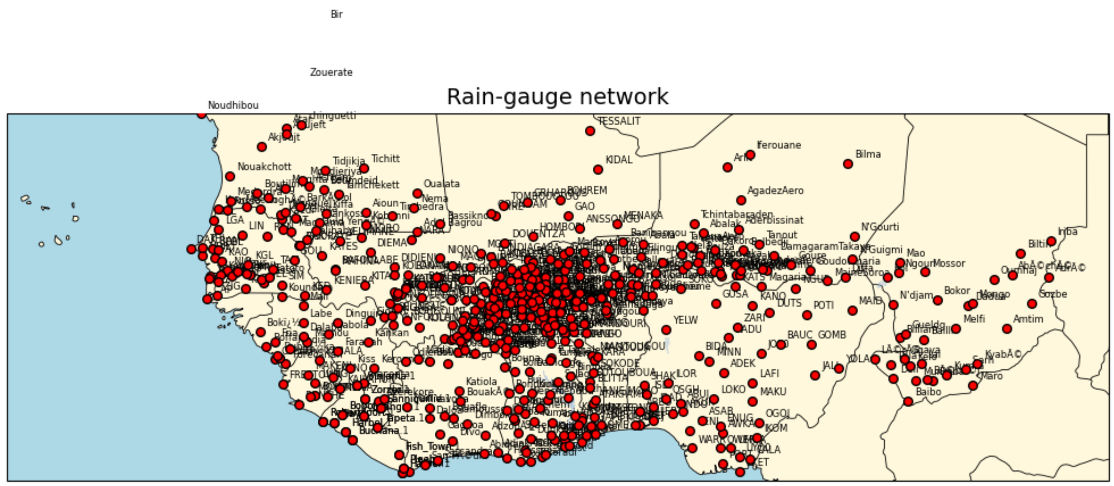
Process Obs in combining with Seasonal cumulative Ground-Observation

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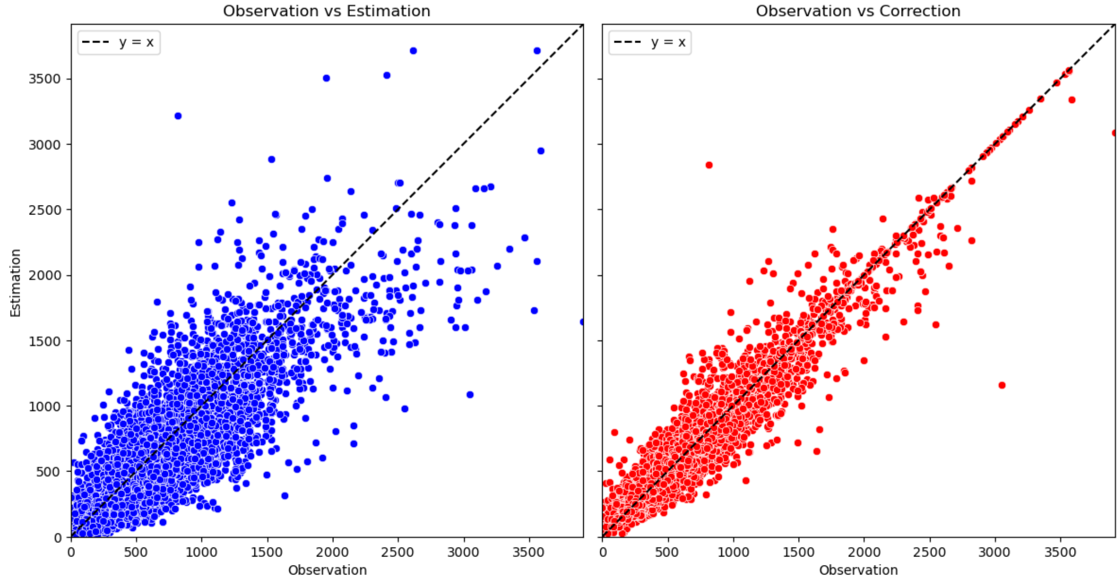

1 1st - Approach: Choice of Best Dynamical models and weighted Average

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Validation of GCM

```
Pearson Correlation
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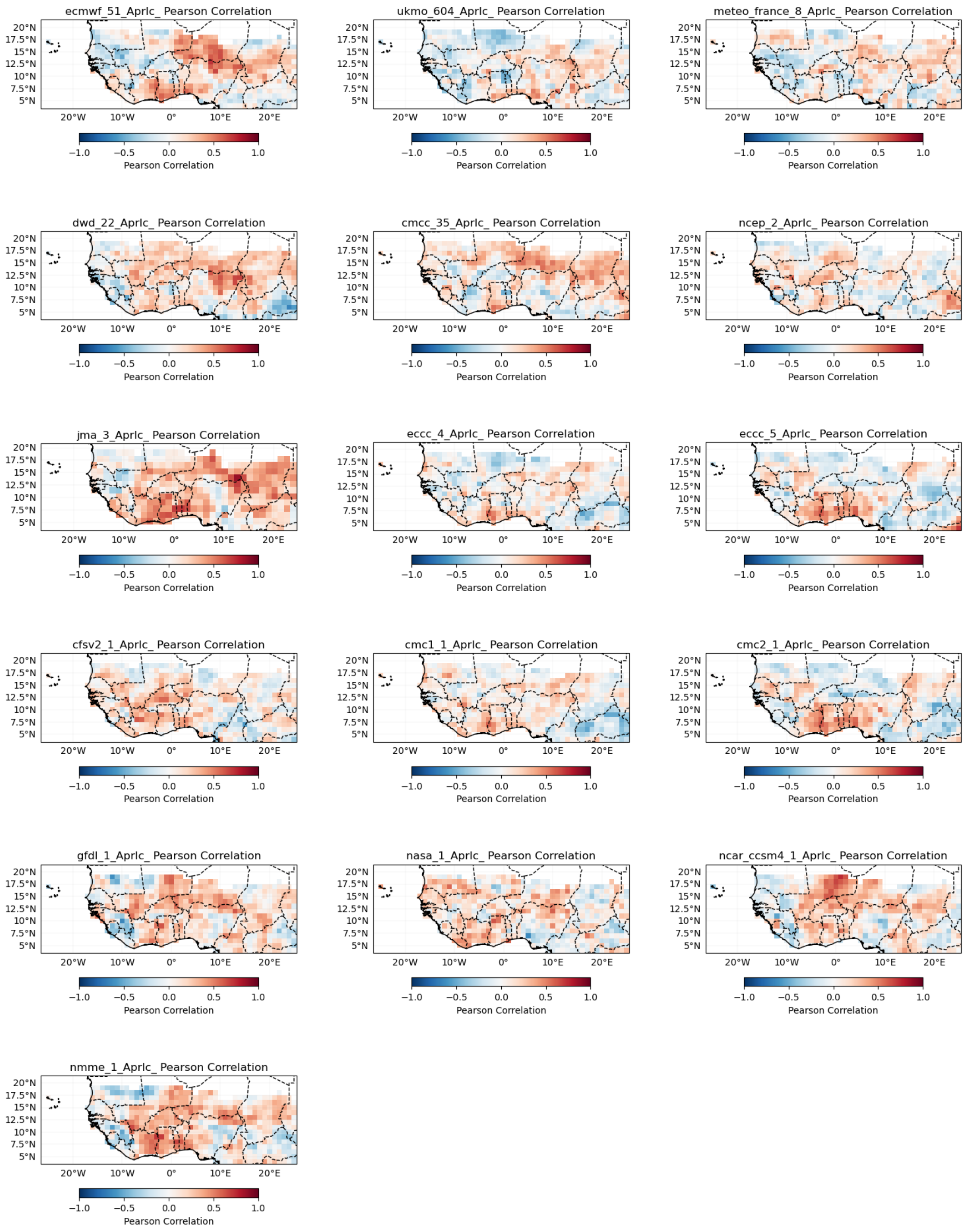
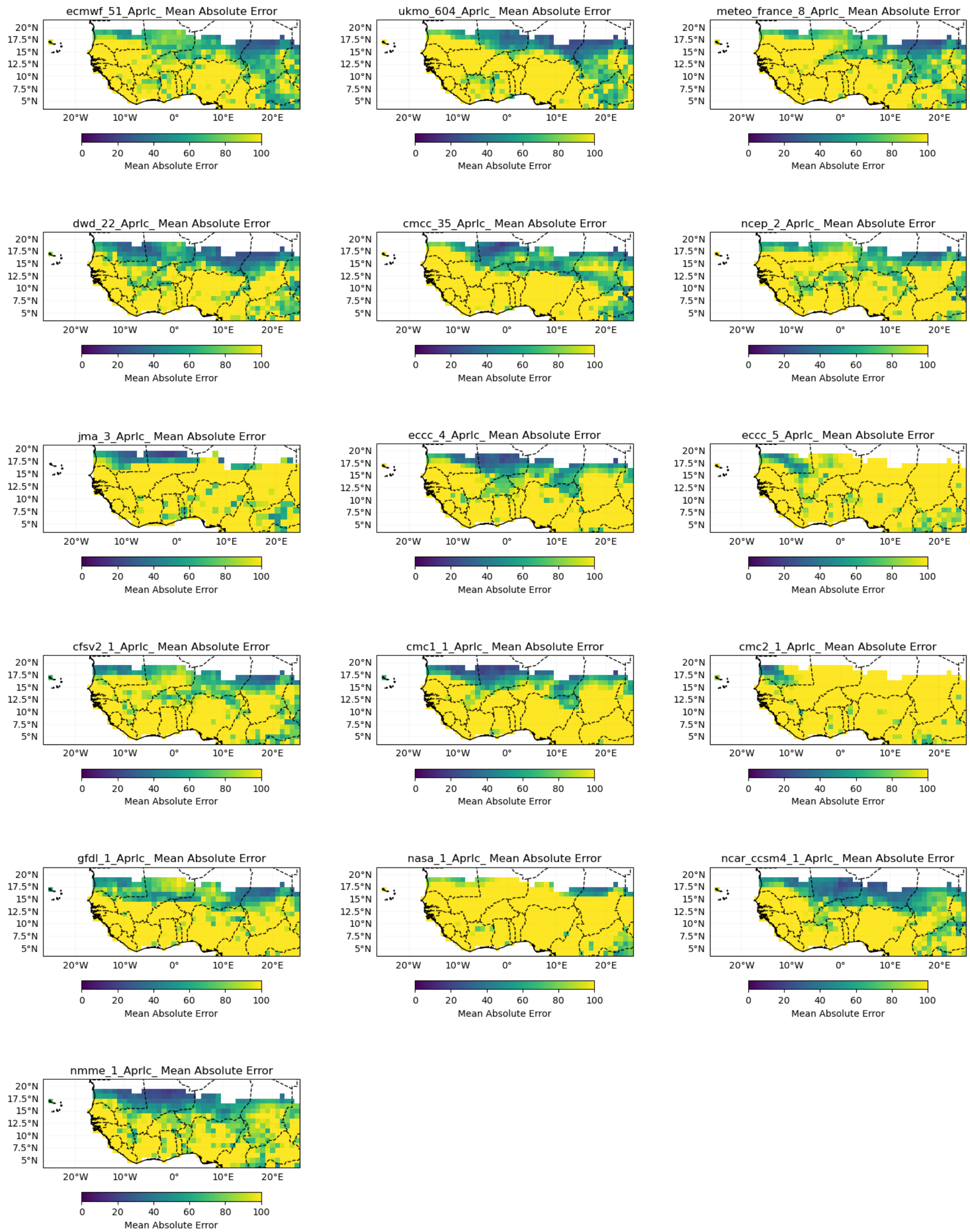
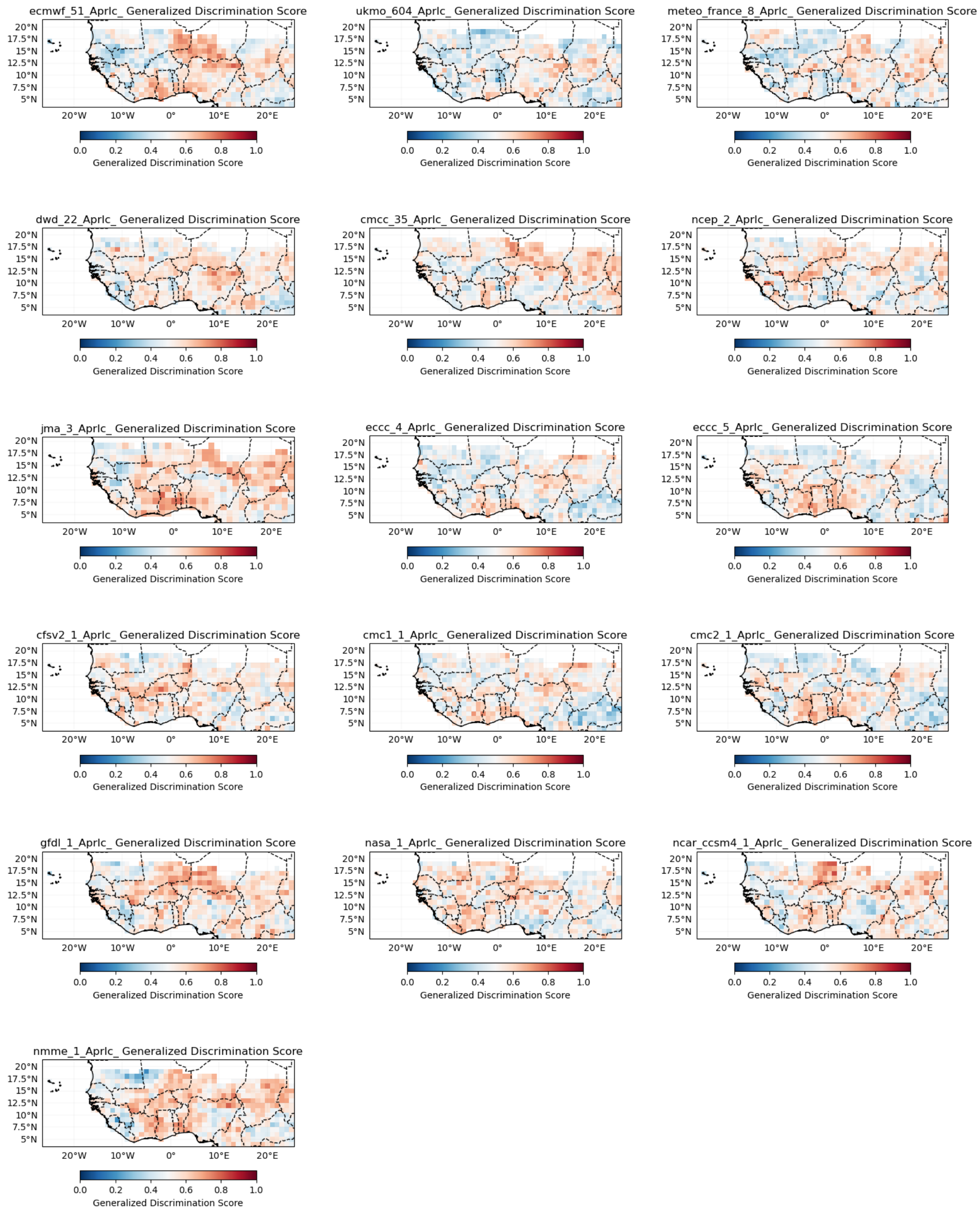


Mean Absolute Error



Generalized Discrimination Score



1.1 Choice of best GCM models

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['GFDL_1.PRCP', 'NMME_1.PRCP', 'DWD_22.PRCP', 'CMCC_35.PRCP', 'CFSV2_1.PRCP']
```

1.2 MME for best GCM models

Cross-validation

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Cross-validation ongoing
100%|                              | 24/24 [08:26<00:00, 21.11s/it]
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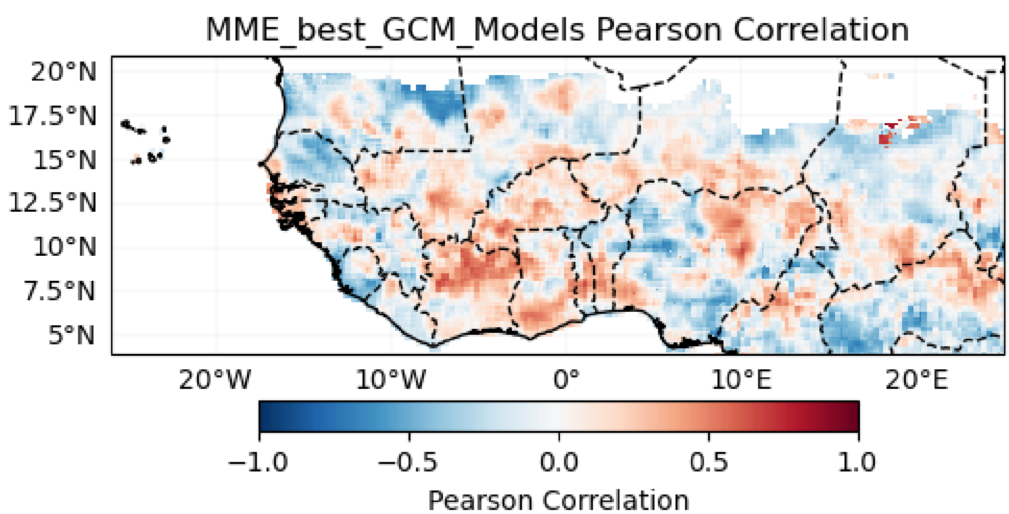
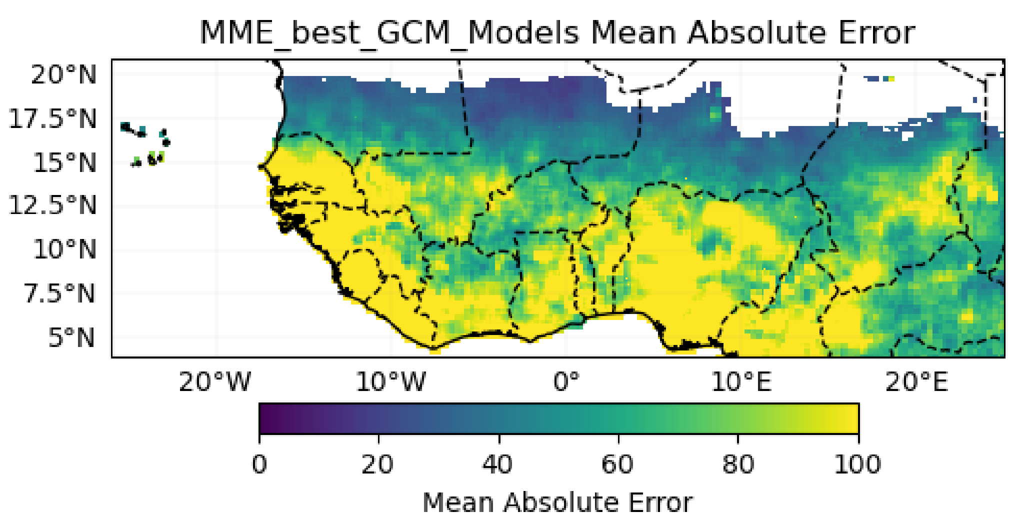
Verification





Probabilistic

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Generalized Discrimination Score
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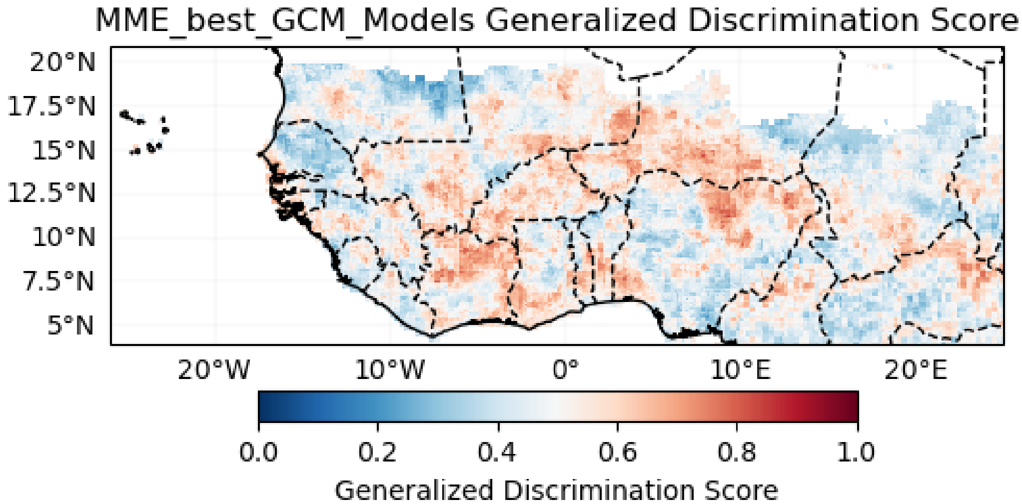
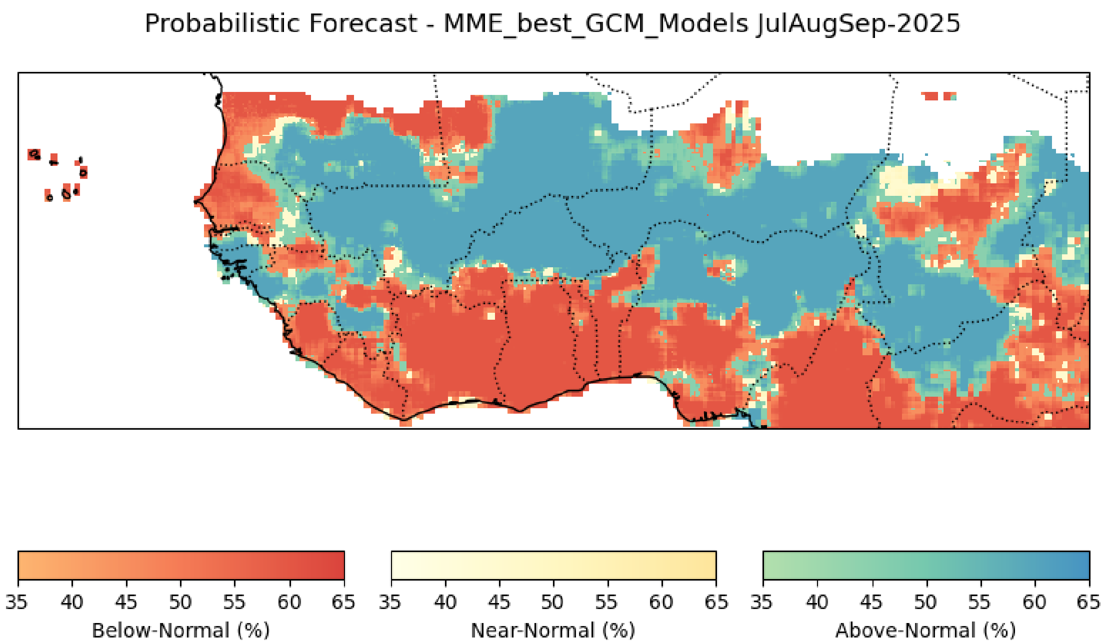


Forecast



8475

2 2nd - Approach: Statistico-Dynamical

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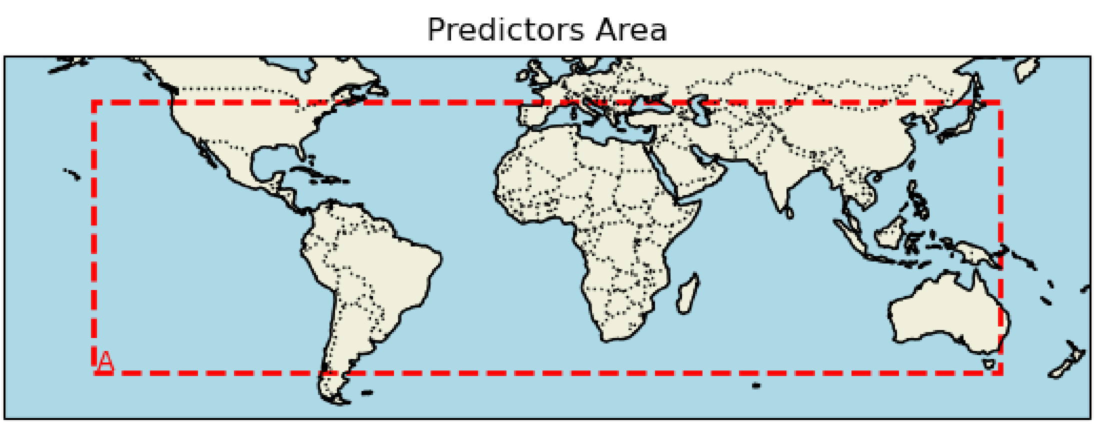
2.0.1 Initialize the WAS_CCA class (default)



Modes of CCA for each model

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CCA Modes for model ECMWF_51.SST
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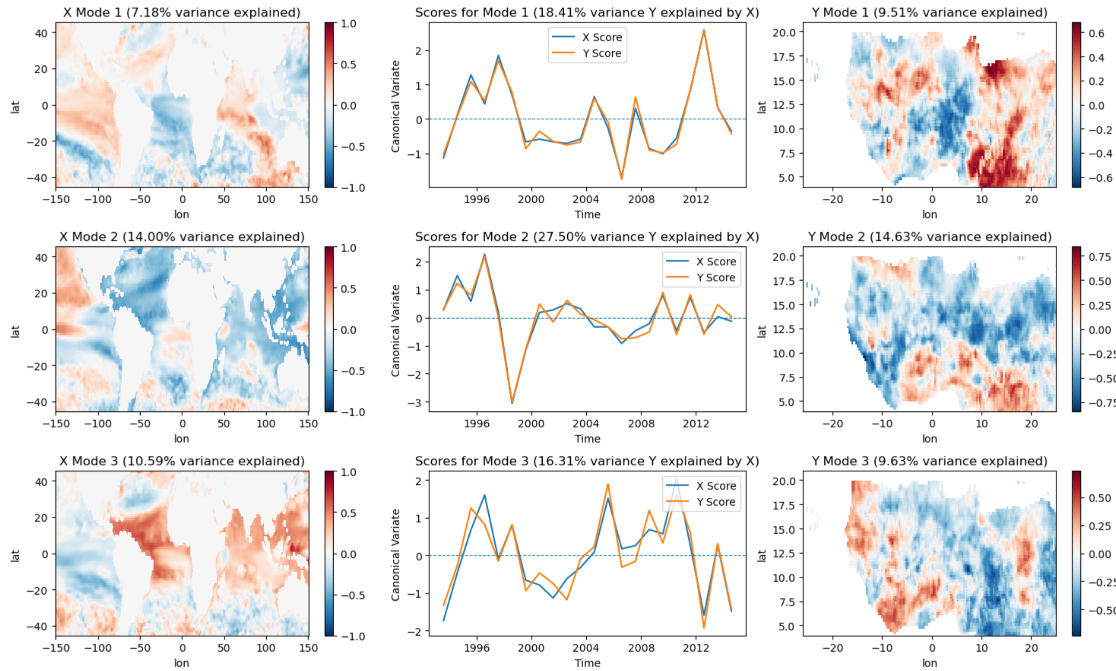
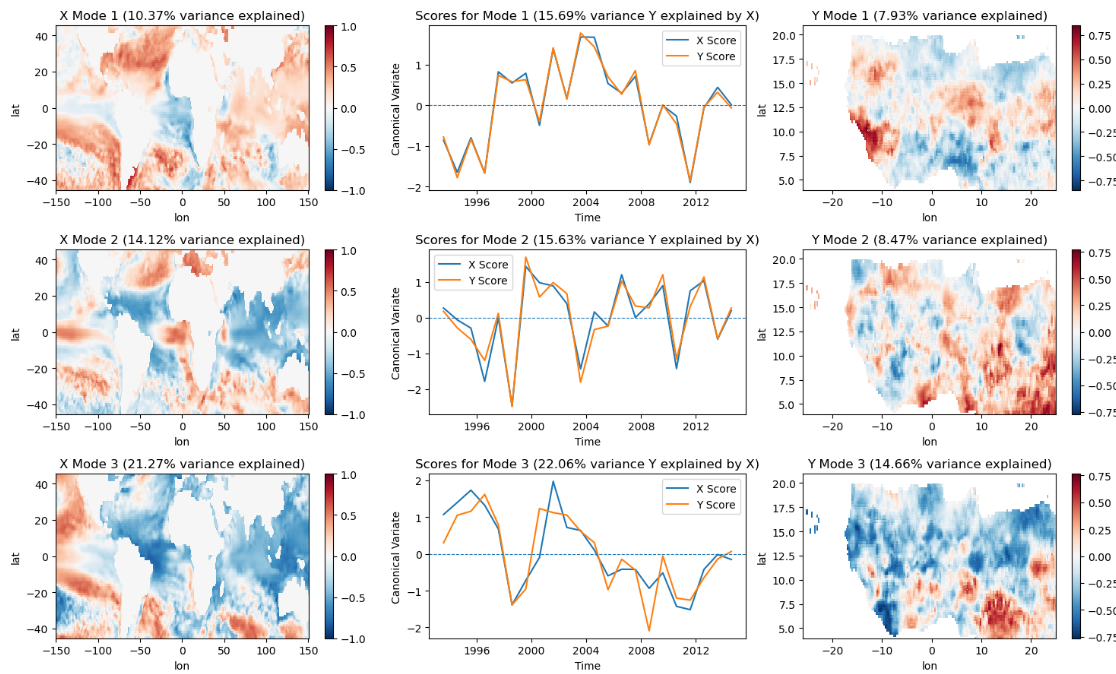
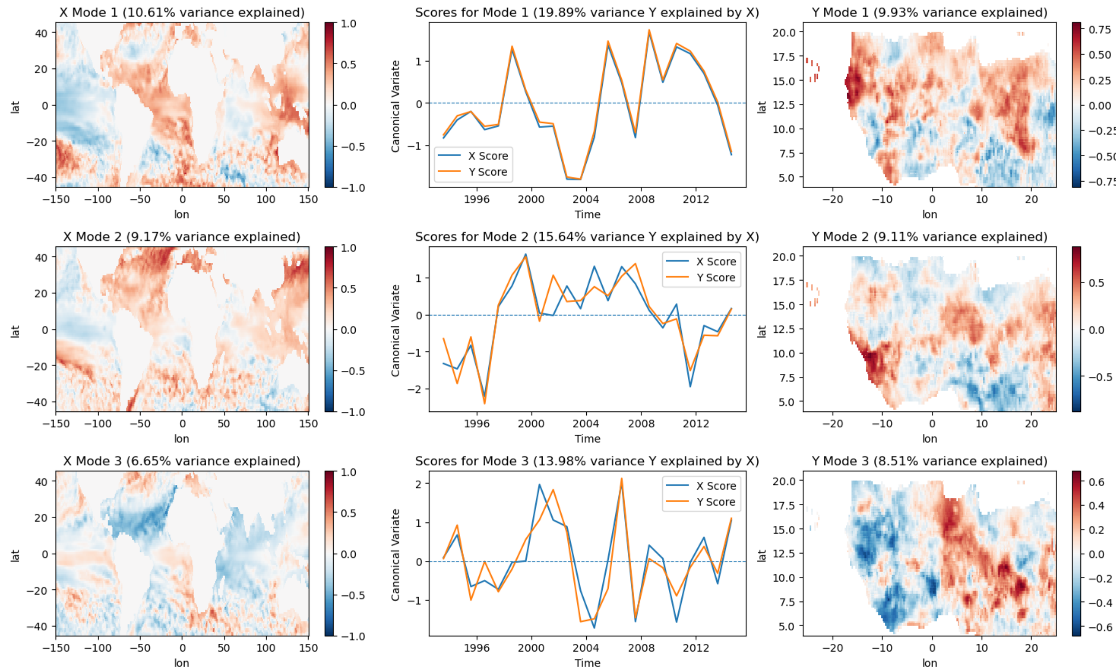


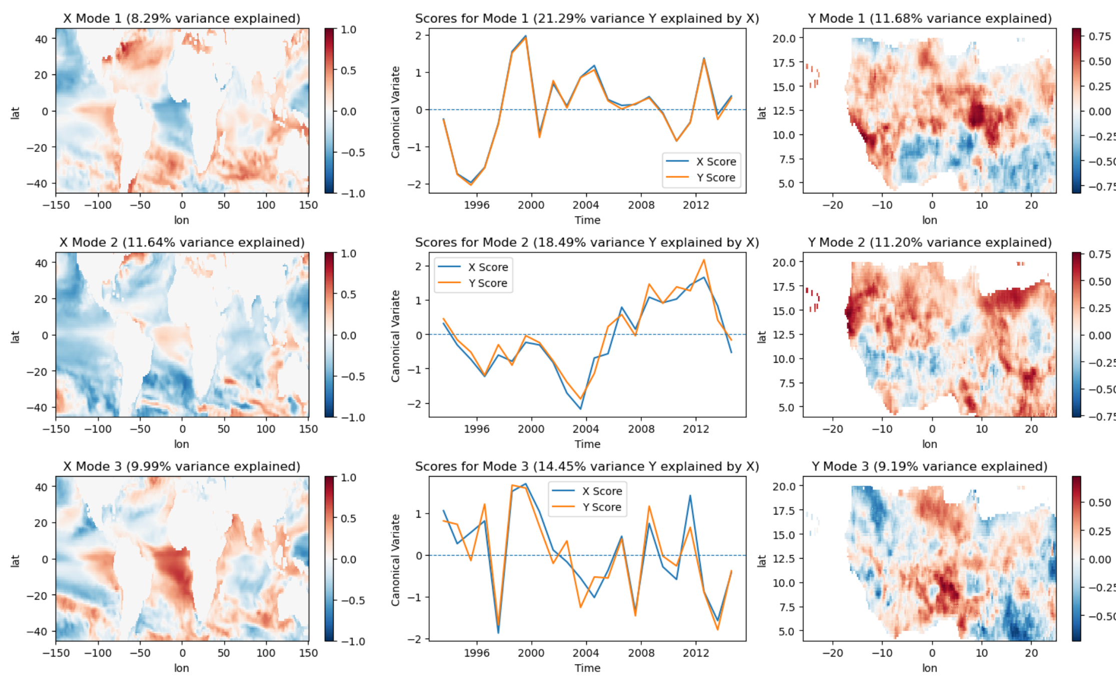
CCA Modes for model UKMO_604.SST



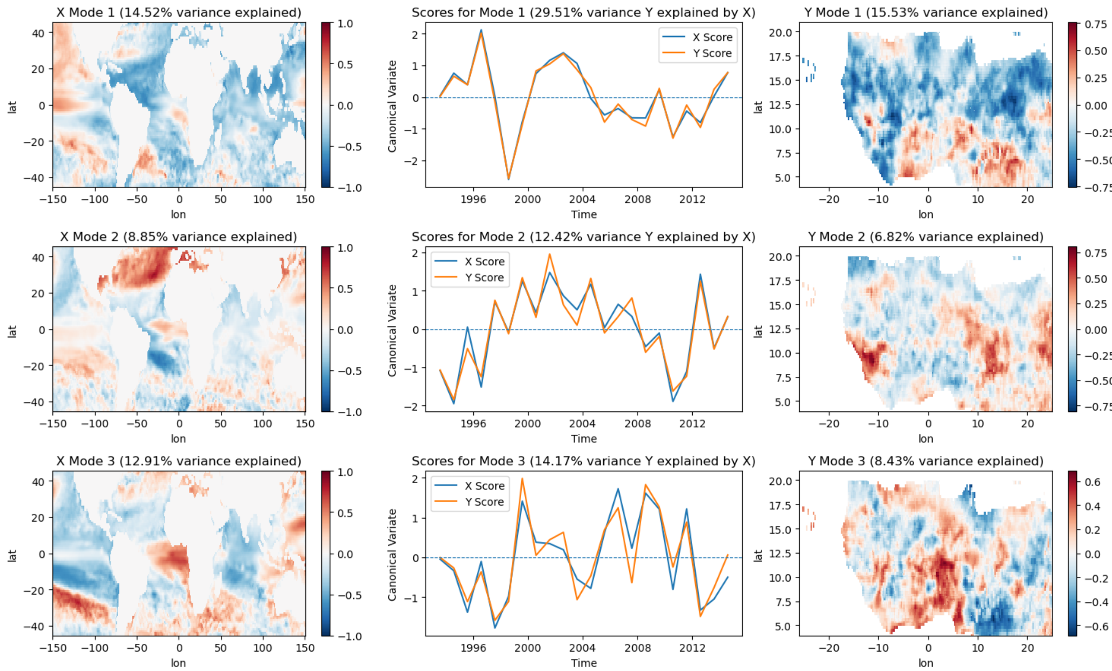
CCA Modes for model METEOFRANCE_8.SST



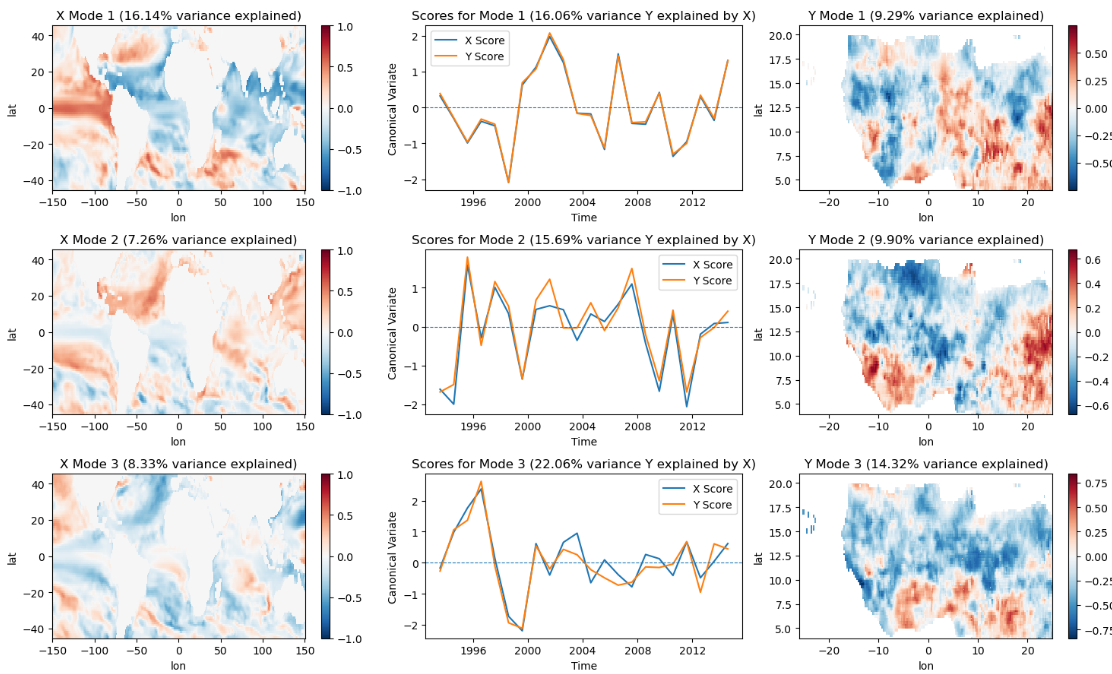
CCA Modes for model DWD_22.SST



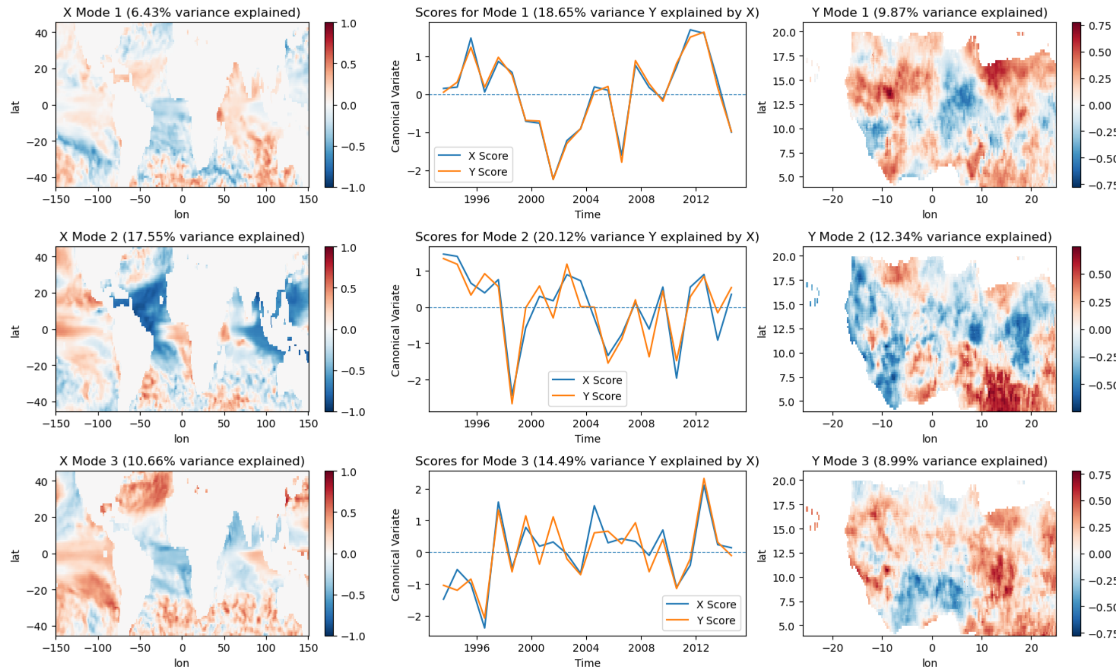
CCA Modes for model CMCC_35.SST



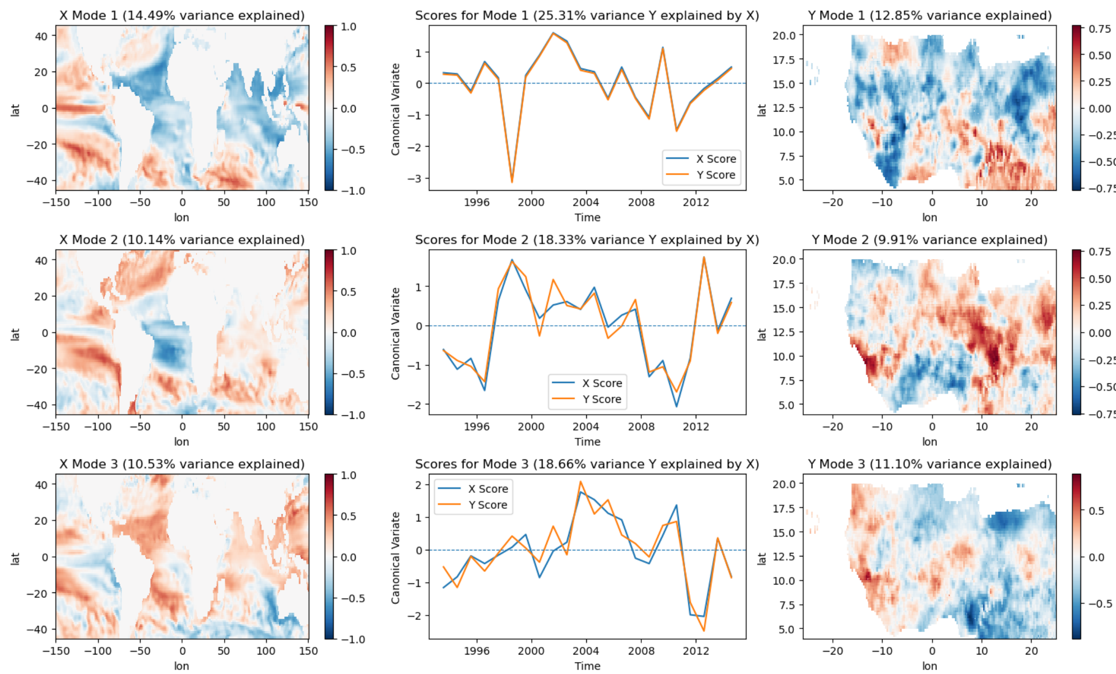
CCA Modes for model NCEP_2.SST



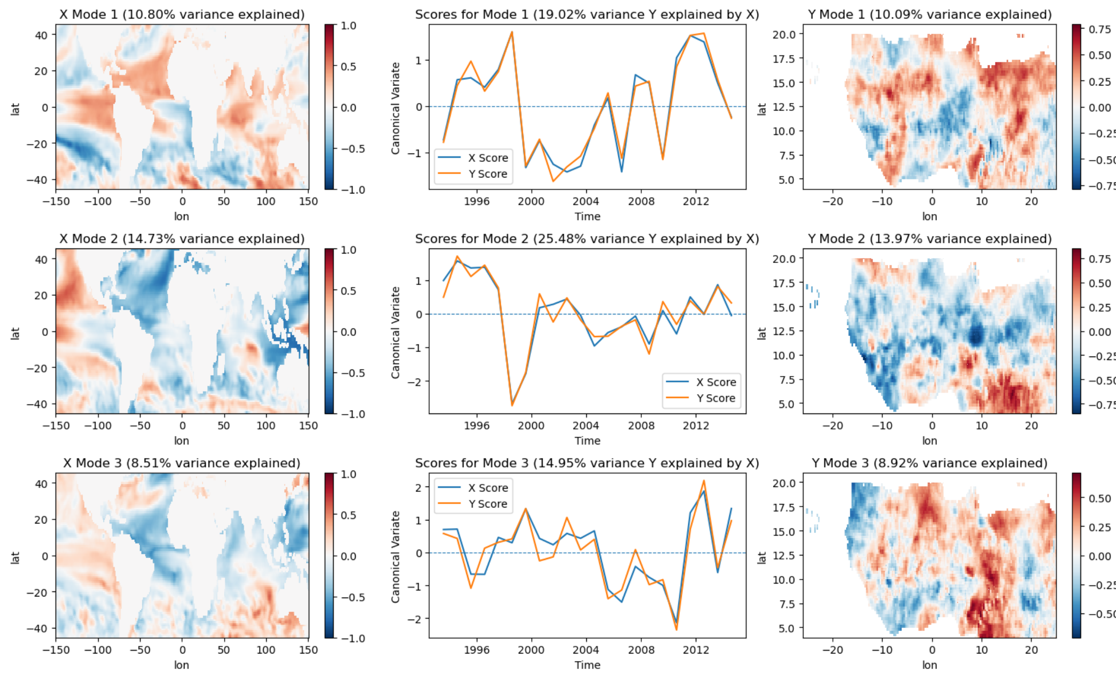
CCA Modes for model JMA_3.SST



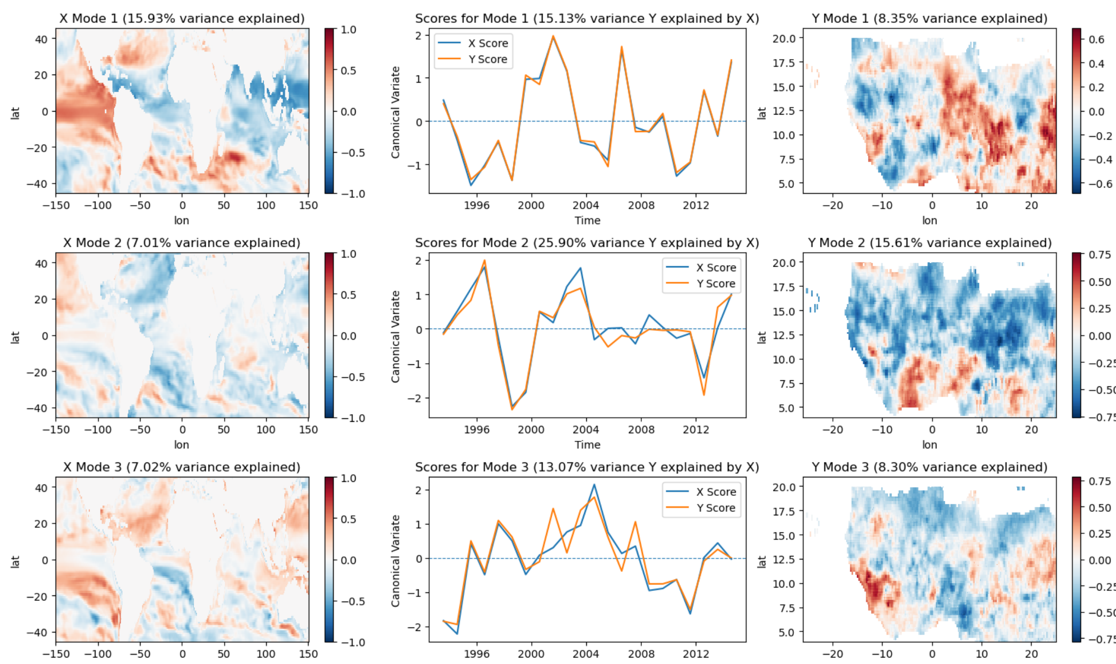
CCA Modes for model ECCC_4.SST



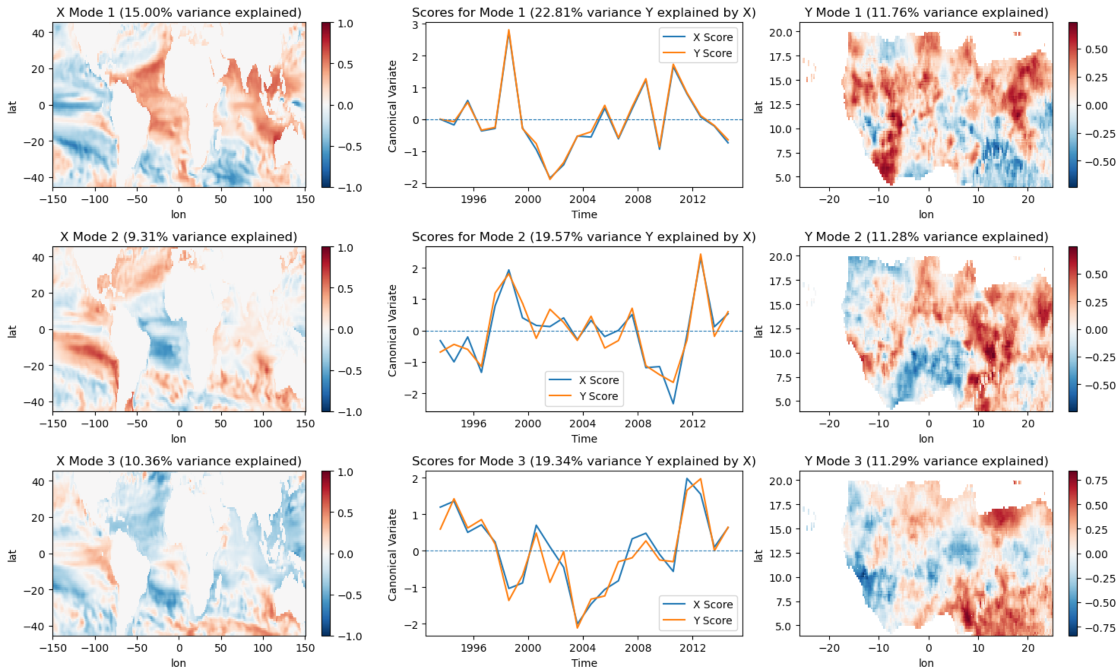
CCA Modes for model ECCC_5.SST



CCA Modes for model CFSV2_1.SST



CCA Modes for model CMC1_1.SST



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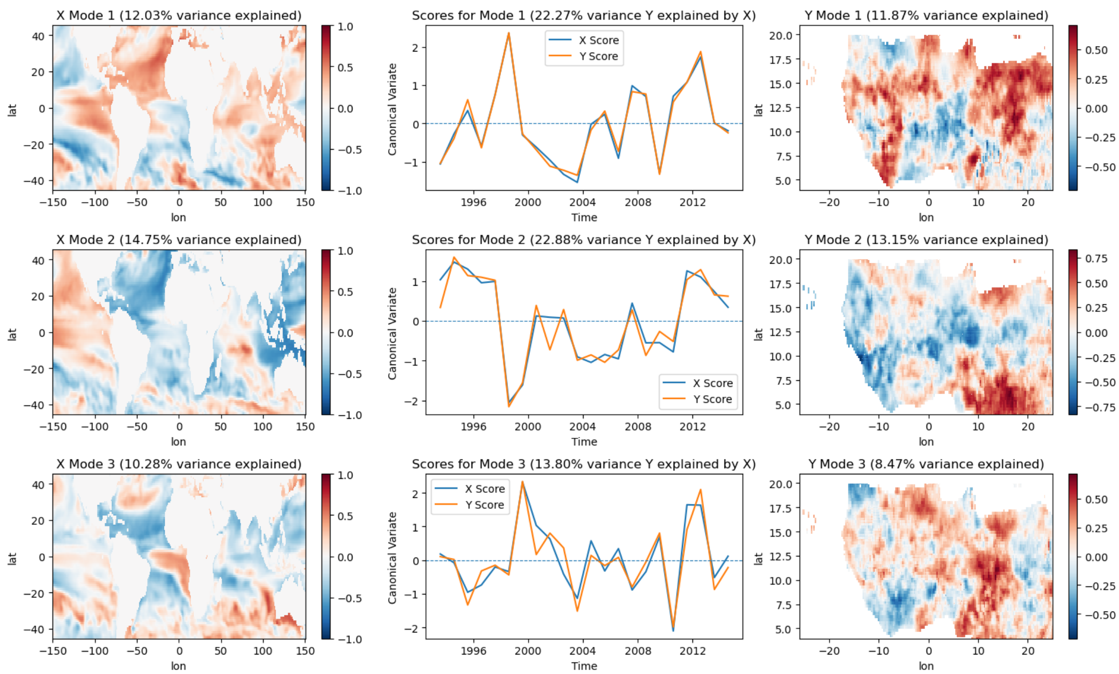

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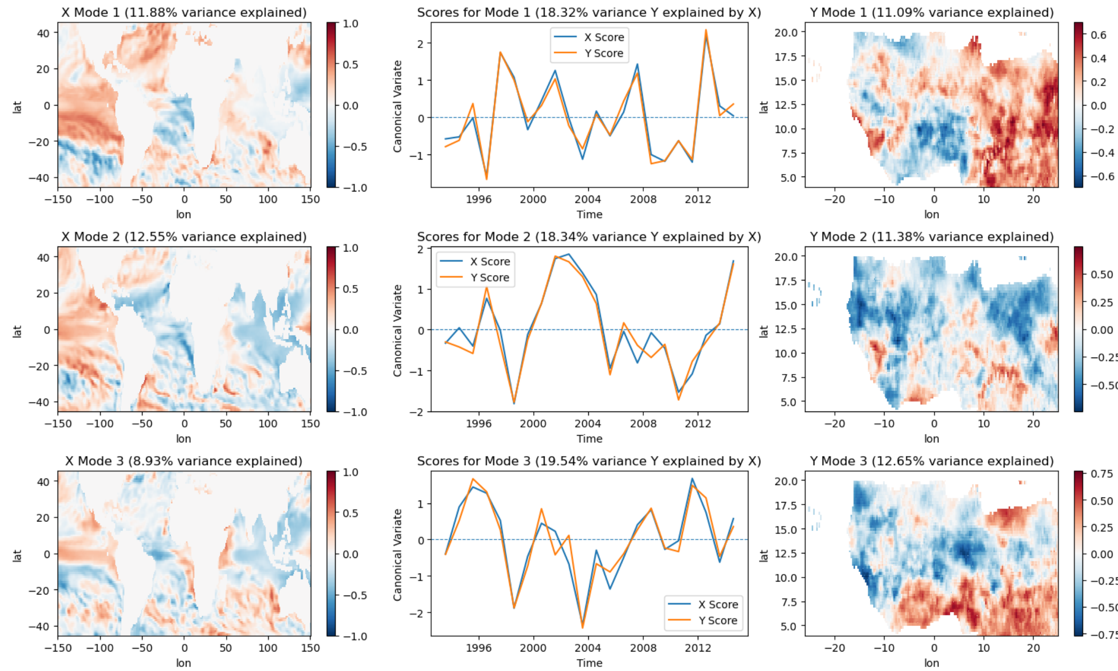

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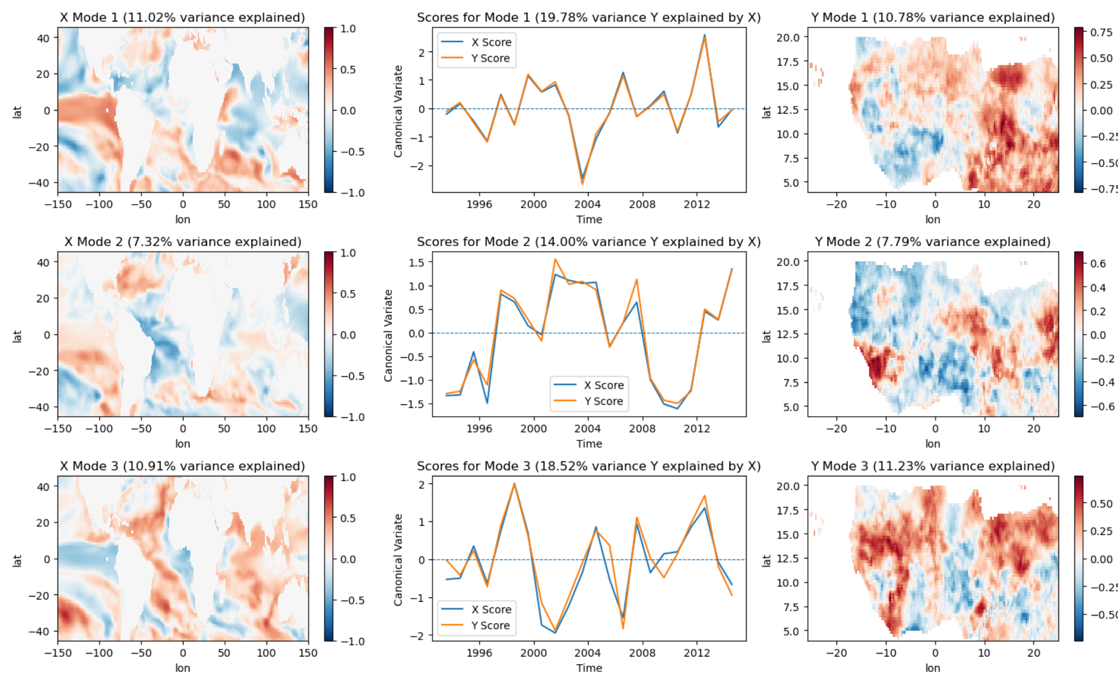

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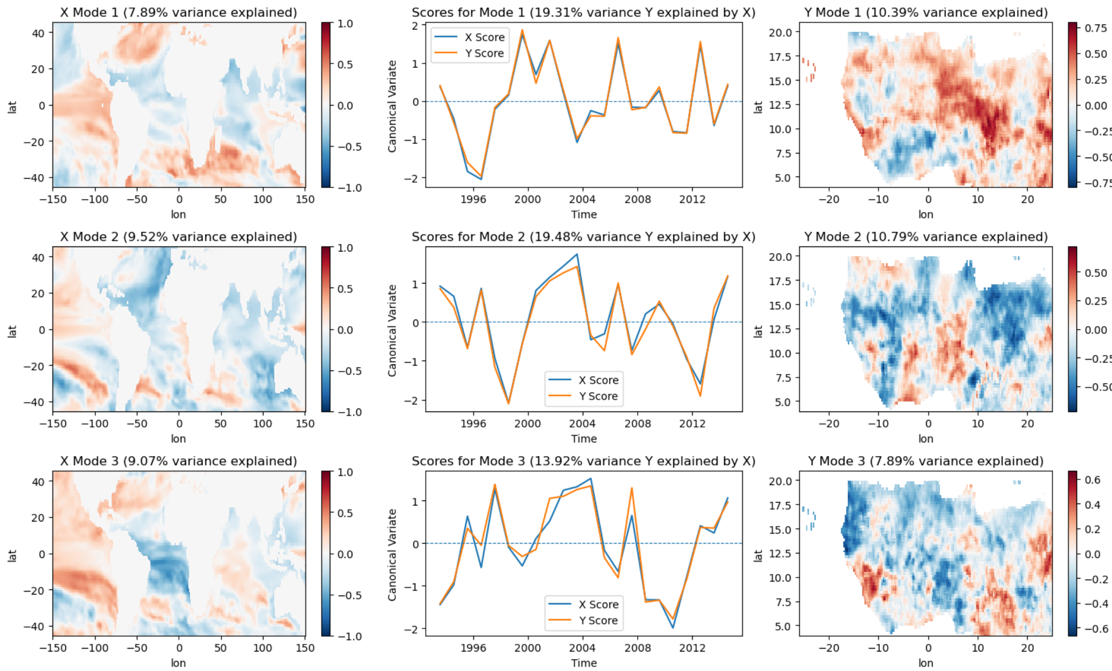

Cross validation

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UKMO_604
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METEOFRANCE_8
Cross-validation ongoing

100%|                    | 24/24 [00:10<00:00,  2.36it/s]

DWD_22
Cross-validation ongoing

100%|                    | 24/24 [00:10<00:00,  2.38it/s]

CMCC_35
Cross-validation ongoing

100%|                    | 24/24 [00:09<00:00,  2.41it/s]

NCEP_2
Cross-validation ongoing

100%|                    | 24/24 [00:09<00:00,  2.43it/s]
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JMA_3
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ECCC_4
Cross-validation ongoing

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ECCC_5
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CFSV2_1
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CMC1_1
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CMC2_1
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GFDL_1
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NCAR_CCSM4_1
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NMME_1
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Verification (individual) Deterministic

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Pearson Correlation
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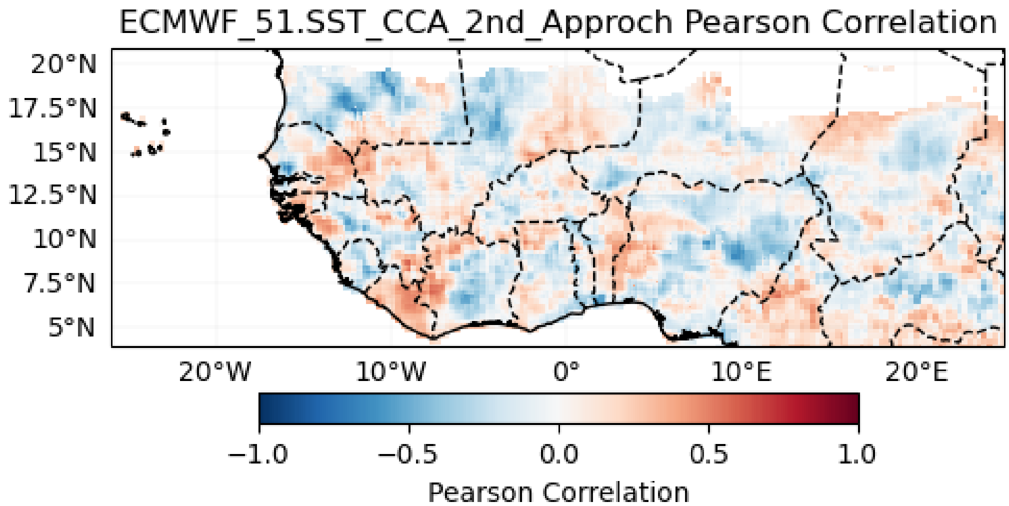
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17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
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10°W
0°
10°E
20°E
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−0.5
0.0
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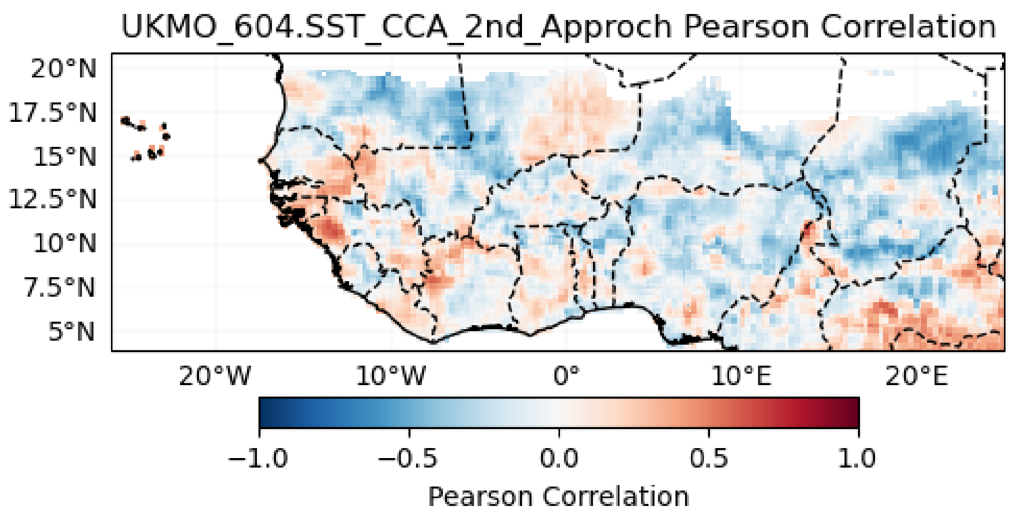
1.0
Pearson Correlation


UKMO_604.SST_CCA_2nd_Approch Pearson Correlation
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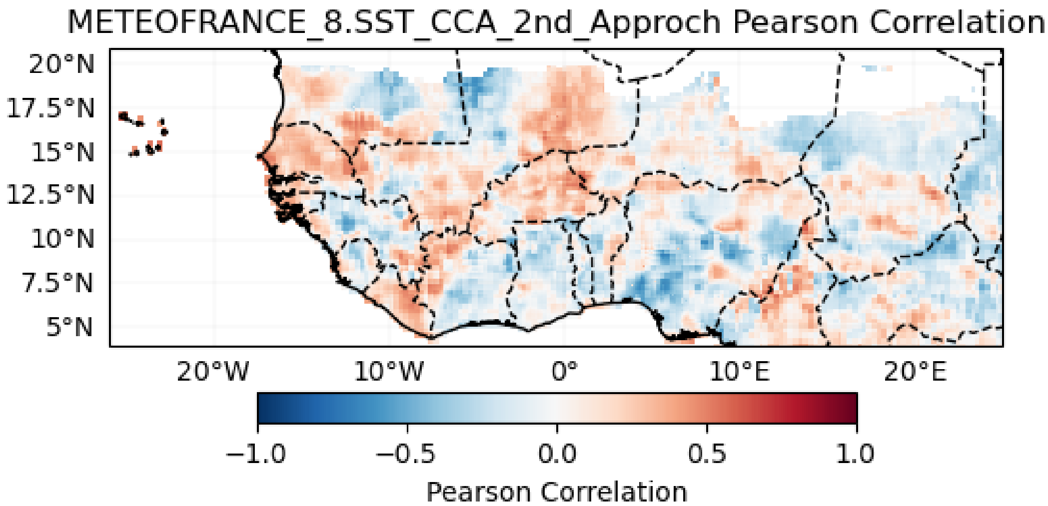
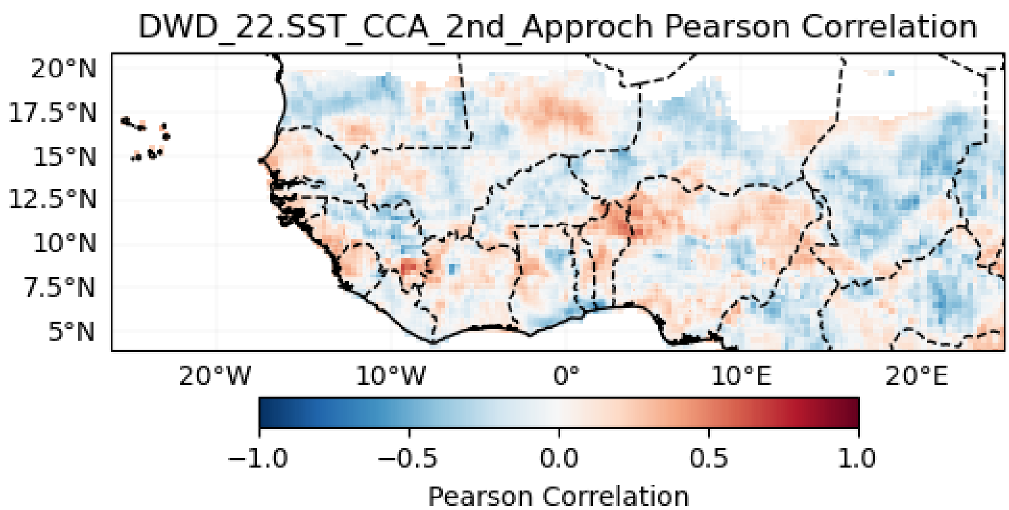
Pearson Correlation


METEOFRANCE_8.SST_CCA_2nd_Approch Pearson Correlation
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
−1.0
−0.5
0.0
0.5
1.0
Pearson Correlation


DWD_22.SST_CCA_2nd_Approch Pearson Correlation
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12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
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−0.5
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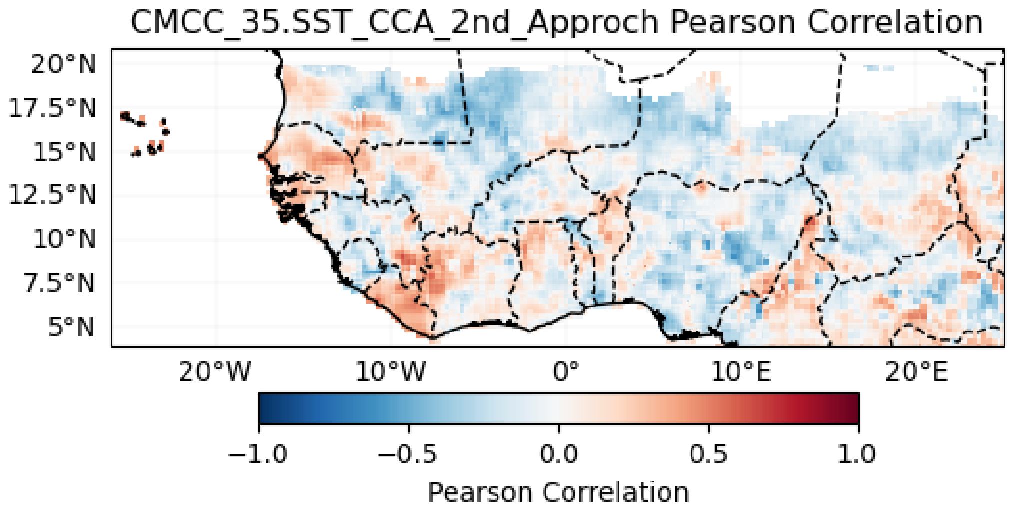
0.5
1.0
Pearson Correlation


CMCC_35.SST_CCA_2nd_Approch Pearson Correlation
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12.5°N
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7.5°N
5°N
20°W
10°W
0°
10°E
20°E
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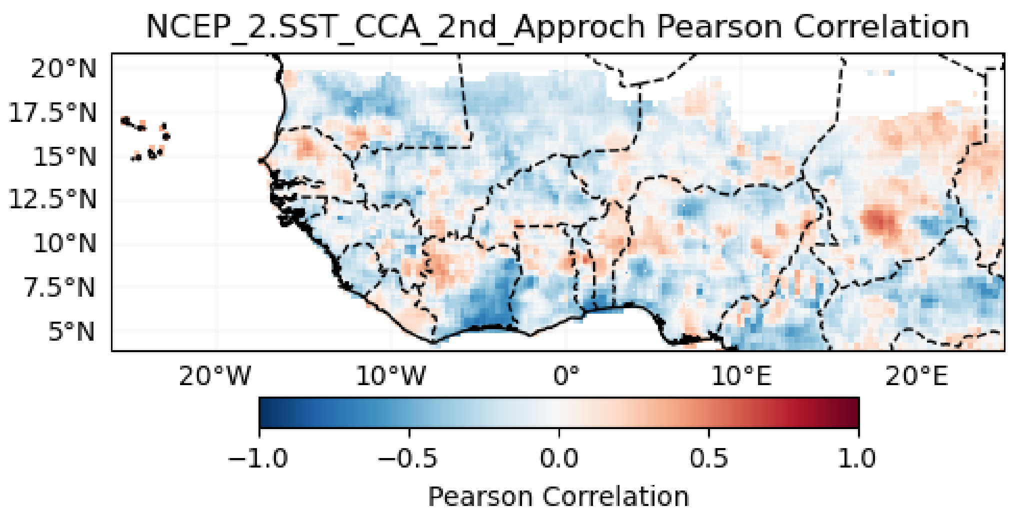
−0.5
0.0
0.5
1.0
Pearson Correlation


NCEP_2.SST_CCA_2nd_Approch Pearson Correlation
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15°N
12.5°N
10°N
7.5°N
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10°W
0°
10°E
20°E
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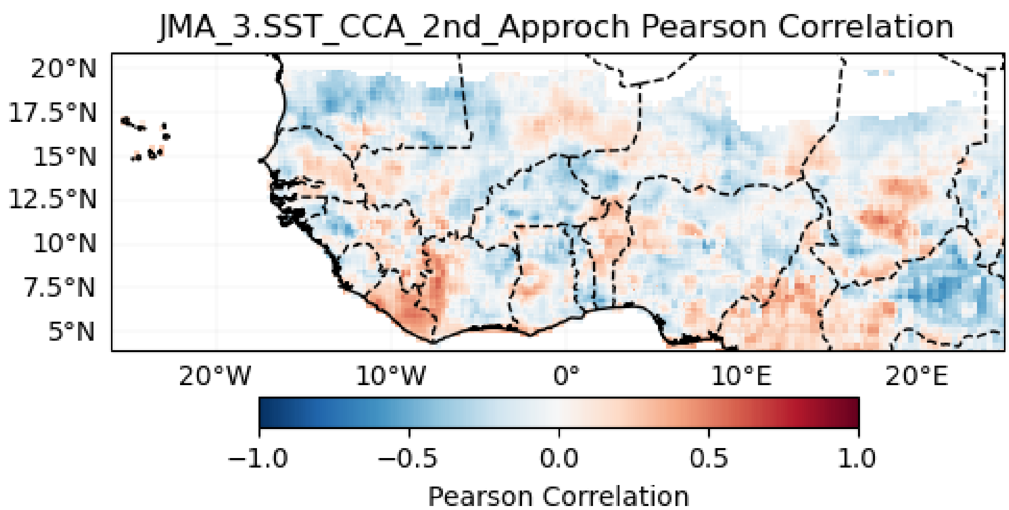
−0.5
0.0
0.5
1.0
Pearson Correlation


JMA_3.SST_CCA_2nd_Approch Pearson Correlation
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7.5°N
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10°W
0°
10°E
20°E
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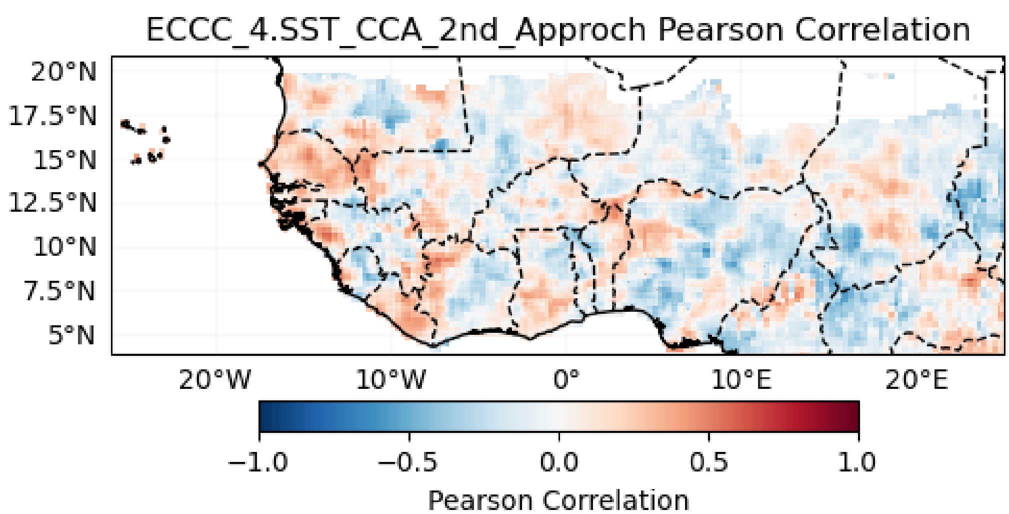
0.0
0.5
1.0
Pearson Correlation


ECCC_4.SST_CCA_2nd_Approch Pearson Correlation
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0°
10°E
20°E
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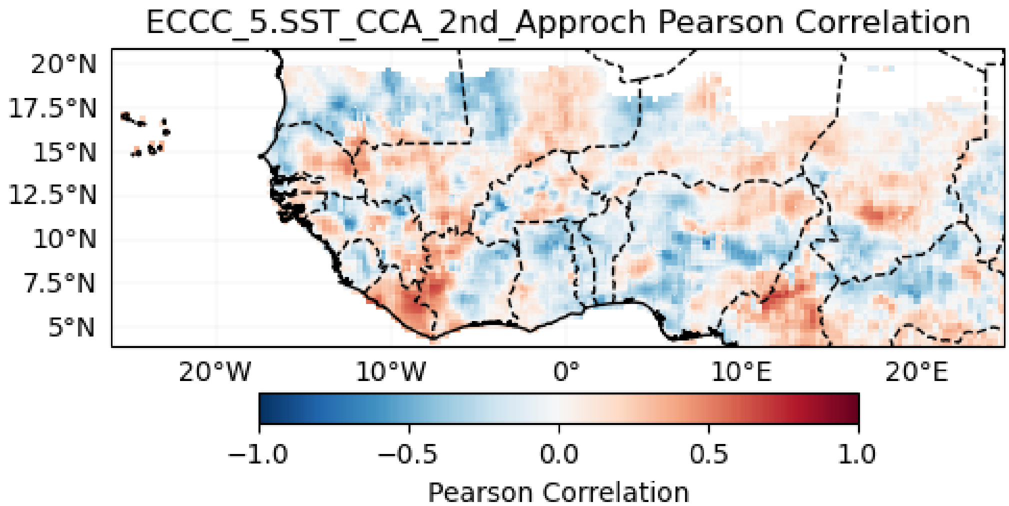
−0.5
0.0
0.5
1.0
Pearson Correlation


ECCC_5.SST_CCA_2nd_Approch Pearson Correlation
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10°E
20°E
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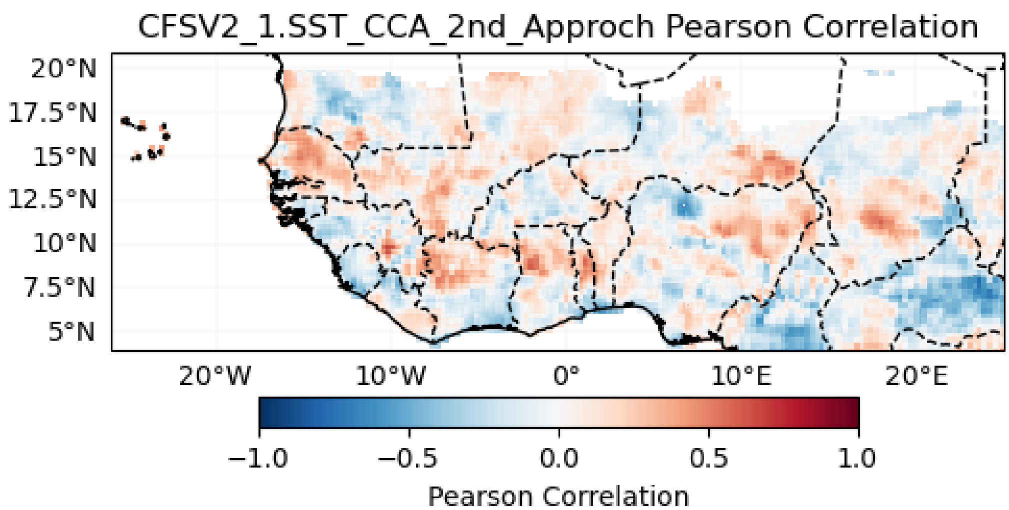
0.5
1.0
Pearson Correlation


CFSV2_1.SST_CCA_2nd_Approch Pearson Correlation
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12.5°N
10°N
7.5°N
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10°W
0°
10°E
20°E
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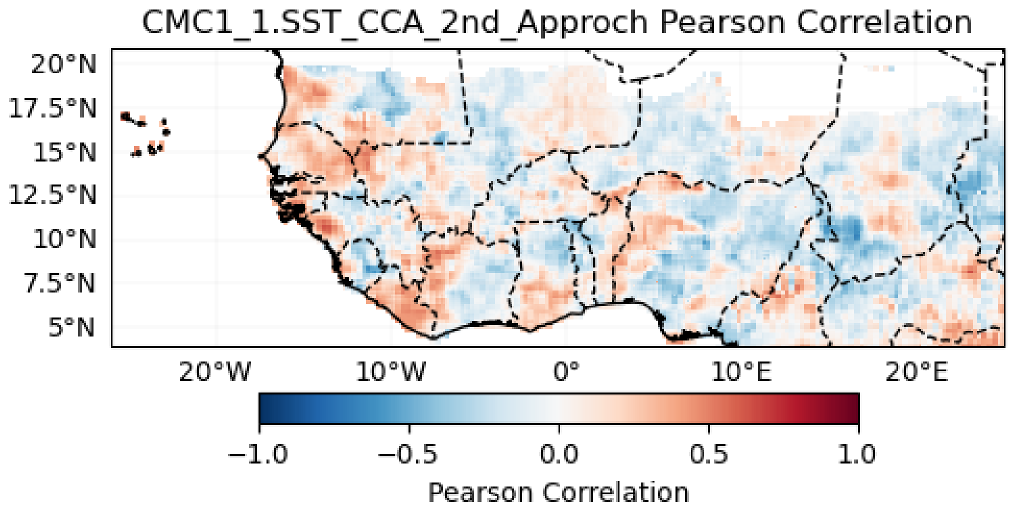
−0.5
0.0
0.5
1.0
Pearson Correlation


CMC1_1.SST_CCA_2nd_Approch Pearson Correlation
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17.5°N
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12.5°N
10°N
7.5°N
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0°
10°E
20°E
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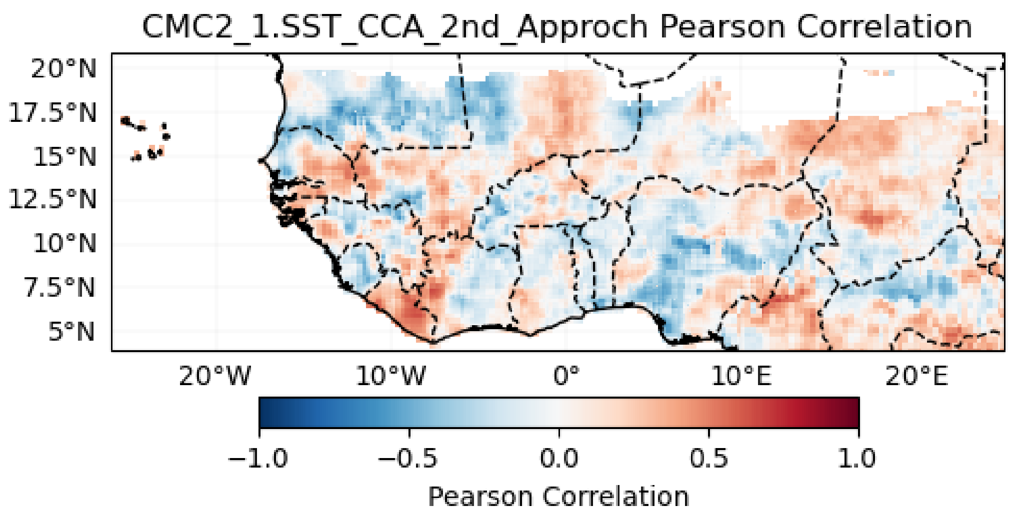
−0.5
0.0
0.5
1.0
Pearson Correlation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12.5°N
10°N
7.5°N
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0°
10°E
20°E
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−0.5
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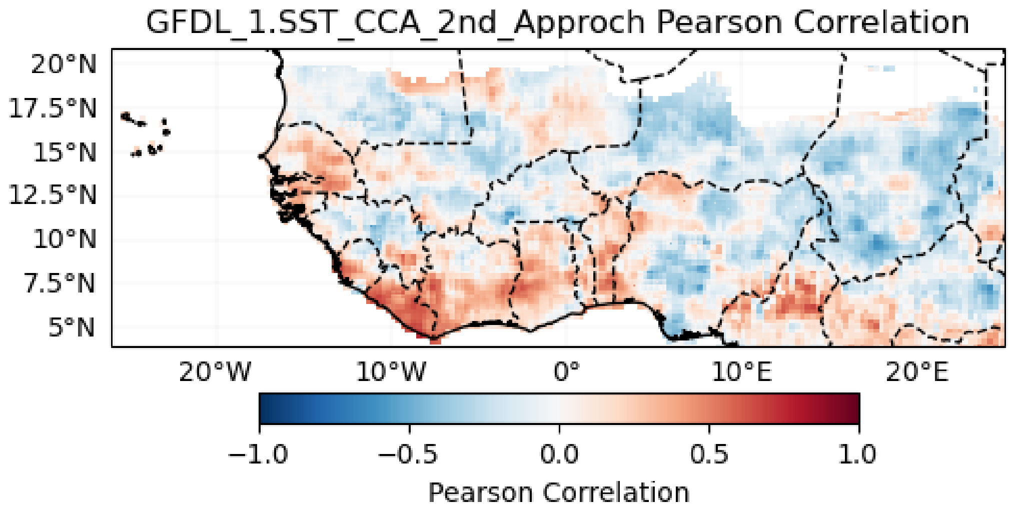
0.5
1.0
Pearson Correlation


GFDL_1.SST_CCA_2nd_Approch Pearson Correlation
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12.5°N
10°N
7.5°N
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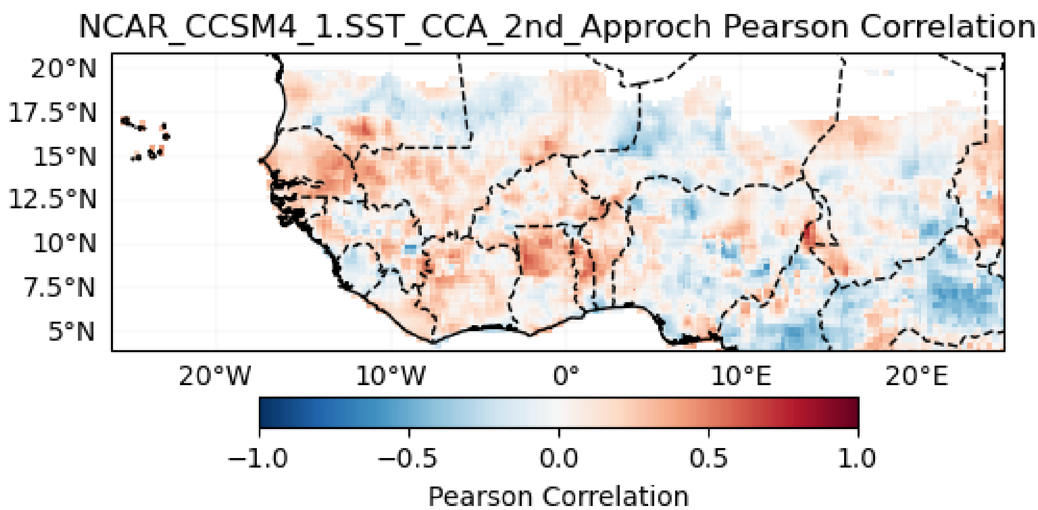
20°W
10°W
0°
10°E
20°E
−1.0
−0.5
0.0
0.5
1.0
Pearson Correlation


NCAR_CCSM4_1.SST_CCA_2nd_Approch Pearson Correlation
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12.5°N
10°N
7.5°N
5°N
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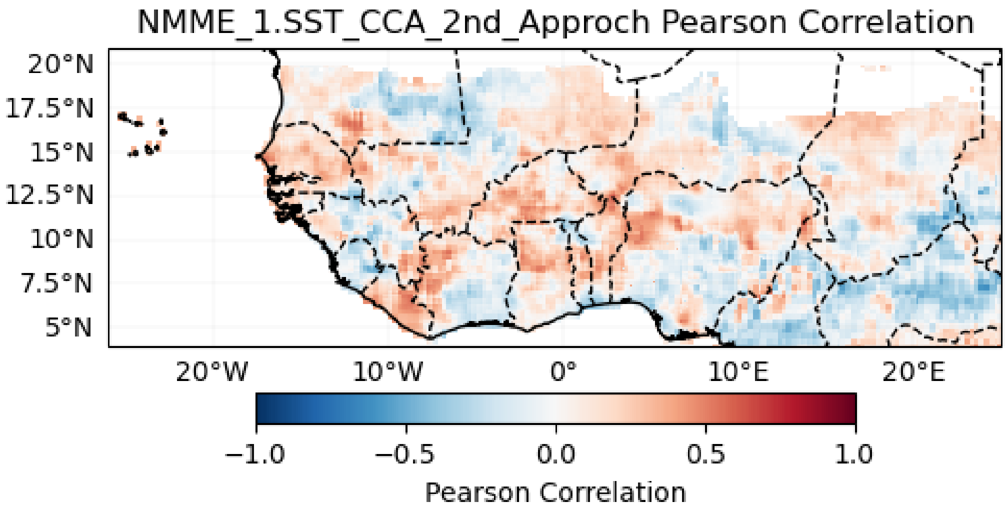
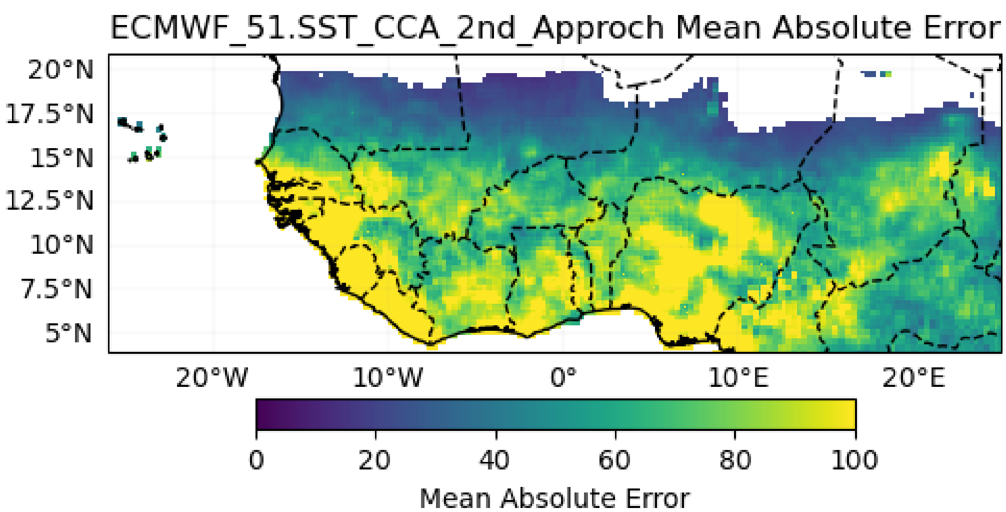
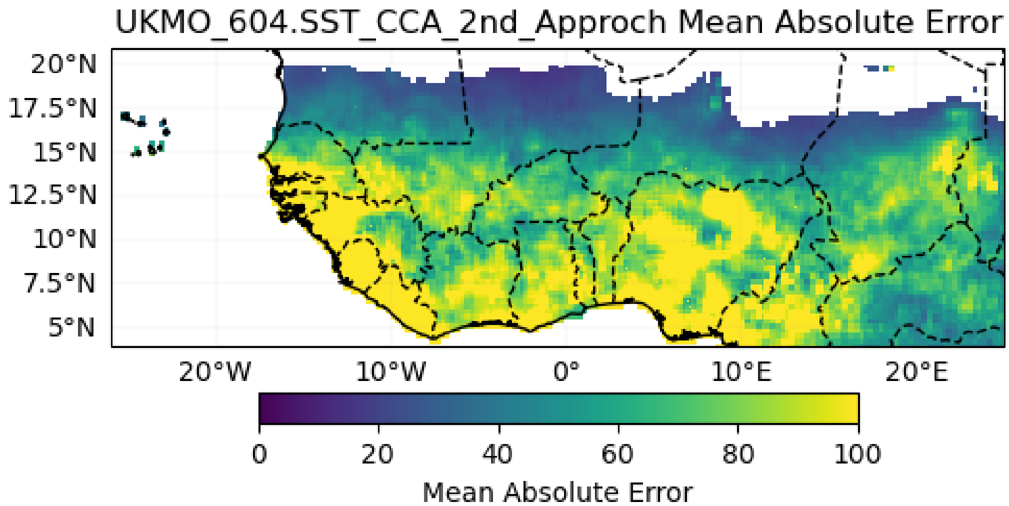
10°W
0°
10°E
20°E
−1.0
−0.5
0.0
0.5
1.0
Pearson Correlation



Mean Absolute Error




UKMO_604.SST_CCA_2nd_Approch Mean Absolute Error
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12.5°N
10°N
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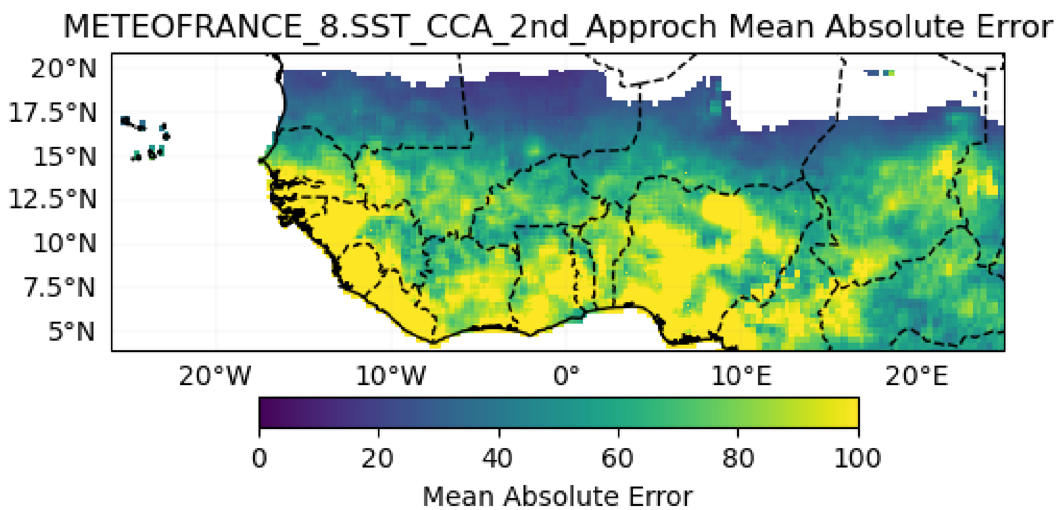
20°W
10°W
0°
10°E
20°E
0
20
40
60
80
100
Mean Absolute Error


METEOFRANCE_8.SST_CCA_2nd_Approch Mean Absolute Error
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15°N
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10°N
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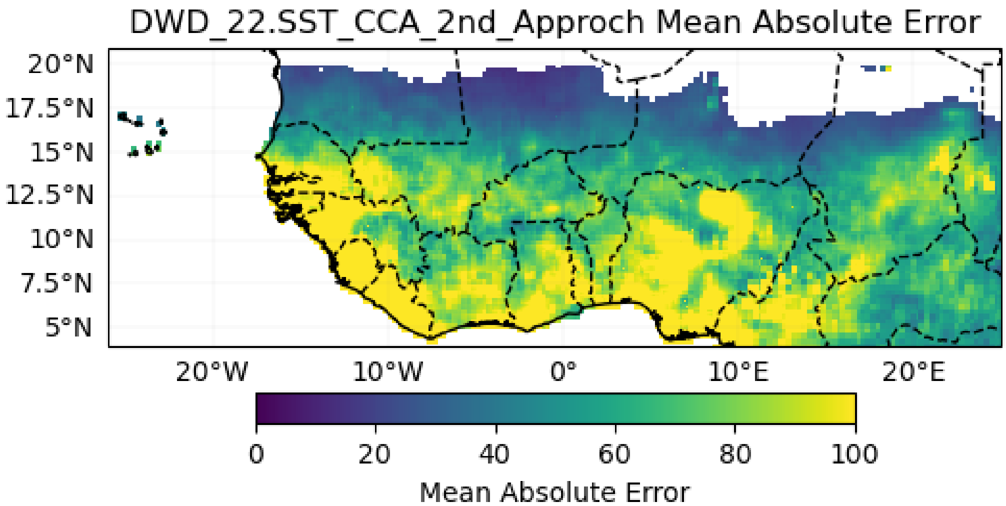
5°N
20°W
10°W
0°
10°E
20°E
0
20
40
60
80
100
Mean Absolute Error


DWD_22.SST_CCA_2nd_Approch Mean Absolute Error
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0°
10°E
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0
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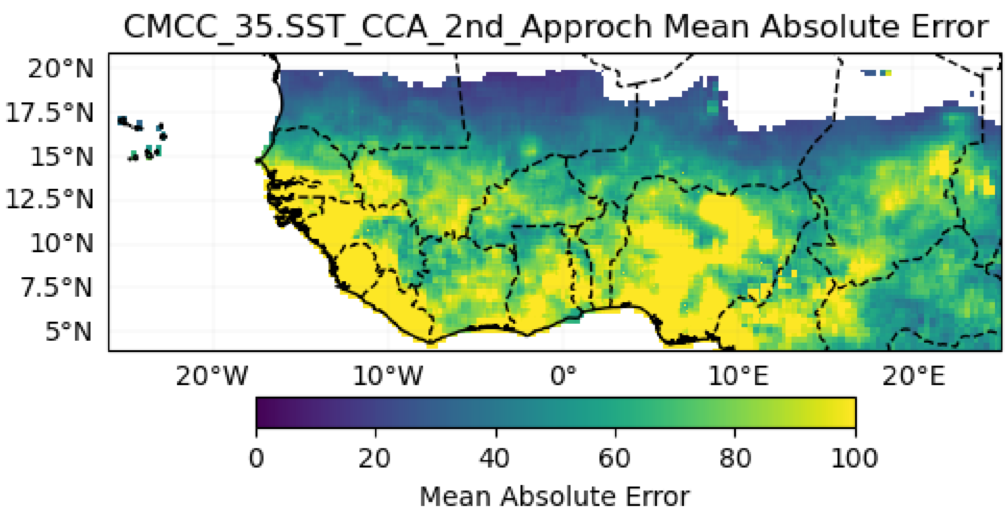
80
100
Mean Absolute Error


CMCC_35.SST_CCA_2nd_Approch Mean Absolute Error
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10°E
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0
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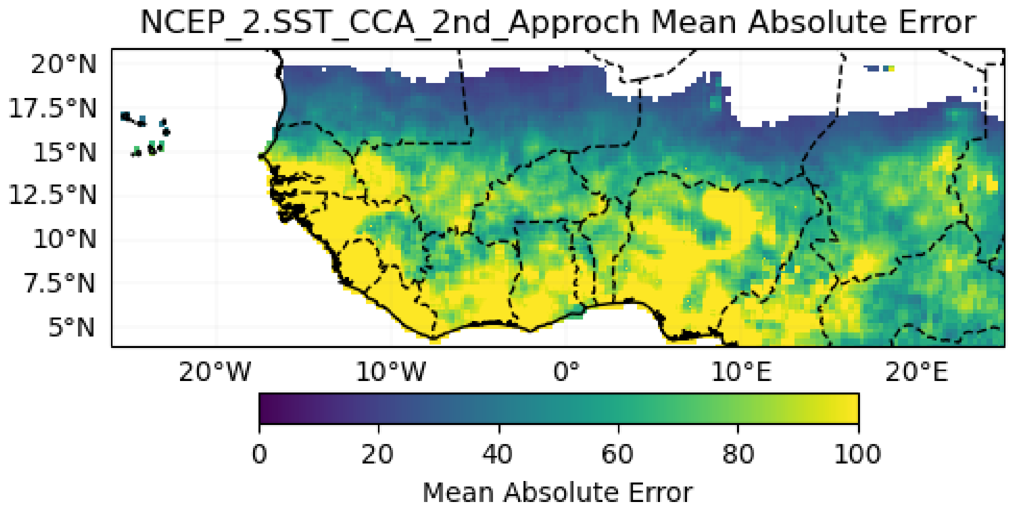
Mean Absolute Error


NCEP_2.SST_CCA_2nd_Approch Mean Absolute Error
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10°N
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10°W
0°
10°E
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0
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40
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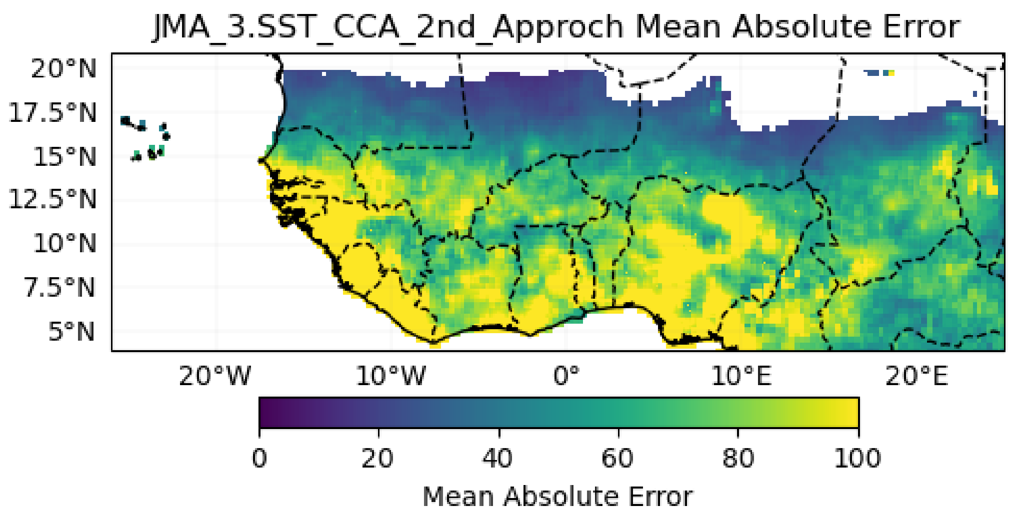
80
100
Mean Absolute Error


JMA_3.SST_CCA_2nd_Approch Mean Absolute Error
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10°N
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10°W
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10°E
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0
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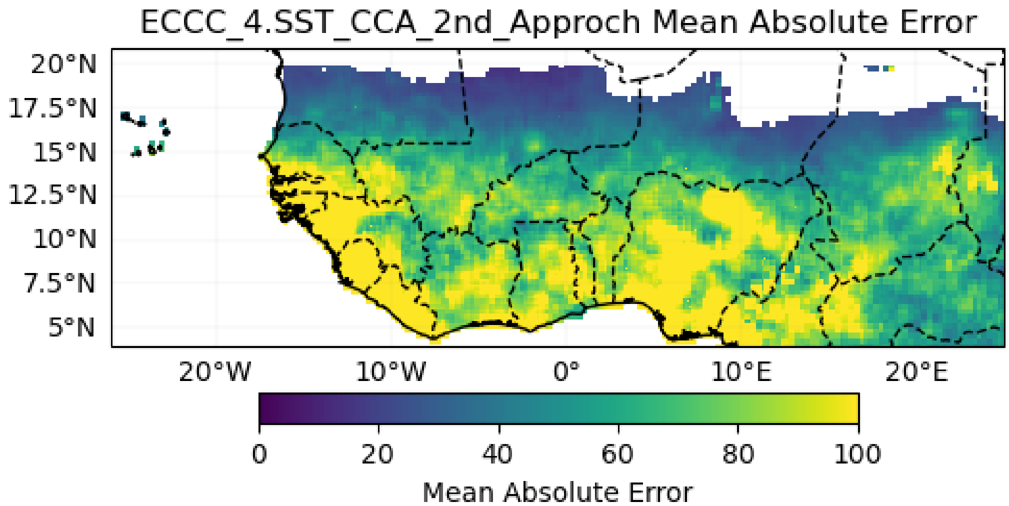
60
80
100
Mean Absolute Error


ECCC_4.SST_CCA_2nd_Approch Mean Absolute Error
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10°W
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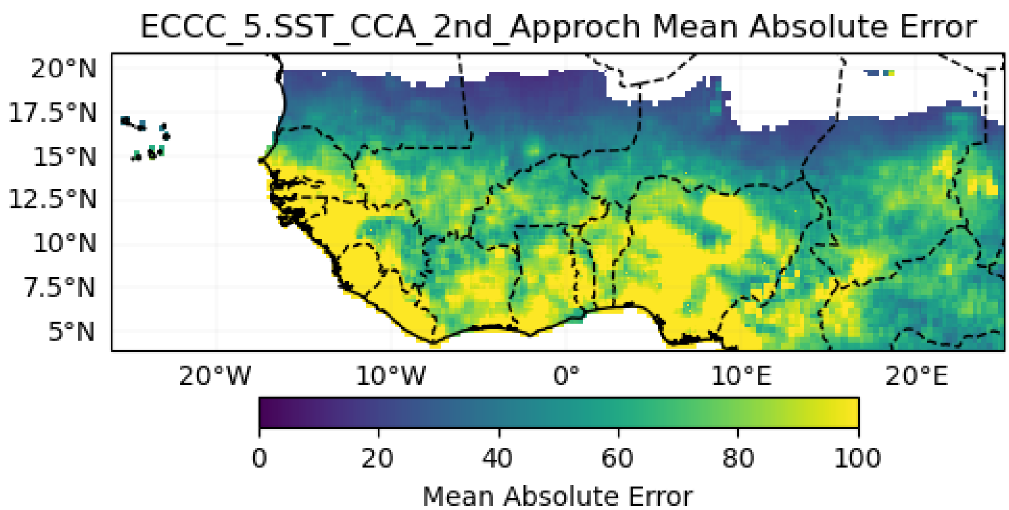
10°E
20°E
0
20
40
60
80
100
Mean Absolute Error


ECCC_5.SST_CCA_2nd_Approch Mean Absolute Error
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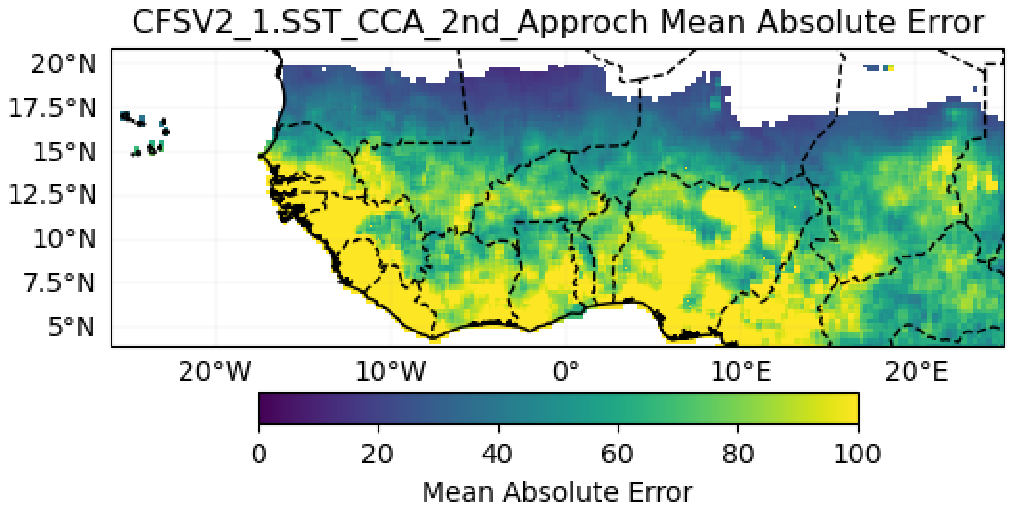
80
100
Mean Absolute Error


CFSV2_1.SST_CCA_2nd_Approch Mean Absolute Error
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10°N
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10°W
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0
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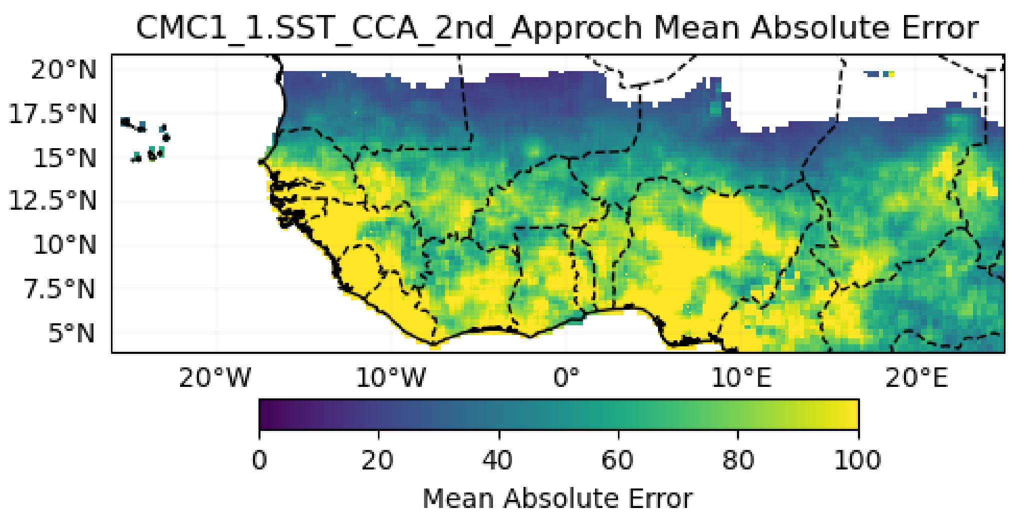
60
80
100
Mean Absolute Error


CMC1_1.SST_CCA_2nd_Approch Mean Absolute Error
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10°W
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0
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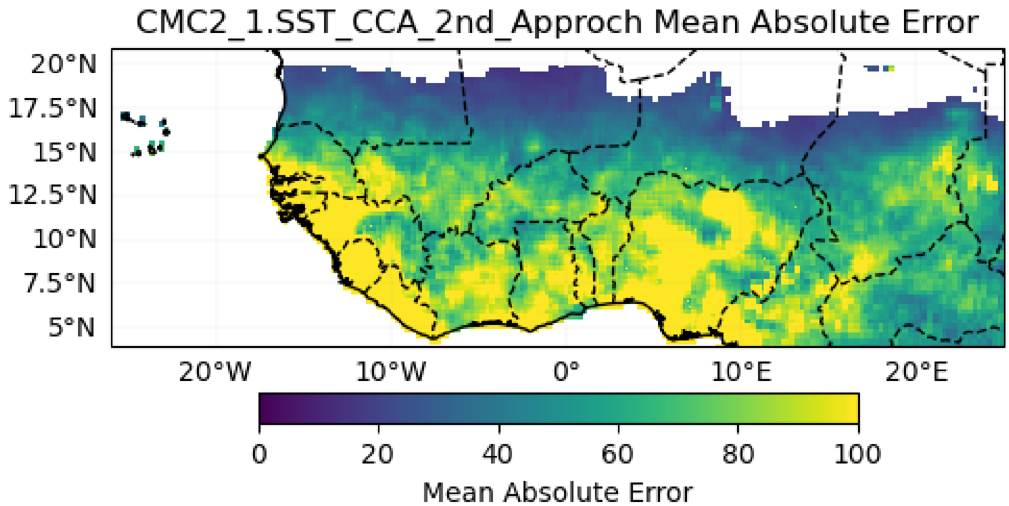
80
100
Mean Absolute Error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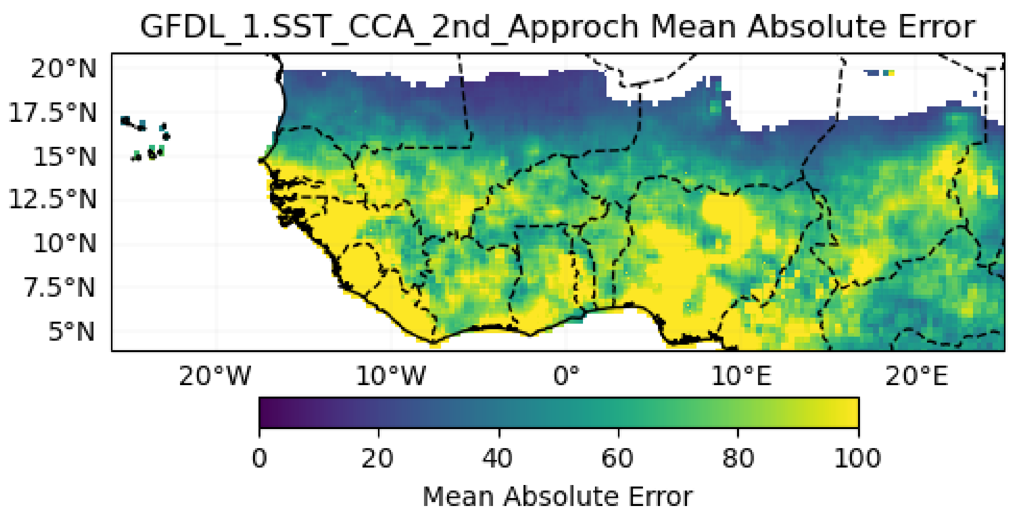
80
100
Mean Absolute Error


GFDL_1.SST_CCA_2nd_Approch Mean Absolute Error
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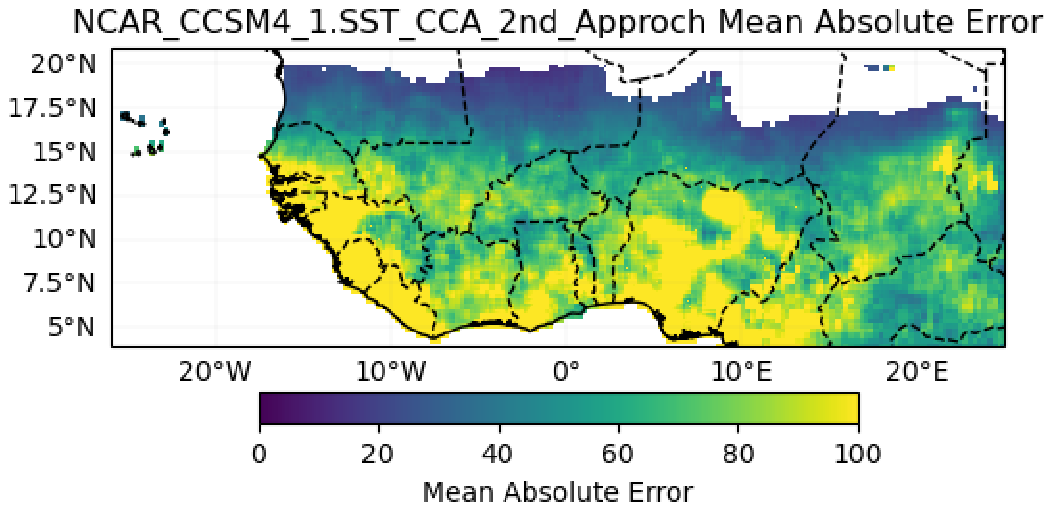
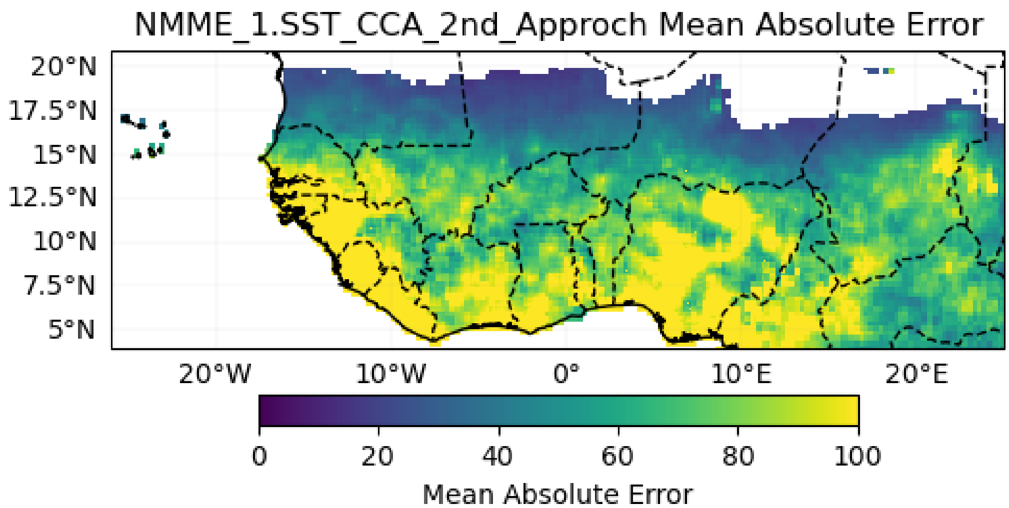
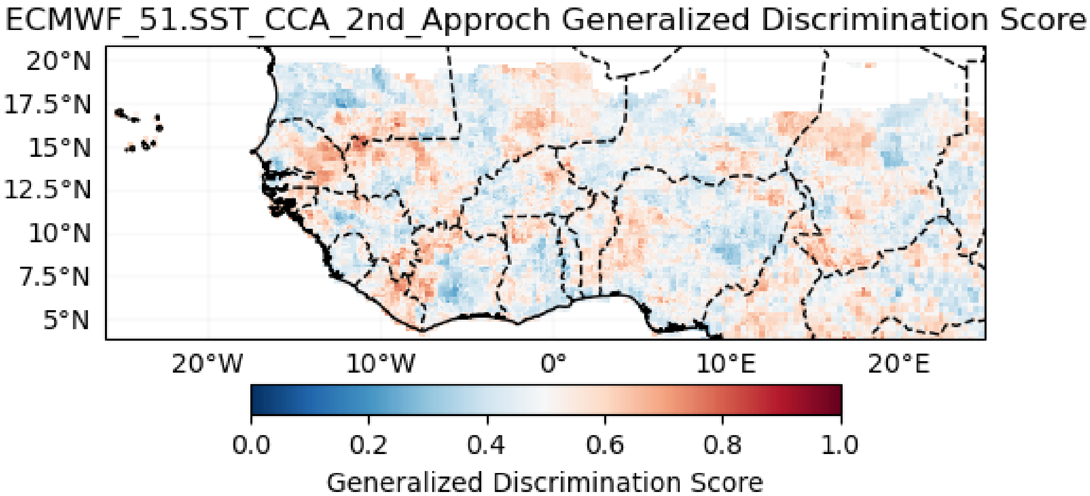
60
80
100
Mean Absolute Error





Probabilistic

`Generalized Discrimination Score`


ECMWF_51.SST_CCA_2nd_Approch Generalized Discrimination Score
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10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
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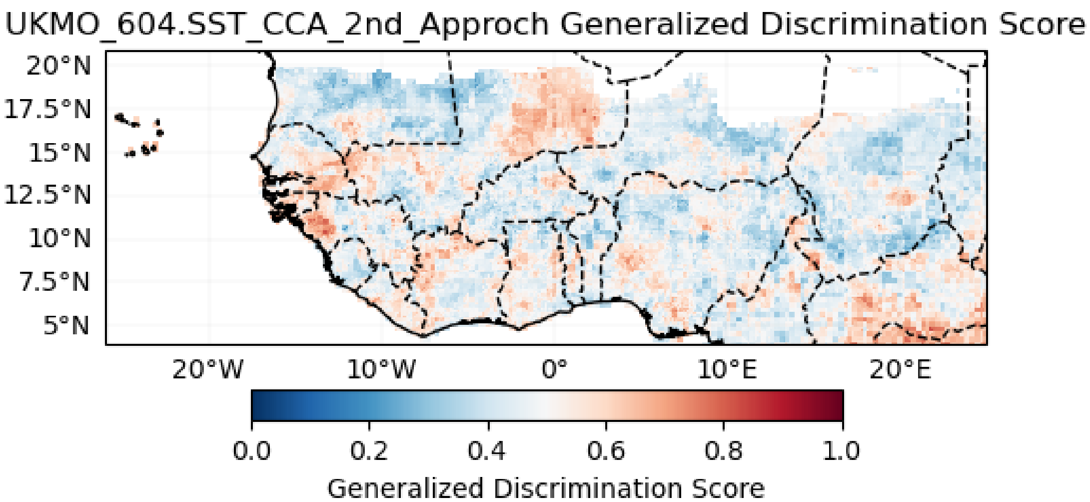
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


UKMO_604.SST_CCA_2nd_Approch Generalized Discrimination Score
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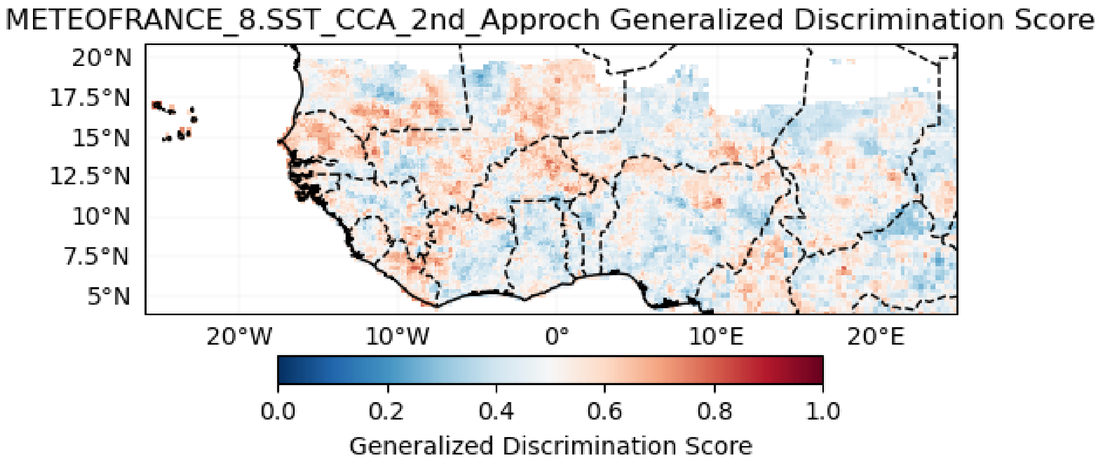
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


METEOFRANCE_8.SST_CCA_2nd_Approch Generalized Discrimination Score
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7.5°N
5°N
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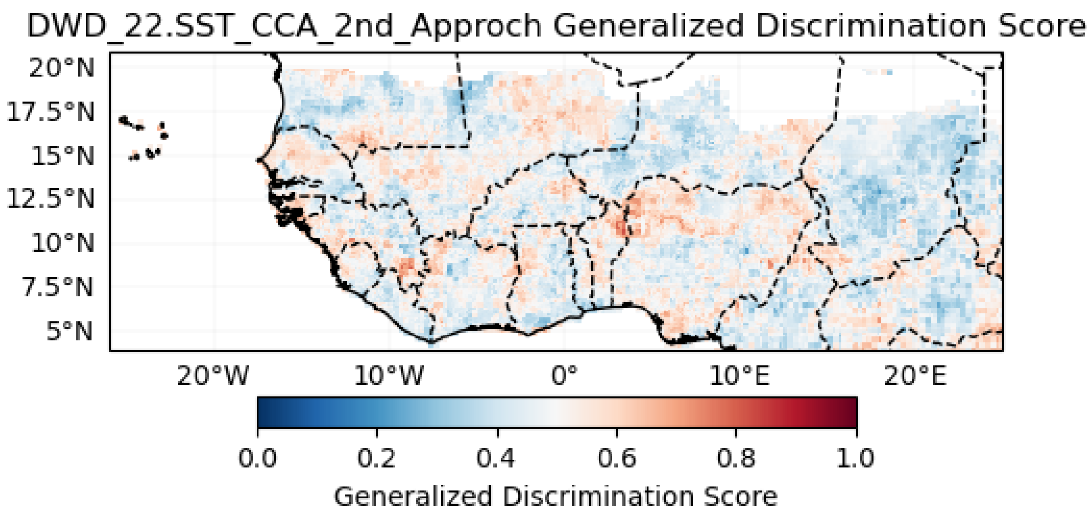
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


DWD_22.SST_CCA_2nd_Approch Generalized Discrimination Score
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7.5°N
5°N
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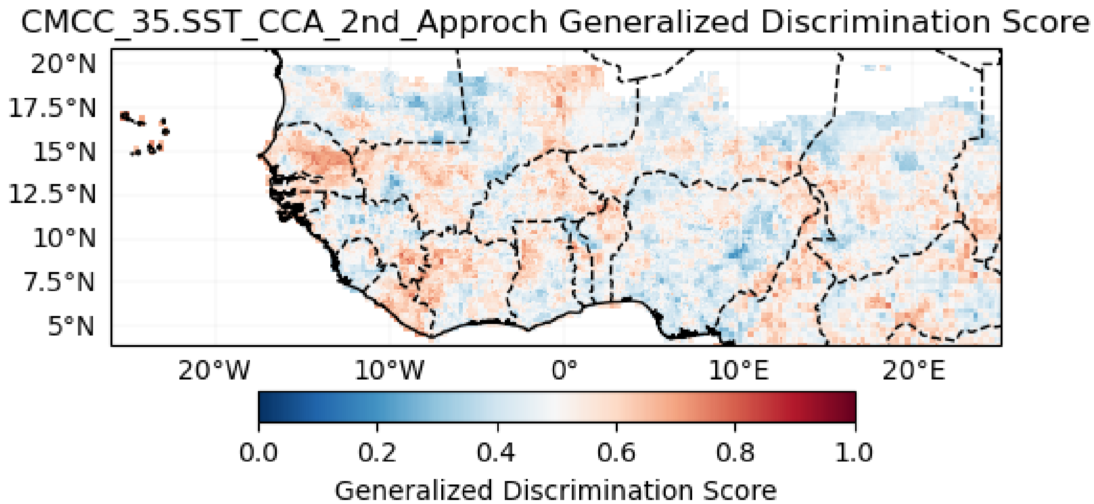
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


CMCC_35.SST_CCA_2nd_Approch Generalized Discrimination Score
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5°N
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10°E
20°E
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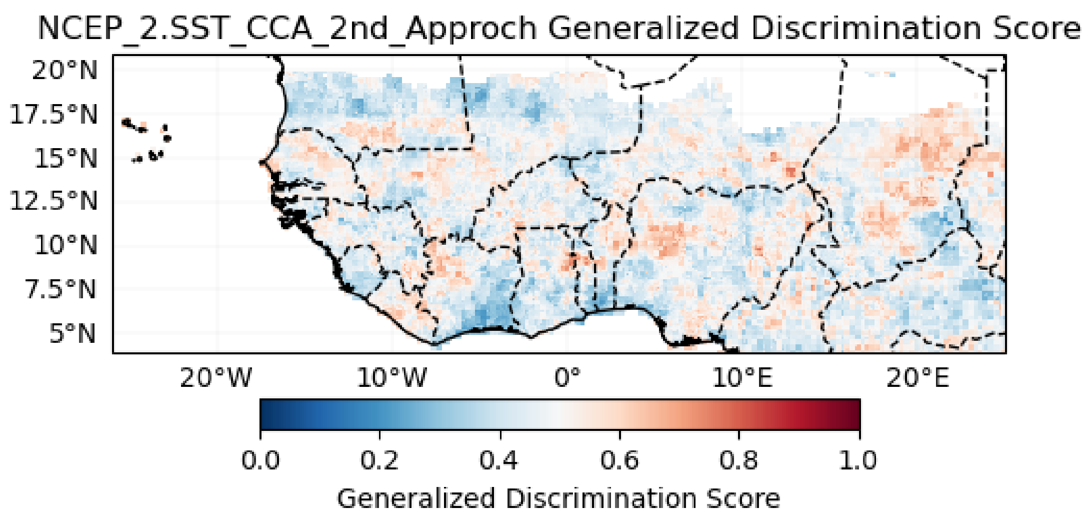
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


NCEP_2.SST_CCA_2nd_Approch Generalized Discrimination Score
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15°N
12.5°N
10°N
7.5°N
5°N
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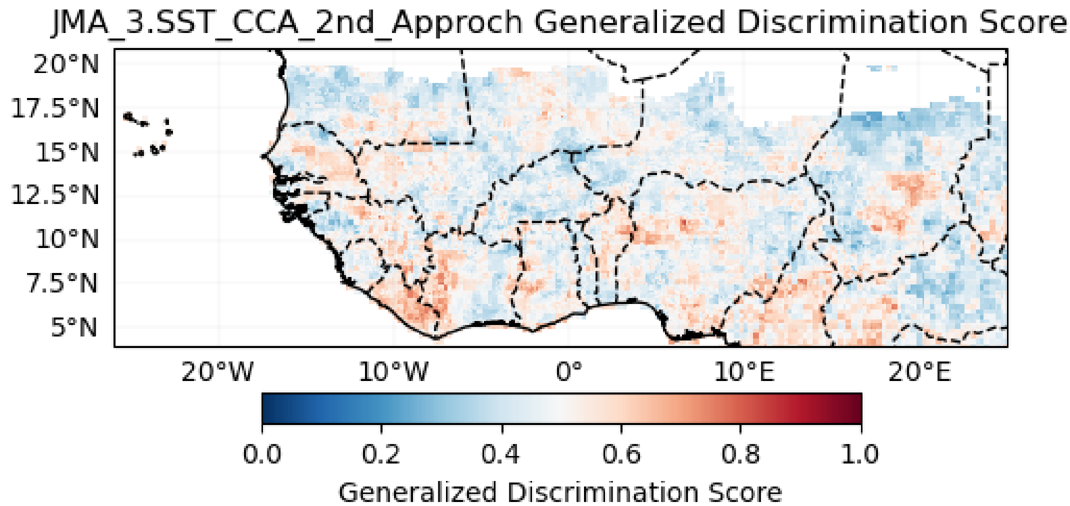
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


JMA_3.SST_CCA_2nd_Approch Generalized Discrimination Score
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7.5°N
5°N
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10°W
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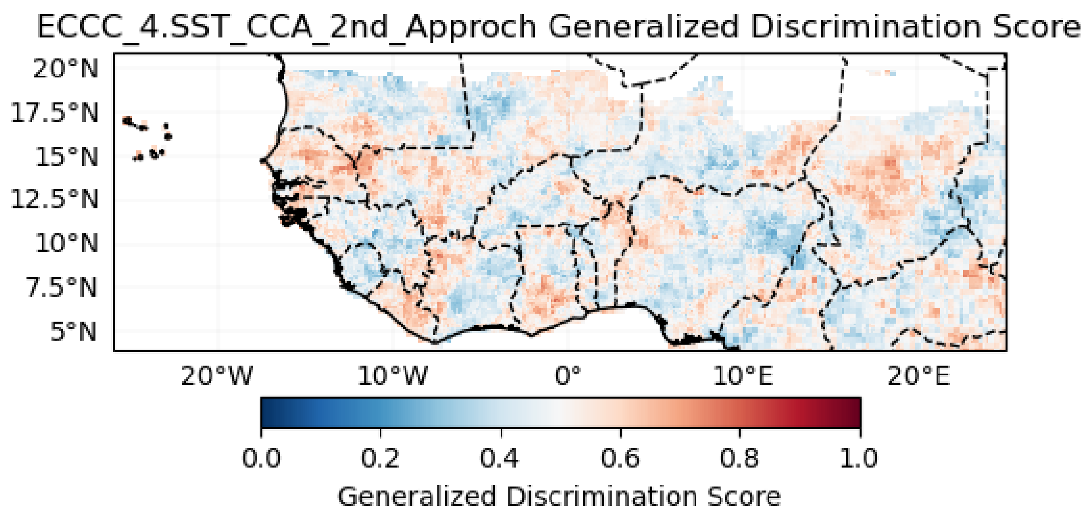
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


ECCC_4.SST_CCA_2nd_Approch Generalized Discrimination Score
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15°N
12.5°N
10°N
7.5°N
5°N
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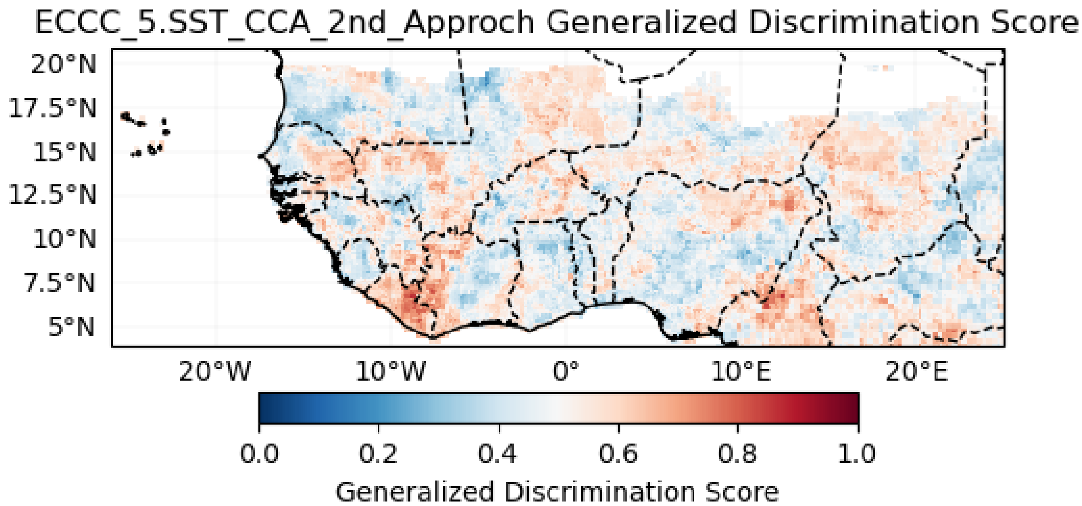
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


ECCC_5.SST_CCA_2nd_Approch Generalized Discrimination Score
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7.5°N
5°N
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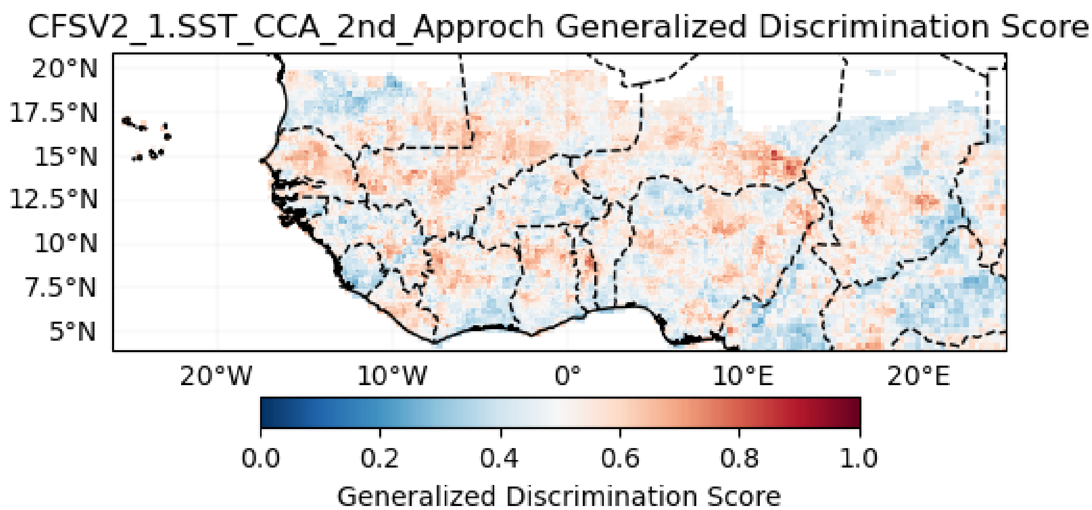
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


CFSV2_1.SST_CCA_2nd_Approch Generalized Discrimination Score
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7.5°N
5°N
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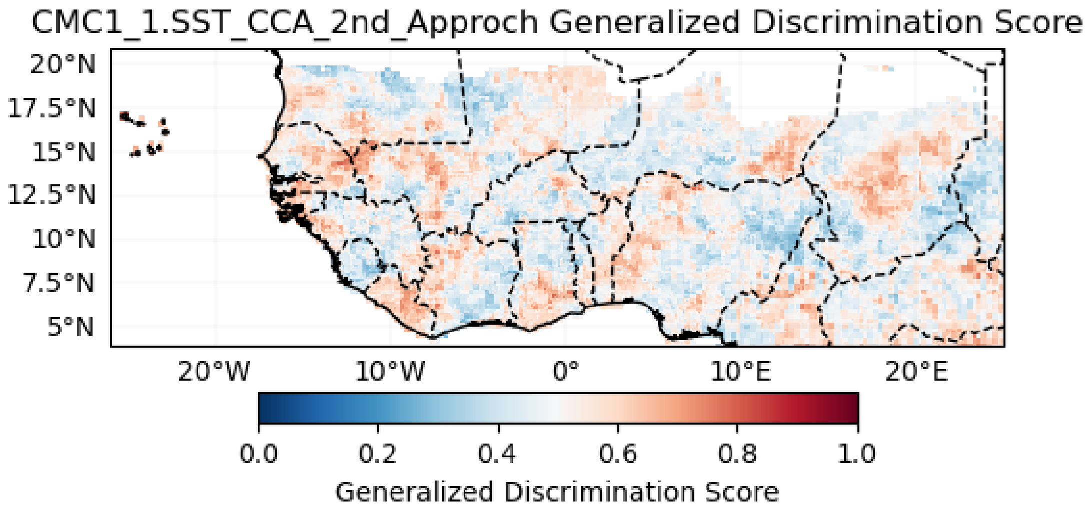
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


CMC1_1.SST_CCA_2nd_Approch Generalized Discrimination Score
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7.5°N
5°N
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10°W
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20°E
0.0
0.2
0.4
0.6
0.8
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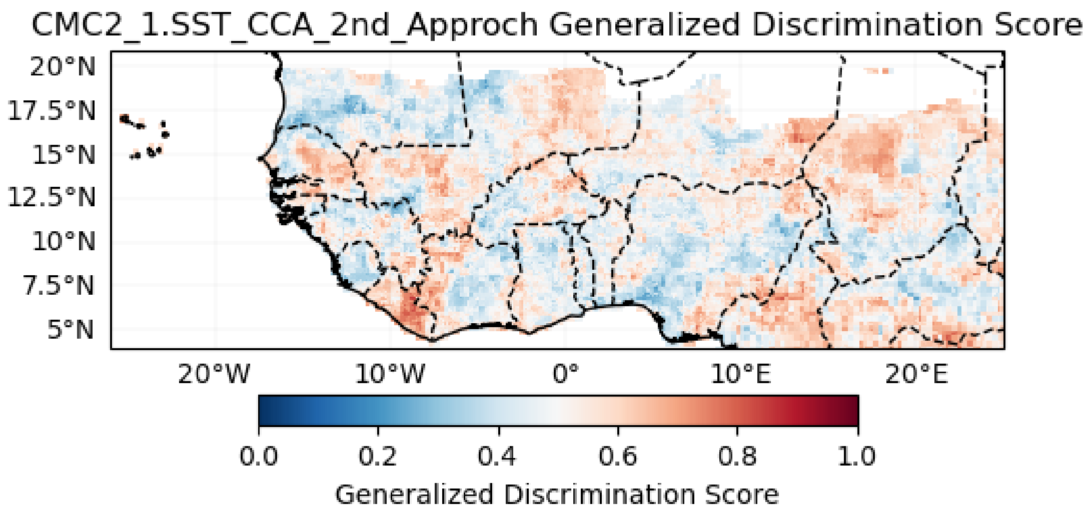
Generalized Discrimination Score


CMC2_1.SST_CCA_2nd_Approch Generalized Discrimination Score
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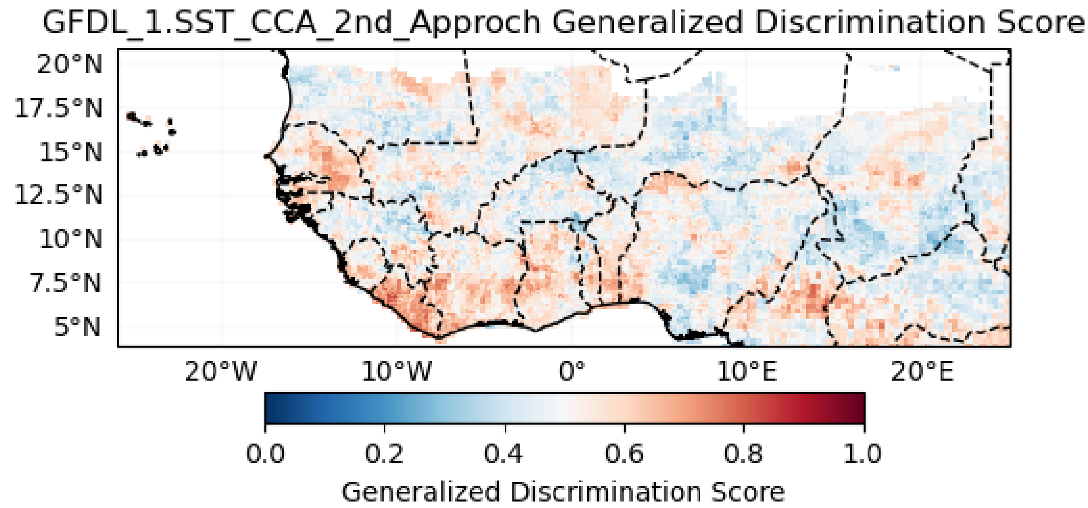
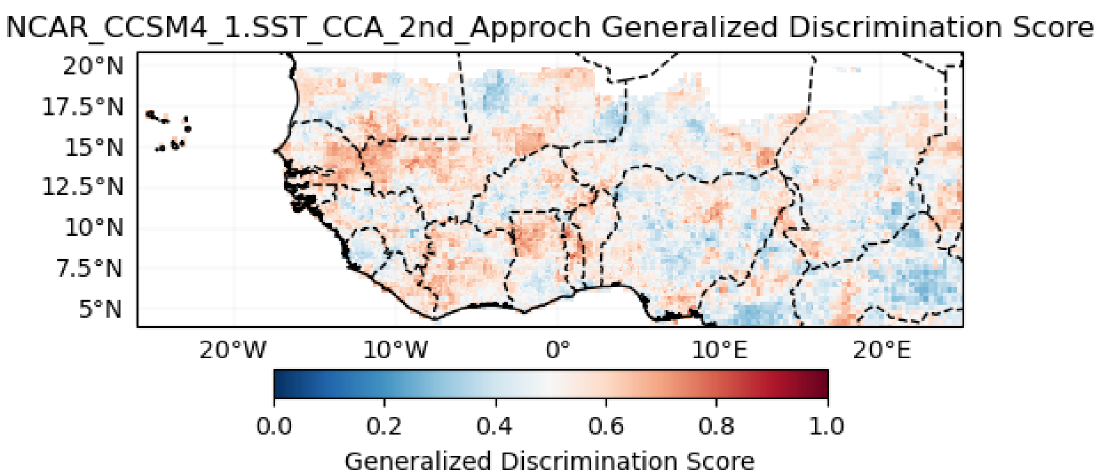
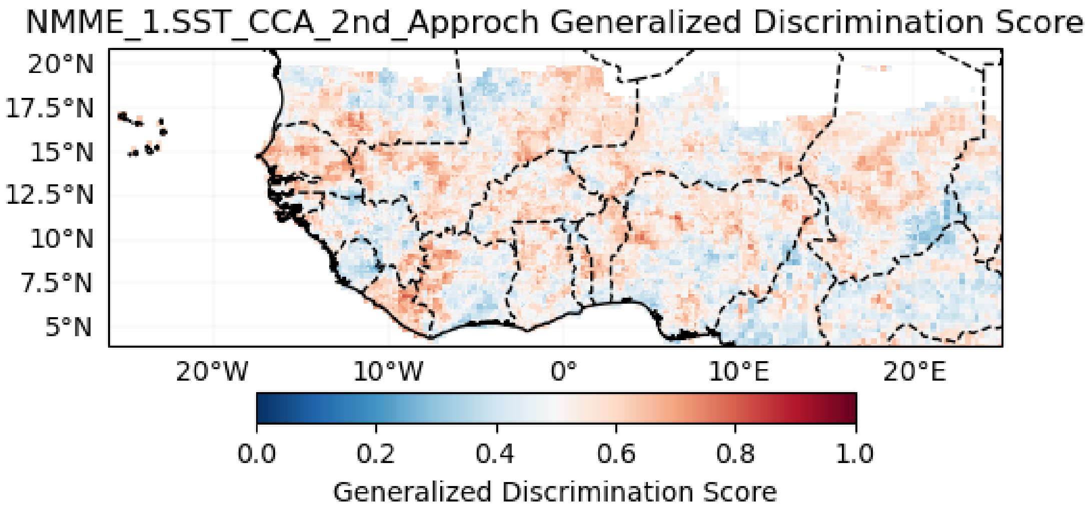
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score

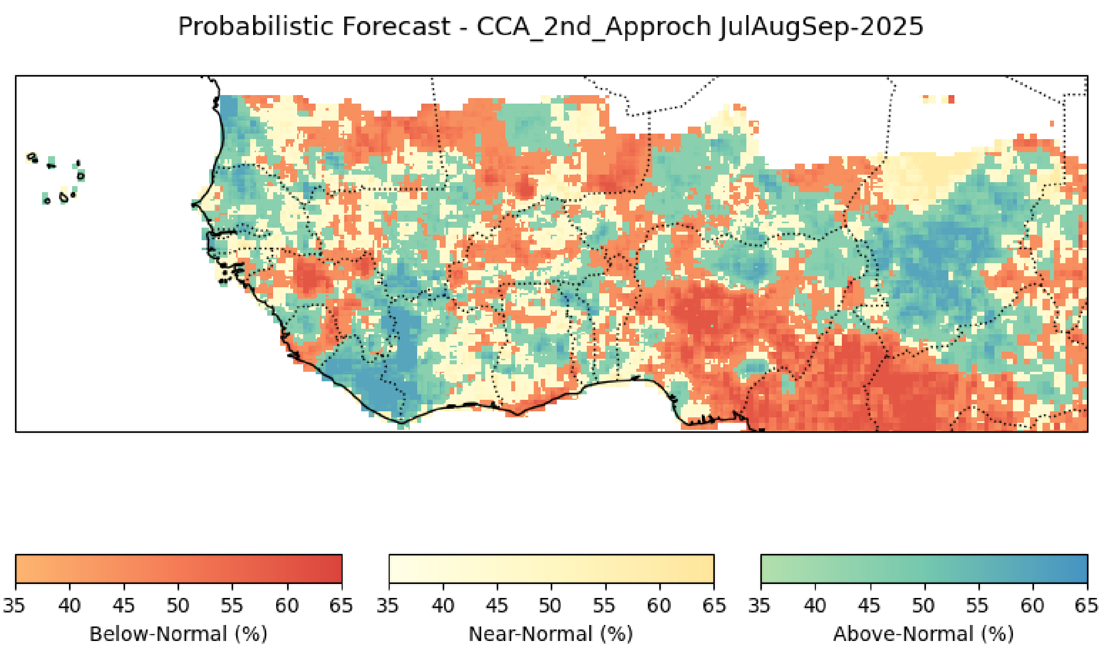

GFDL_1.SST_CCA_2nd_Approch Generalized Discrimination Score
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score


NCAR_CCSM4_1.SST_CCA_2nd_Approch Generalized Discrimination Score
20°N
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
20°W
10°W
0°
10°E
20°E
0.0
0.2
0.4
0.6
0.8
1.0
Generalized Discrimination Score

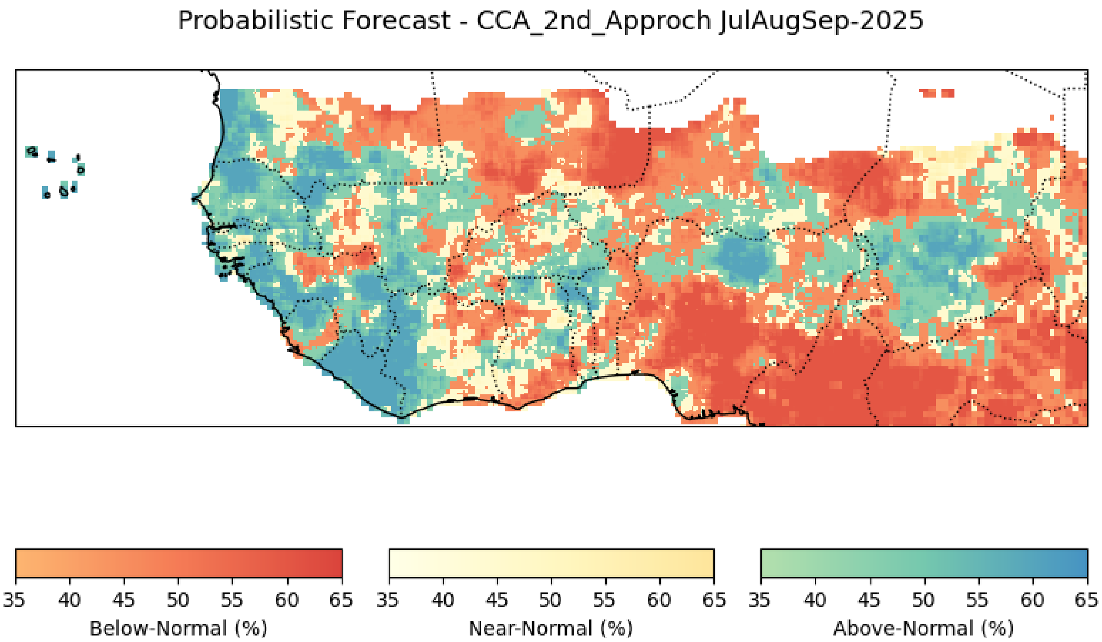


Forecast (individual)

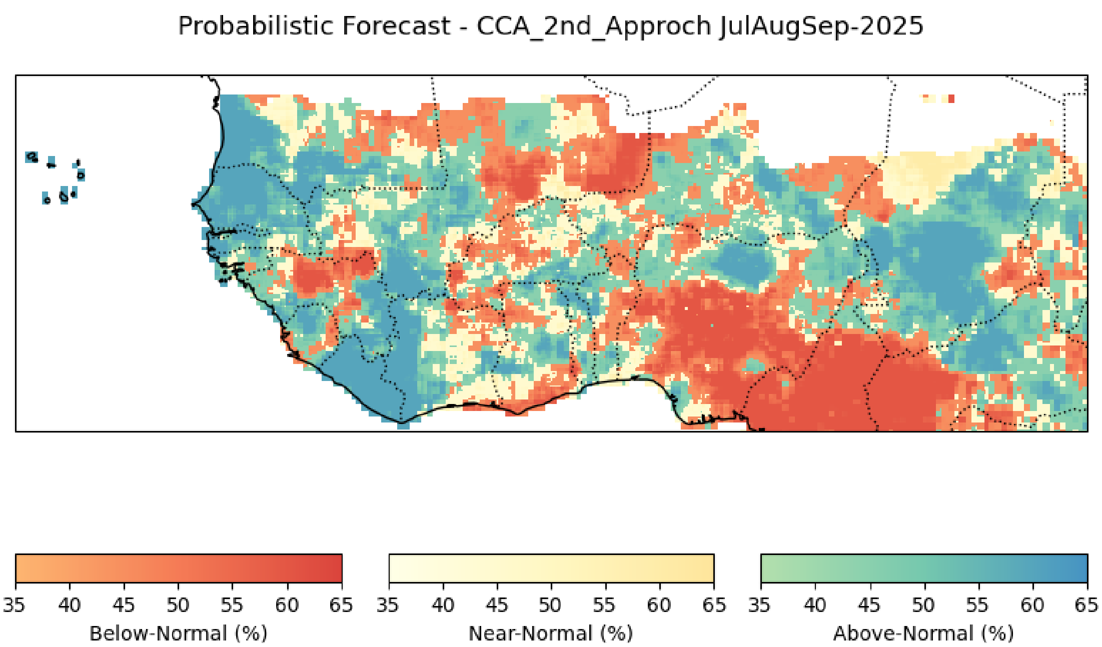
`ECMWF_51`



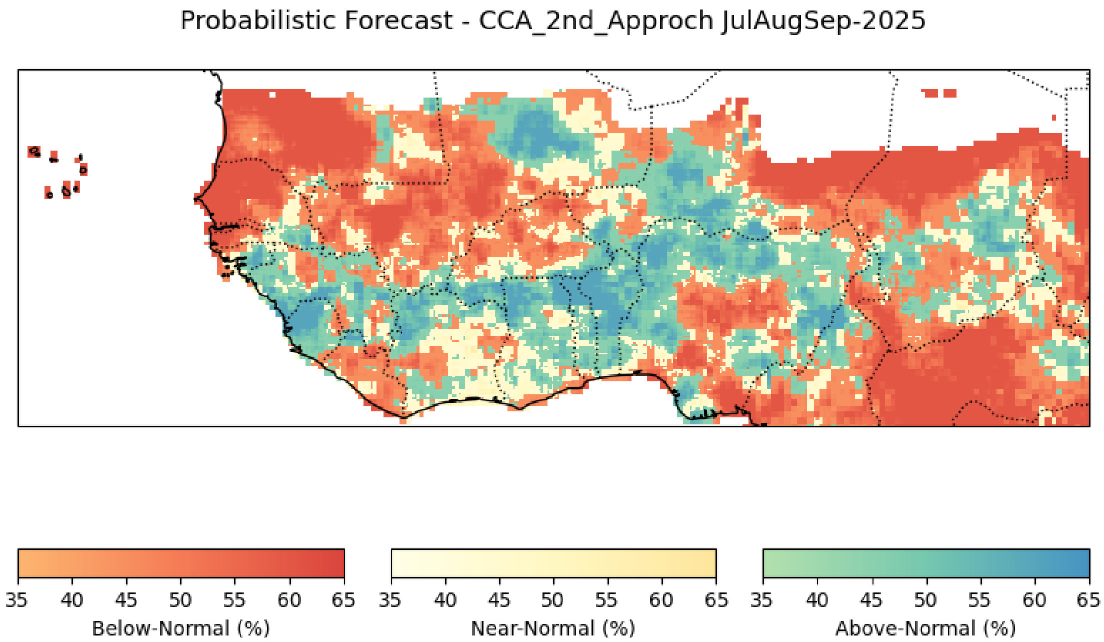
`UKMO_604`



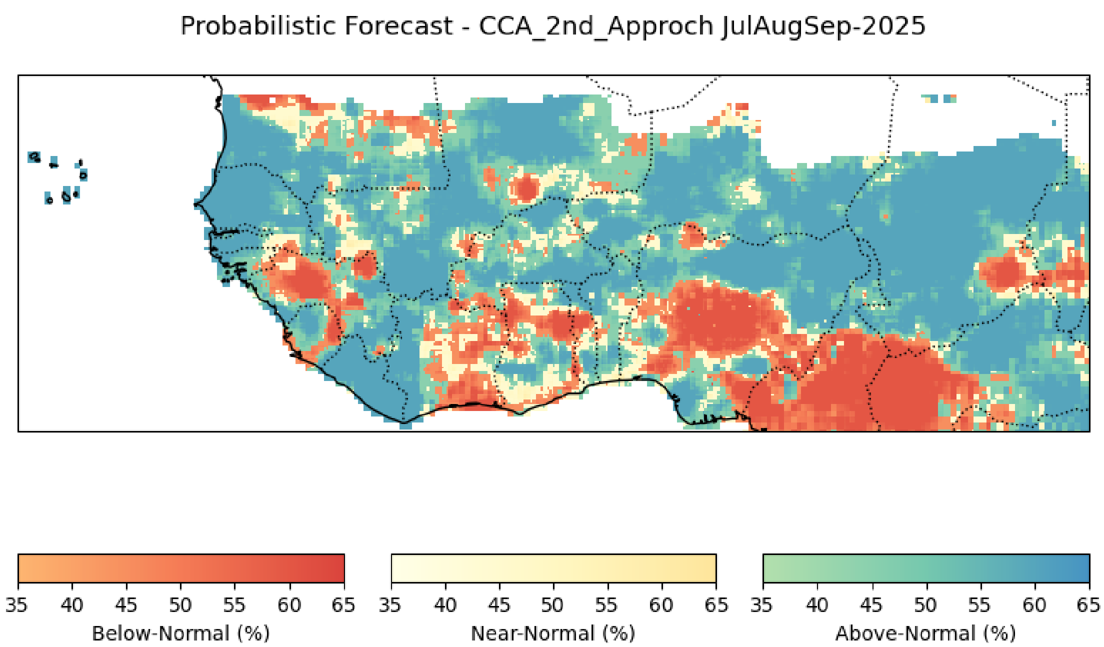
METEOFRANCE_8



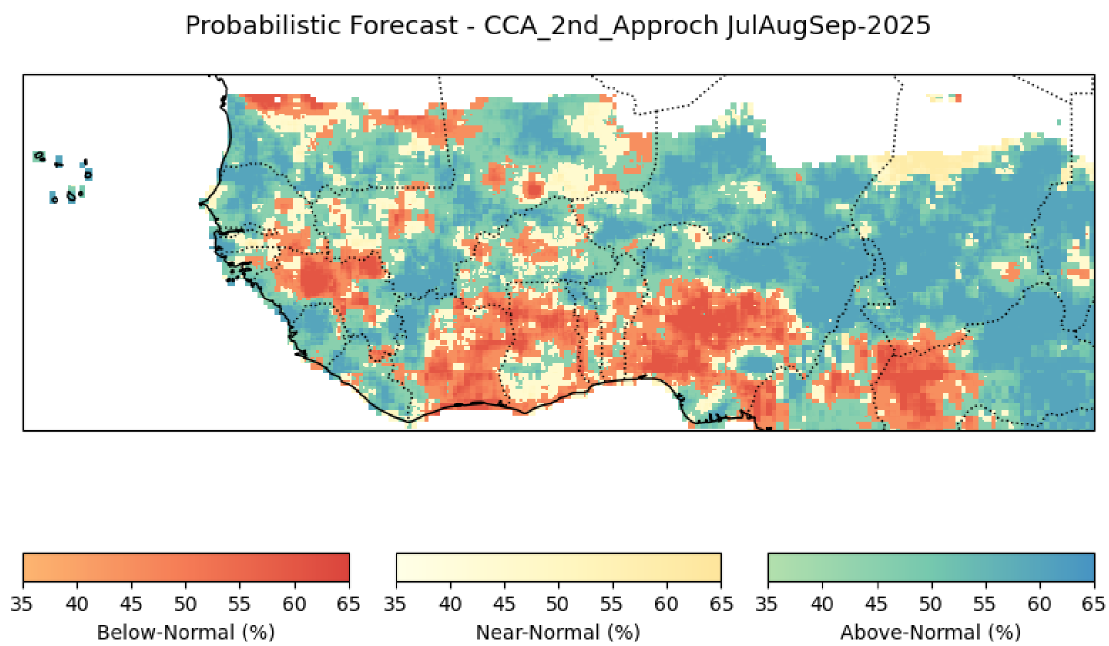
DWD_22



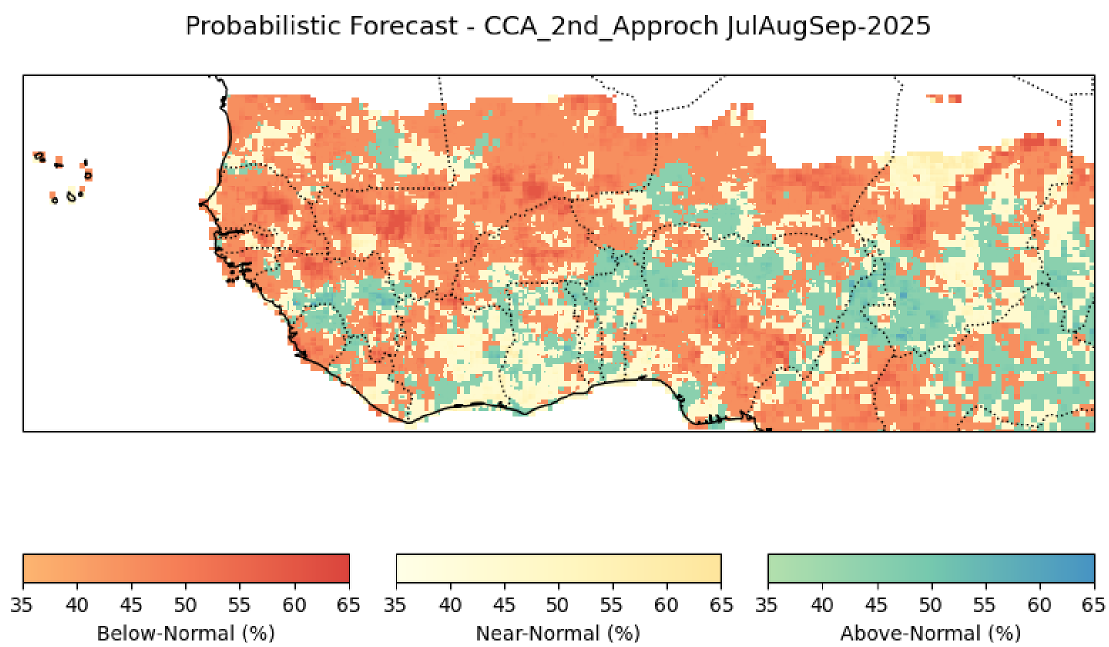
CMCC_35



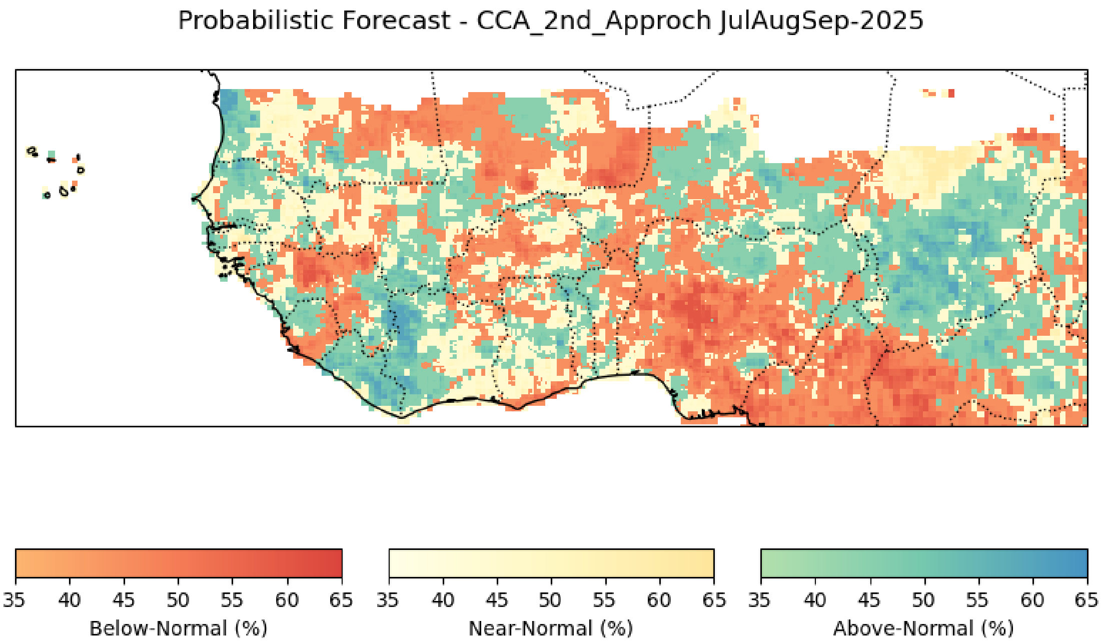
NCEP_2


Probabilistic Forecast - CCA_2nd_Approch JulAugSep-2025
35 40 45 50 55 60 65
Below-Normal (%)
35 40 45 50 55 60 65
Near-Normal (%)
35 40 45 50 55 60 65
Above-Normal (%)

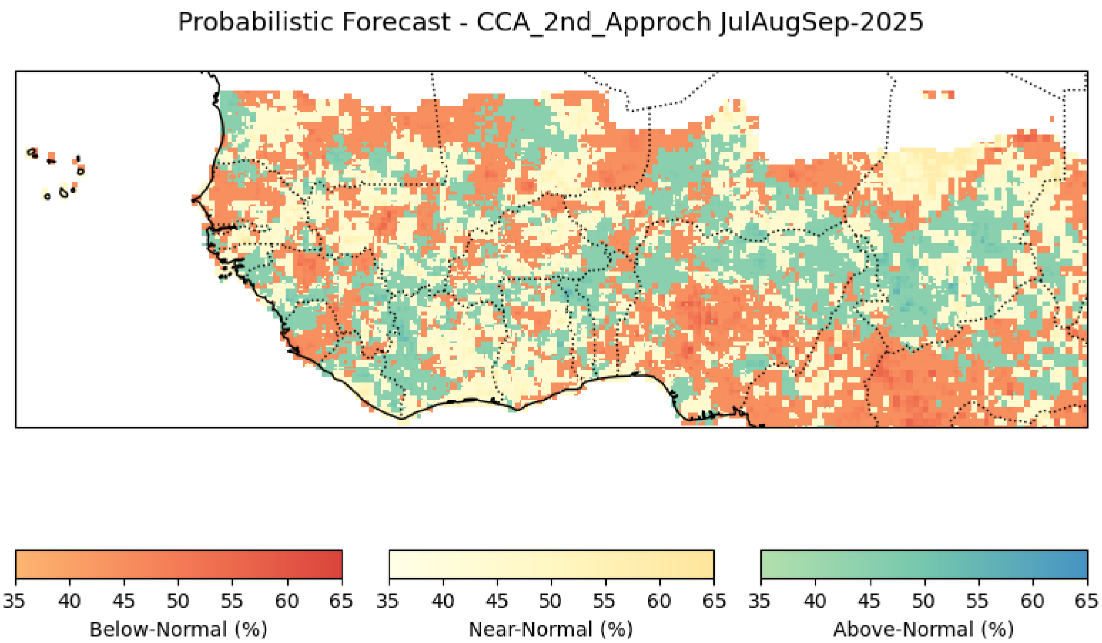
JMA_3


Probabilistic Forecast - CCA_2nd_Approch JulAugSep-2025
35 40 45 50 55 60 65
Below-Normal (%)
35 40 45 50 55 60 65
Near-Normal (%)
35 40 45 50 55 60 65
Above-Normal (%)

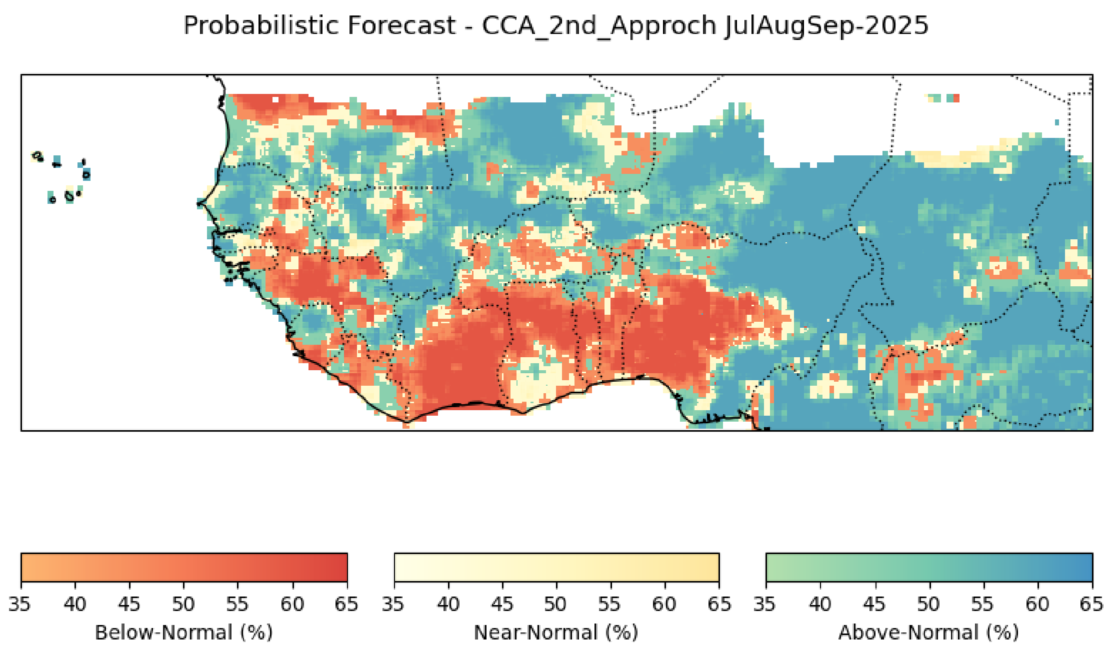
ECCC_4



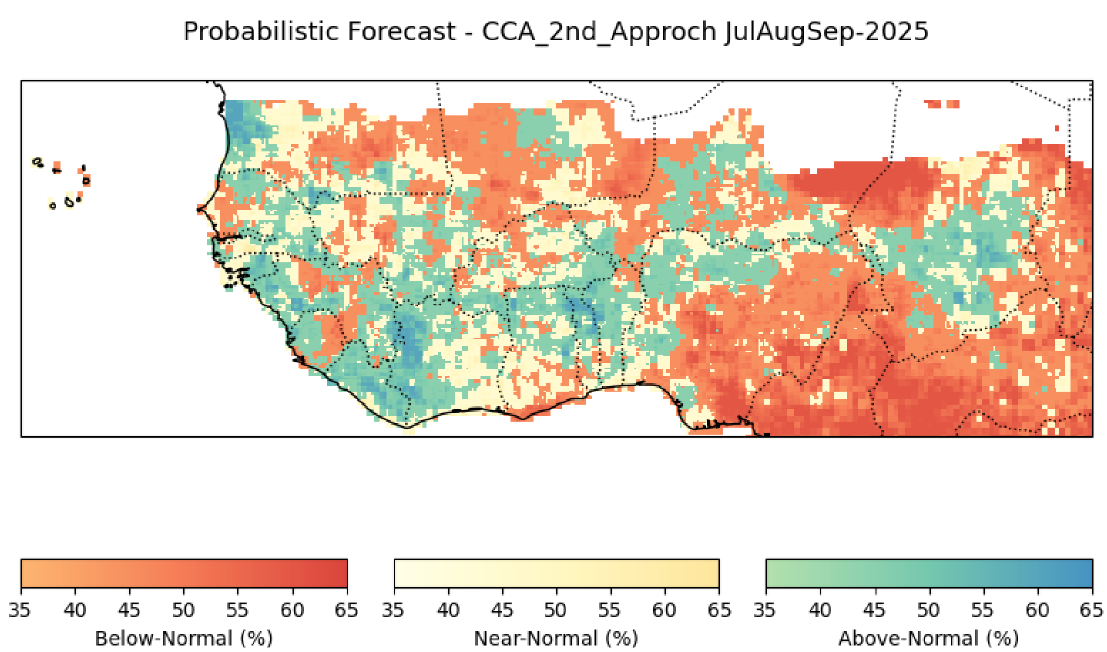
ECCC_5



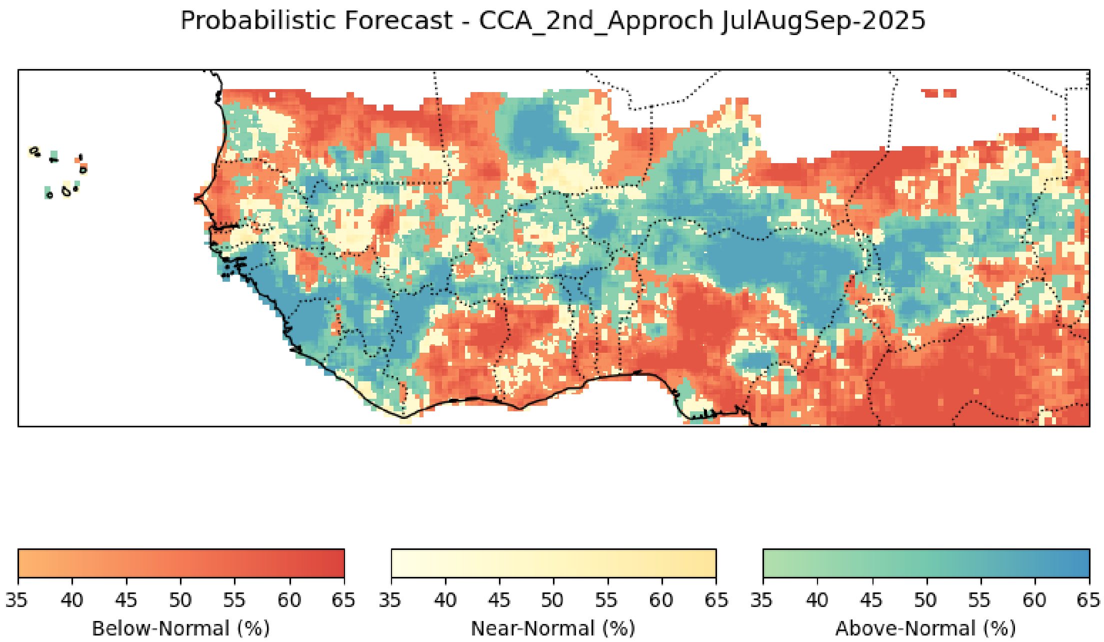
CFSV2_1


Probabilistic Forecast - CCA_2nd_Approch JulAugSep-2025
35 40 45 50 55 60 65
Below-Normal (%)
35 40 45 50 55 60 65
Near-Normal (%)
35 40 45 50 55 60 65
Above-Normal (%)

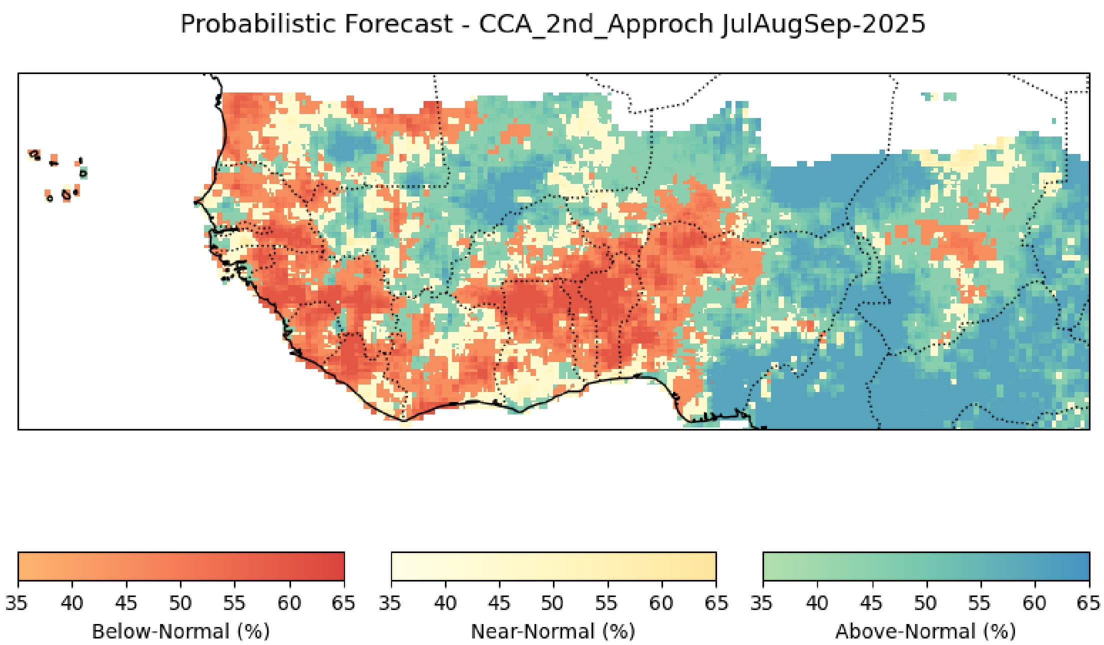
CMC1_1


Probabilistic Forecast - CCA_2nd_Approch JulAugSep-2025
35 40 45 50 55 60 65
Below-Normal (%)
35 40 45 50 55 60 65
Near-Normal (%)
35 40 45 50 55 60 65
Above-Normal (%)

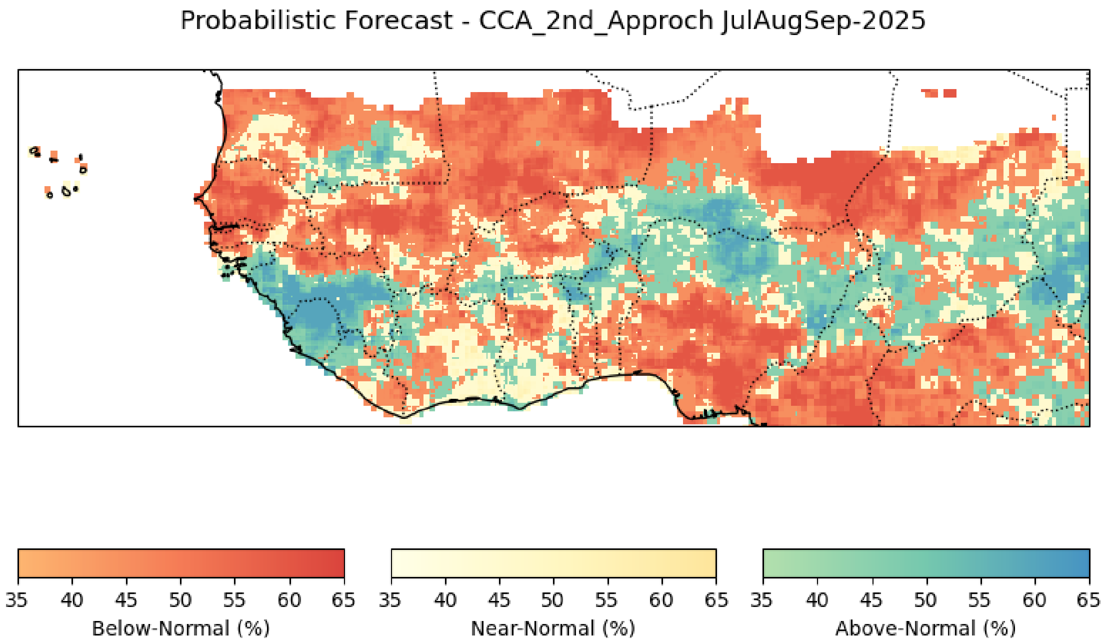
CMC2_1



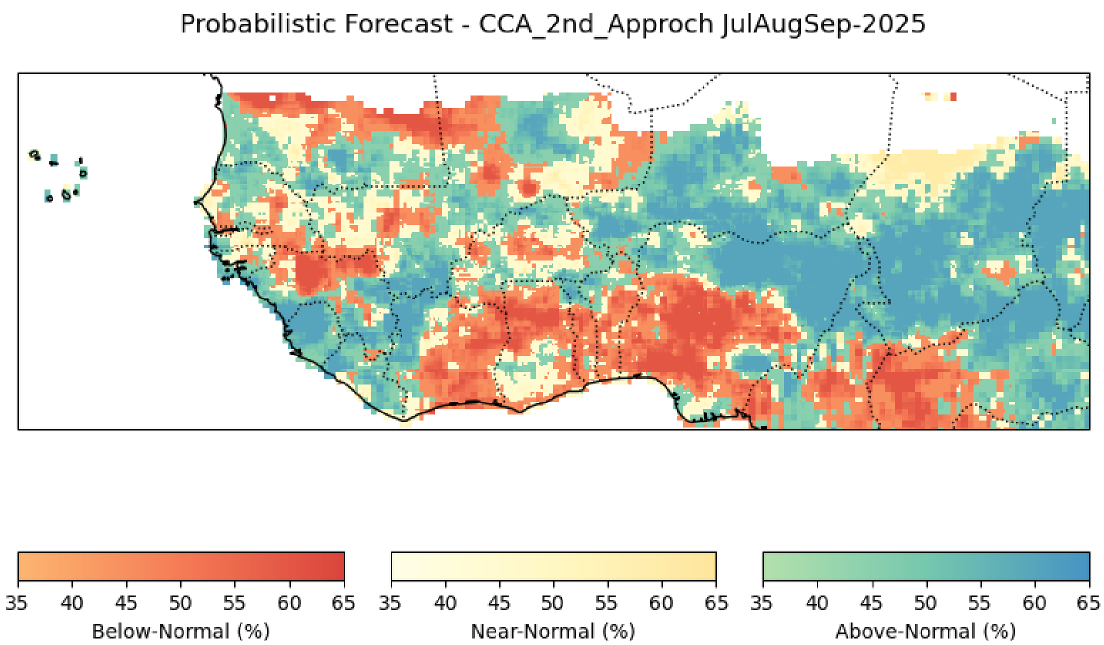
GFDL_1



NCAR_CCSM4_1


Probabilistic Forecast - CCA_2nd_Approch JulAugSep-2025
35 40 45 50 55 60 65
Below-Normal (%)
35 40 45 50 55 60 65
Near-Normal (%)
35 40 45 50 55 60 65
Above-Normal (%)

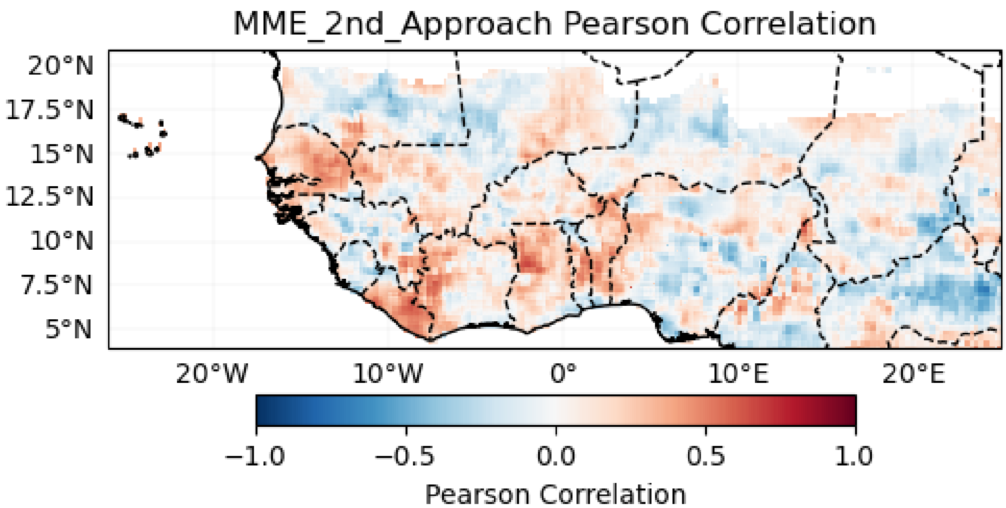
NMME_1

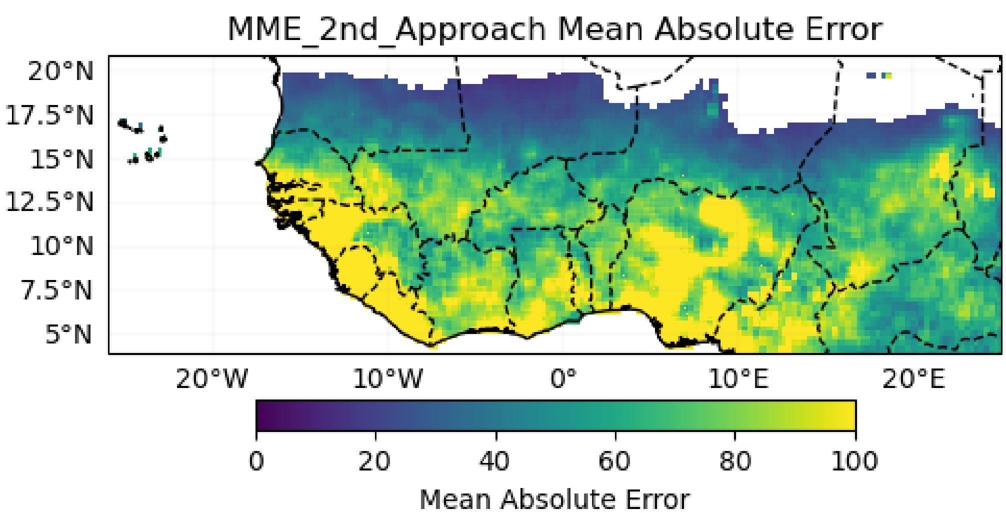

Probabilistic Forecast - CCA_2nd_Approch JulAugSep-2025
35 40 45 50 55 60 65
Below-Normal (%)
35 40 45 50 55 60 65
Near-Normal (%)
35 40 45 50 55 60 65
Above-Normal (%)

2.0.2 MME 2nd-Approach: Statistico-Dynamical

`['NMME_1.SST', 'GFDL_1.SST', 'ECMWF_51.SST', 'CFSV2_1.SST', 'NCAR_CCSM4_1.SST']`

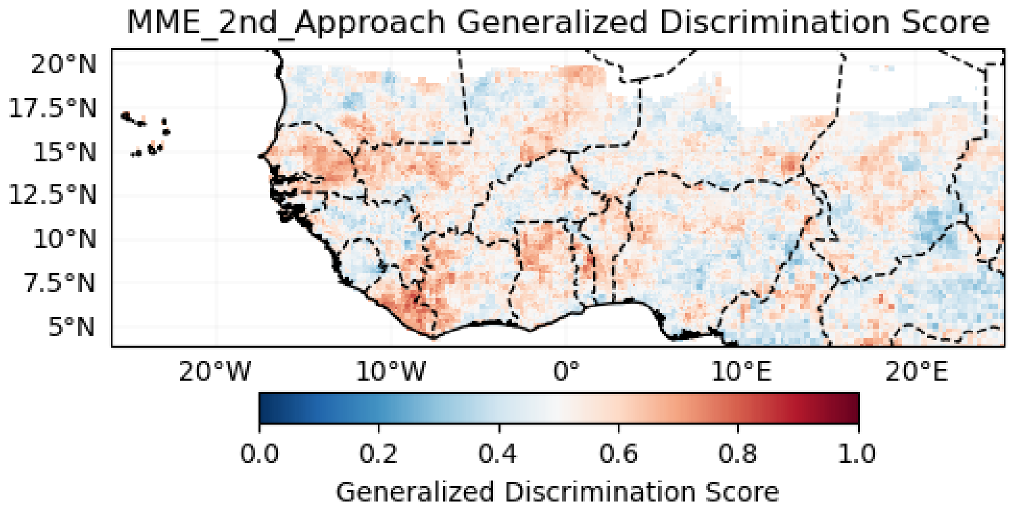
Verification Deterministic



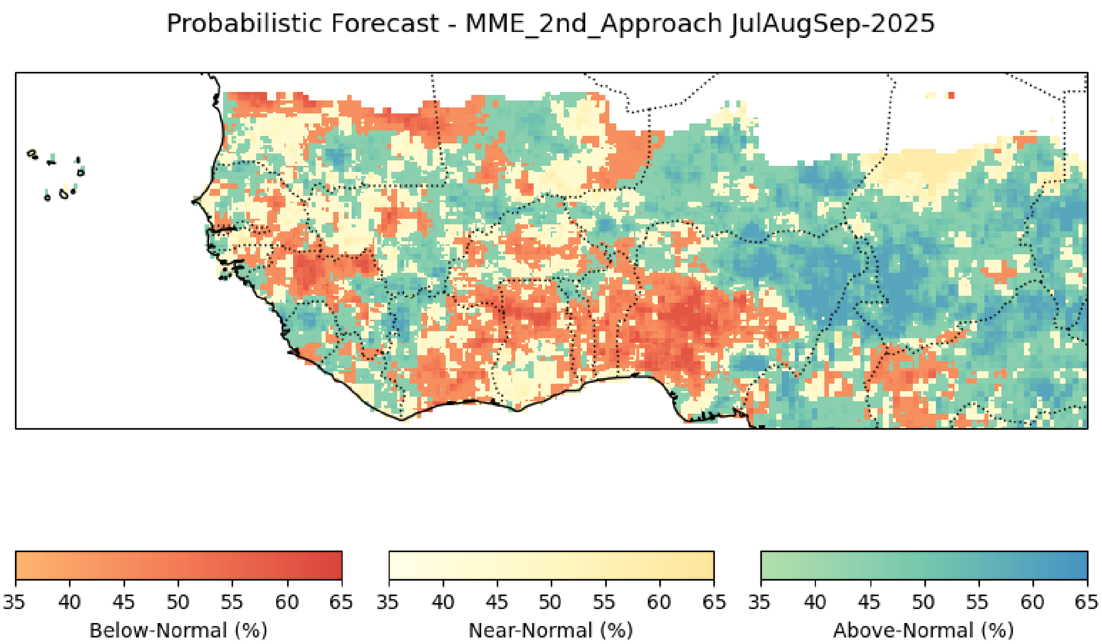


Probabilistic

`Generalized Discrimination Score`



Forecast



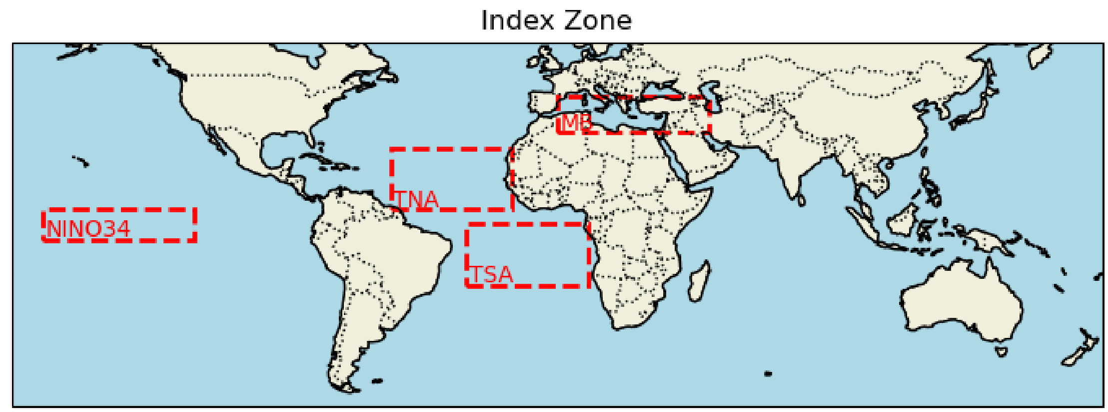

```
139734
```

3 3rd - Approach: Observational-based seasonal forecast

```
JAS_2025_bis/reanalysis_data/NOAA_SST_1991_2025_JanFebMar.nc exists. Skipping.
```

Process SST index

['NINO34', 'NINO12', 'NINO3', 'NINO4', 'NINO_Global', 'TNA', 'TSA', 'NAT', 'SAT', 'TASI', 'WTIO', 'SETIO', 'DMI', 'MB']



NOAA SST

Calculate variance inflation factor to see multicolinearity predictor

	feature	VIF
0	NINO34	1.268913
1	TNA	1.698945
2	TSA	2.233701
3	DMI	1.298347
4	MB	1.718974

['NINO34', 'TNA', 'TSA', 'DMI', 'MB']

3.1 Initialize WAS_LinearRegression_Model

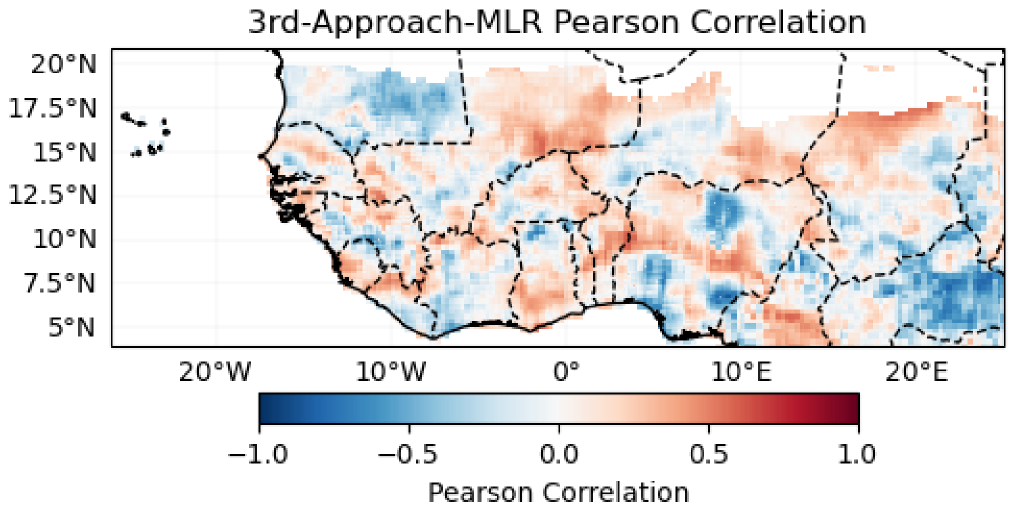
8

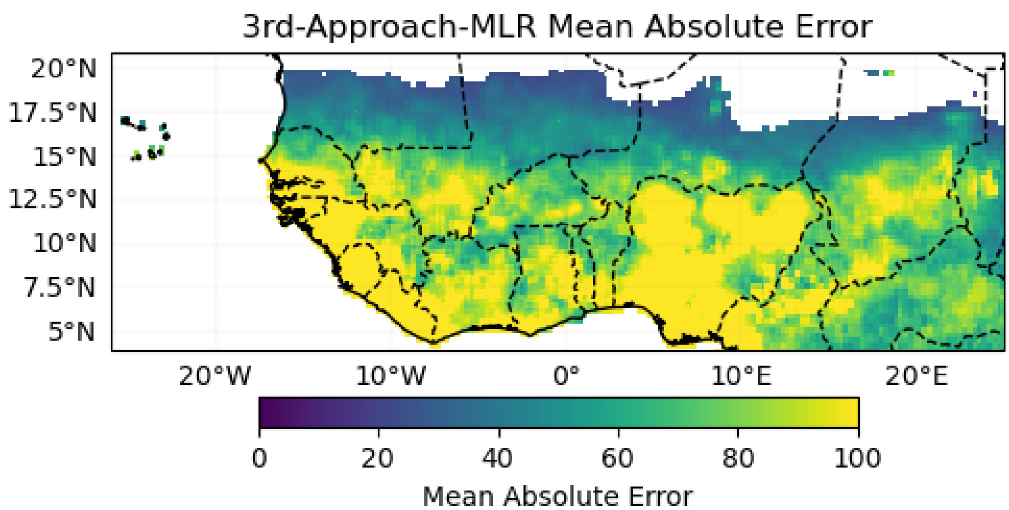
Cross validation

Cross-validation ongoing

100%| | 34/34 [09:57<00:00, 17.56s/it]

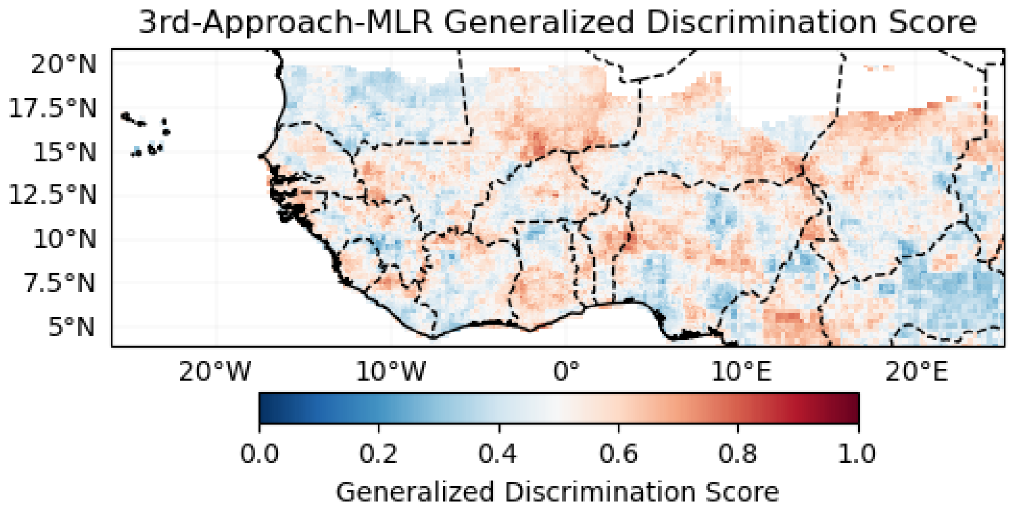
Verification Deterministic



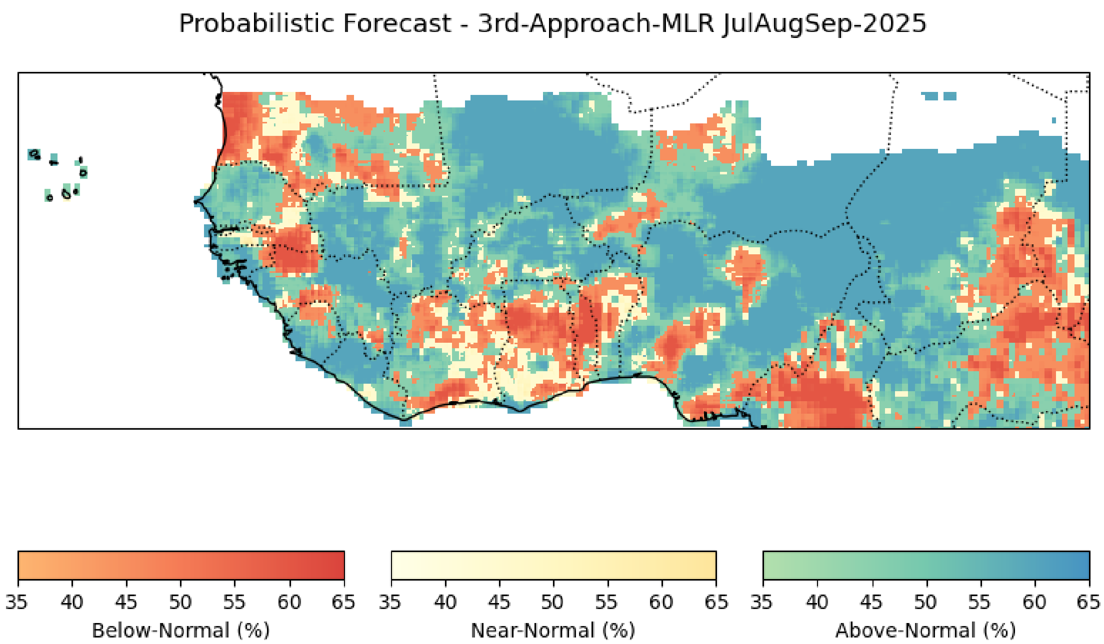


Probabilistic

`Generalized Discrimination Score`



Forecast



4 4th - Analog-based seasonal forecast

4.1 Initialize WAS_Analog class

Cross-validation

```
JAS_2025_bis/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc
already exists. Skipping download.
Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization
month
JAS_2025_bis/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc
File 'JAS_2025_bis/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc' already
exists. Skipping download.
All requested SST data has been processed.
Cross-validation ongoing

  0%|                                                     | 0/34 [00:00<?, ?it/s]

similar years for 1991 are [2002 1997 2023]

  6%|                                             | 2/34 [00:00<00:06,  5.15it/s]

similar years for 1992 are [2020 2019 2005]

  9%|                                            | 3/34 [00:00<00:05,  5.18it/s]

similar years for 1993 are [2020 1991 1996]

 12%|                                           | 4/34 [00:00<00:05,  5.18it/s]

similar years for 1994 are [2003 1991 2009]

 15%|                                          | 5/34 [00:00<00:05,  5.18it/s]

similar years for 1995 are [2016 1998 2020]

 18%|                                         | 6/34 [00:01<00:05,  5.19it/s]

similar years for 1996 are [1993 2018 2002]

 21%|                                        | 7/34 [00:01<00:05,  5.20it/s]

similar years for 1997 are [2023 2015 2009]

 24%|                                       | 8/34 [00:01<00:04,  5.22it/s]

similar years for 1998 are [2010 2016 2024]

 26%|                                      | 9/34 [00:01<00:04,  5.21it/s]

similar years for 1999 are [2021 2011 2008]

 29%|                                    | 10/34 [00:01<00:04,  5.15it/s]

similar years for 2000 are [2012 2011 2009]

 32%|                                   | 11/34 [00:02<00:04,  5.15it/s]

similar years for 2001 are [2023 2012 2017]

 35%|                                  | 12/34 [00:02<00:04,  5.17it/s]

similar years for 2002 are [2023 1991 2018]

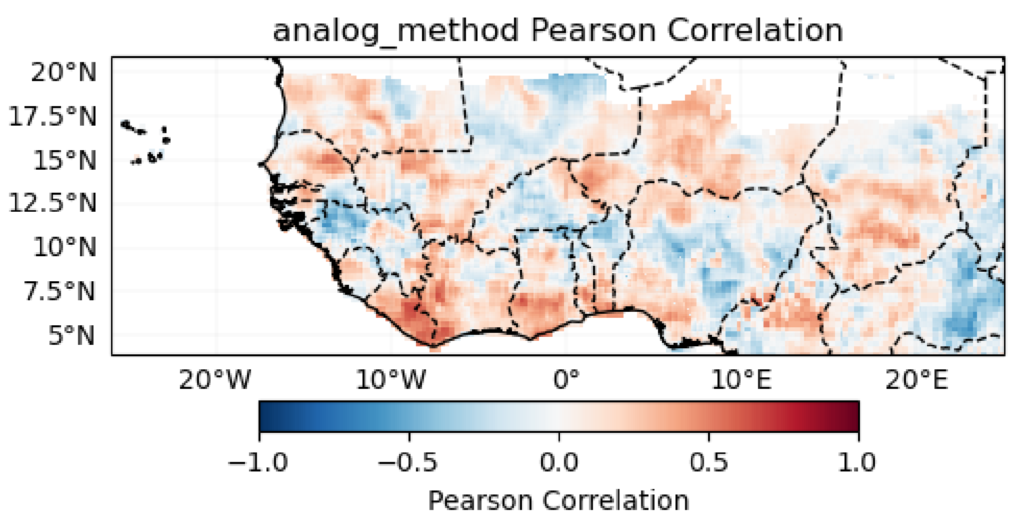
 38%|                                 | 13/34 [00:02<00:04,  5.19it/s]

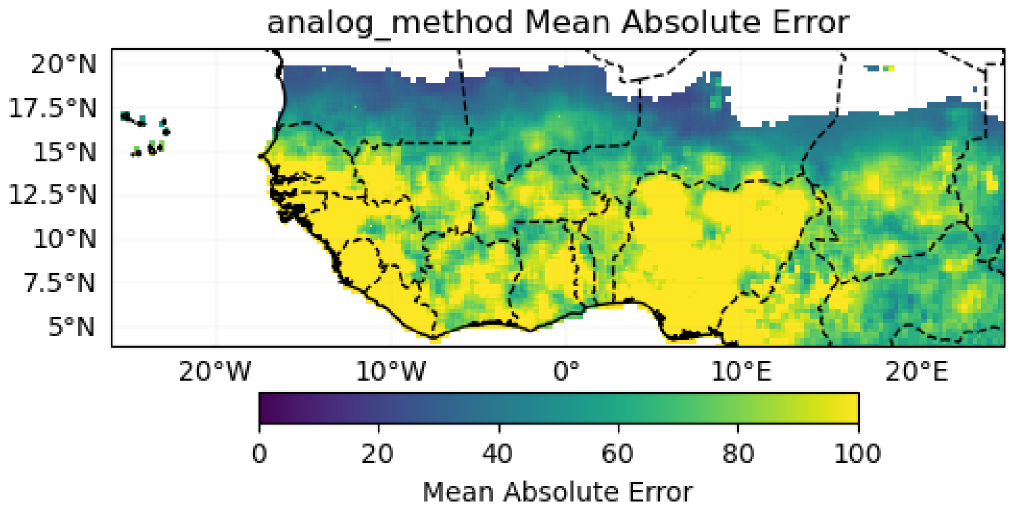
similar years for 2003 are [2024 1998 2016]
```

```
 41%|                              | 14/34 [00:02<00:03,  5.22it/s]
similar years for 2004 are [2007 2002 2018]
 44%|                              | 15/34 [00:02<00:03,  5.23it/s]
similar years for 2005 are [2010 2016 2020]
 47%|                              | 16/34 [00:03<00:03,  5.25it/s]
similar years for 2006 are [2009 2017 2011]
 50%|                              | 17/34 [00:03<00:03,  5.25it/s]
similar years for 2007 are [2016 2010 2024]
 53%|                              | 18/34 [00:03<00:03,  5.22it/s]
similar years for 2008 are [2021 2011 2018]
 56%|                              | 19/34 [00:03<00:02,  5.23it/s]
similar years for 2009 are [2023 1997 2018]
 59%|                              | 20/34 [00:03<00:02,  5.27it/s]
similar years for 2010 are [1998 2007 2016]
 62%|                              | 21/34 [00:04<00:02,  5.29it/s]
similar years for 2011 are [2008 2023 2021]
 65%|                              | 22/34 [00:04<00:02,  5.31it/s]
similar years for 2012 are [2023 2000 2011]
 68%|                              | 23/34 [00:04<00:02,  5.32it/s]
similar years for 2013 are [2016 2020 2024]
 71%|                              | 24/34 [00:04<00:01,  5.32it/s]
similar years for 2014 are [2023 2009 1991]
 74%|                              | 25/34 [00:04<00:01,  5.27it/s]
similar years for 2015 are [2014 2023 1997]
 76%|                              | 26/34 [00:04<00:01,  5.22it/s]
similar years for 2016 are [1998 1995 2007]
 79%|                              | 27/34 [00:05<00:01,  5.24it/s]
similar years for 2017 are [2001 2023 2011]
 82%|                              | 28/34 [00:05<00:01,  5.28it/s]
similar years for 2018 are [2021 2008 2011]
 85%|                              | 29/34 [00:05<00:00,  5.31it/s]
similar years for 2019 are [1998 1995 2023]
```

```
 88%|                              | 30/34 [00:05<00:00,  5.32it/s]
similar years for 2020 are [1998 2024 2016]
 91%|                             | 31/34 [00:05<00:00,  5.34it/s]
similar years for 2021 are [2008 1999 2018]
 94%|                            | 32/34 [00:06<00:00,  5.37it/s]
similar years for 2022 are [2018 2010 2011]
 97%|                            | 33/34 [00:06<00:00,  5.38it/s]
similar years for 2023 are [1997 2012 2011]
100%|                            | 34/34 [00:06<00:00,  5.26it/s]
similar years for 2024 are [1998 2020 2003]
```

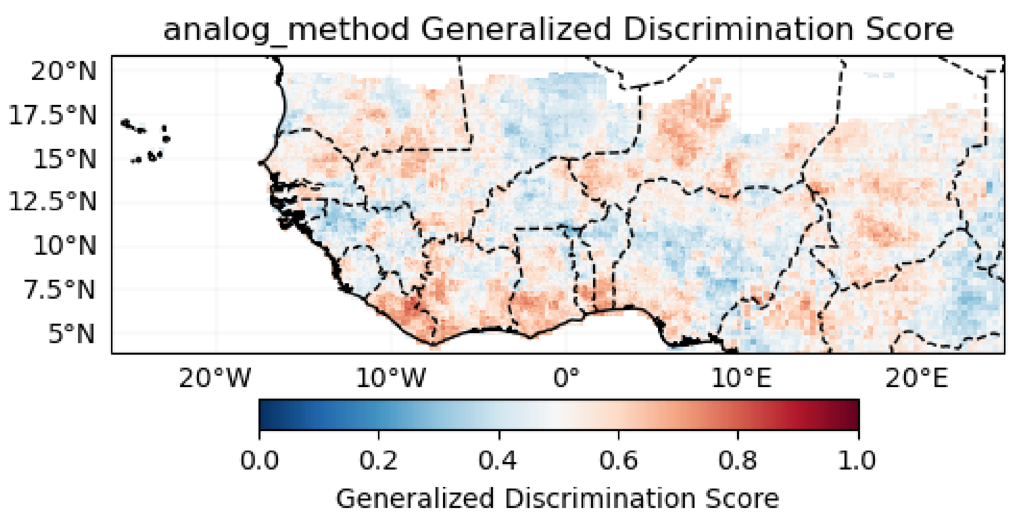
Verification Deterministic





Probabilistic

```
Generalized Discrimination Score
```



Forecast

```
JAS_2025_bis/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc
already exists. Skipping download.
Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization
month
JAS_2025_bis/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc
File 'JAS_2025_bis/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc' already
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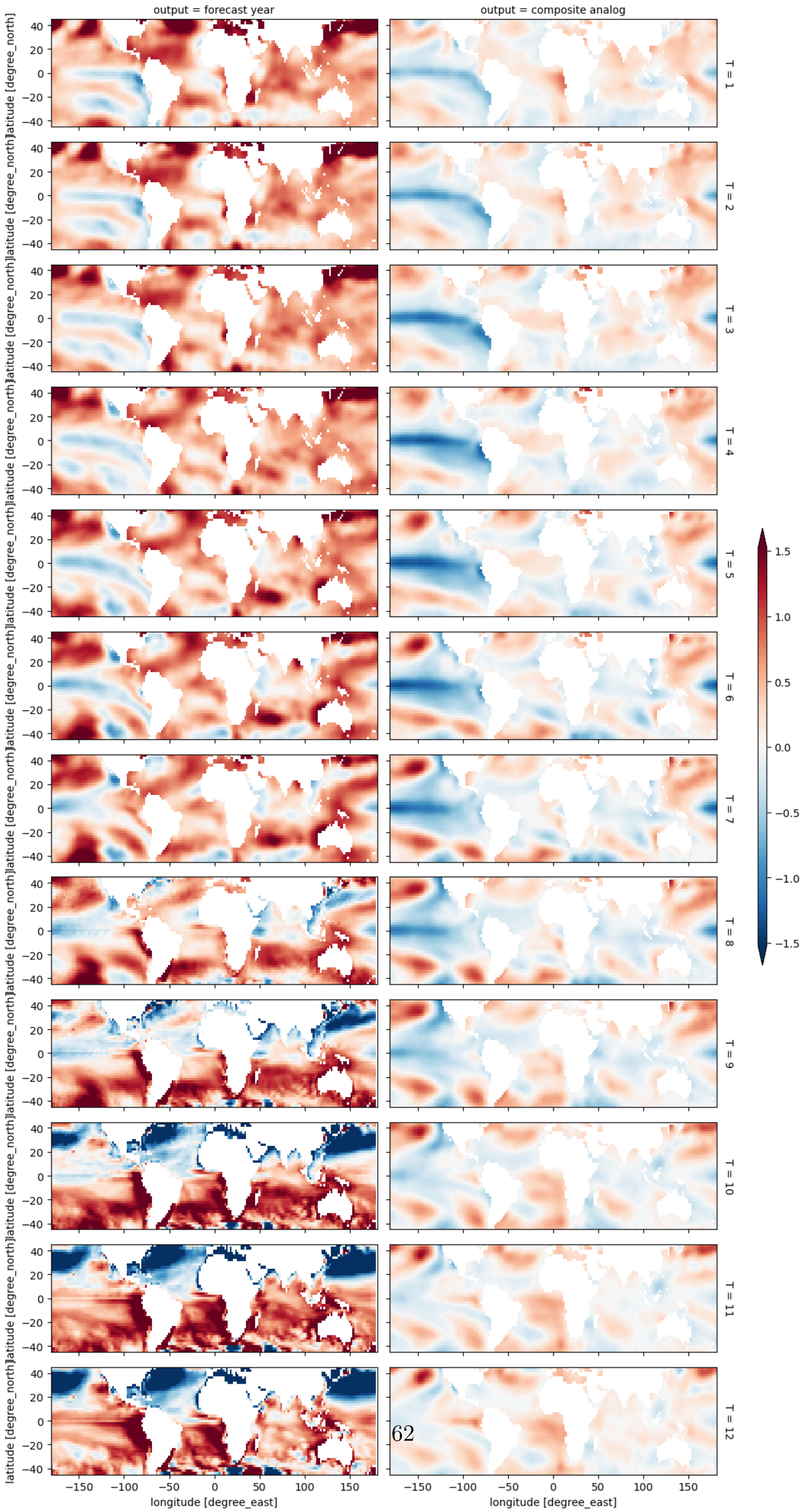
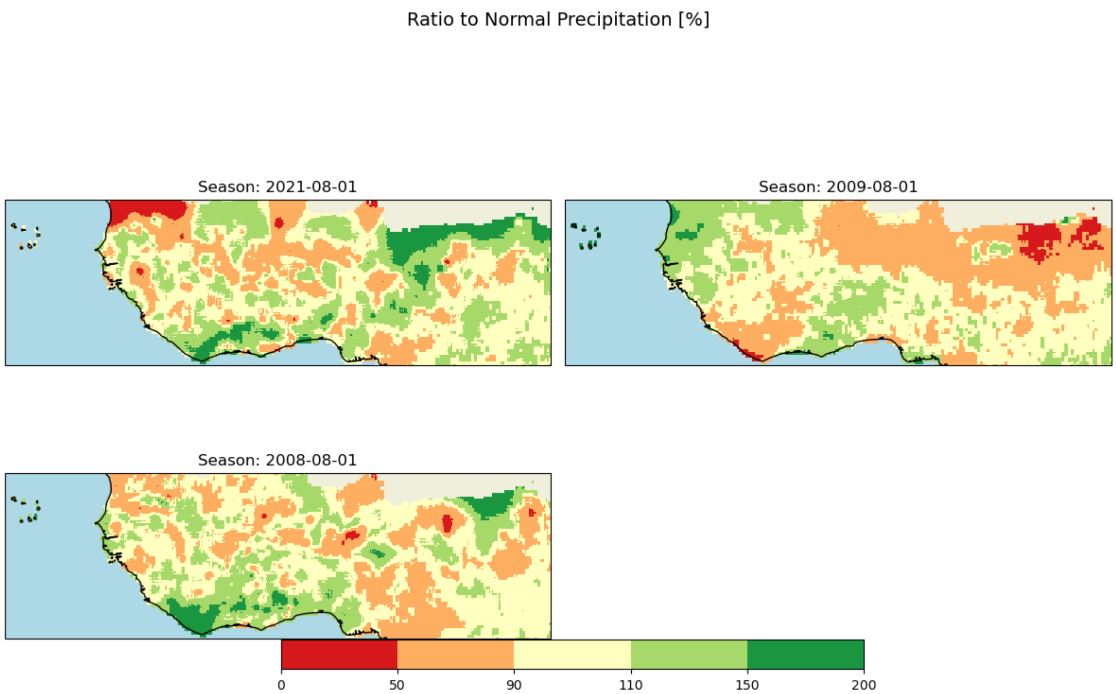
exists. Skipping download.
All requested SST data has been processed.
```

```
array([2021, 2009, 2008])
```



output = forecast year
output = composite analog
T = 1
T = 2
T = 3
T = 4
T = 5
T = 6
T = 7
T = 8
T = 9
T = 10
T = 11
T = 12
latitude [degree_north]
longitude [degree_east]

```
JAS_2025_bis/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc
already exists. Skipping download.
Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization
month
JAS_2025_bis/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc
File 'JAS_2025_bis/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc' already
exists. Skipping download.
All requested SST data has been processed.
```




```
array([2021, 2009, 2008])
```

```
JAS_2025_bis/analog/NOAA_SST_3_1990_2025_JanFebMarAprMayJunJulAugSepOctNovDec.nc
already exists. Skipping download.
Lead time not provided. Using months: [1, 2, 3, 4, 5] with Mar as initialization
month
JAS_2025_bis/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc
File 'JAS_2025_bis/analog/forecast_ncep2_SST_MarIc_AprMayJunJulAug.nc' already
exists. Skipping download.
All requested SST data has been processed.
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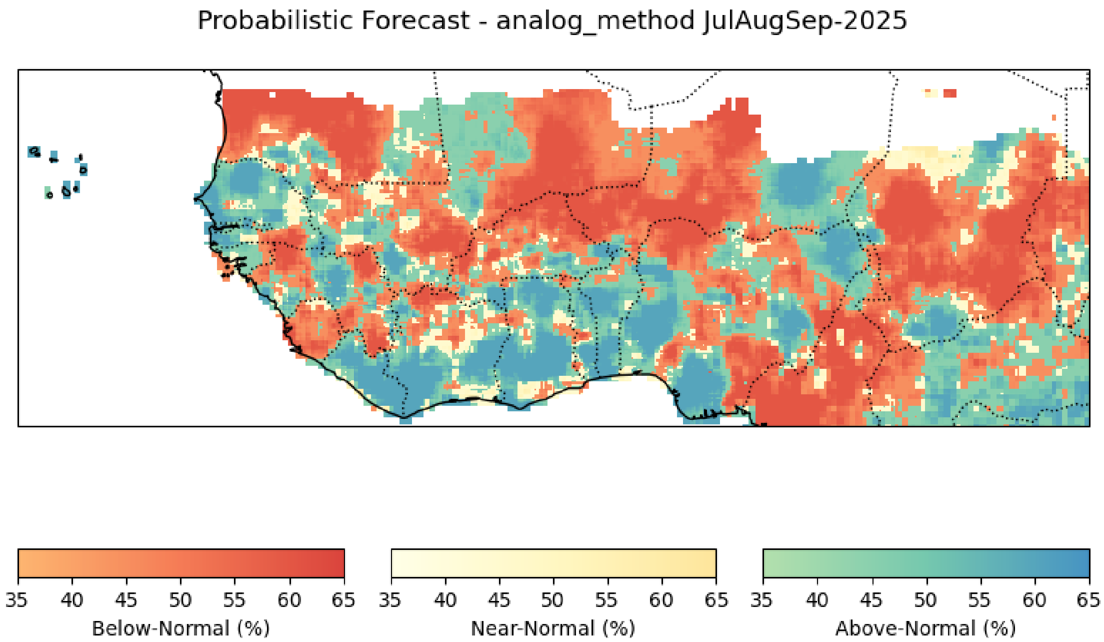
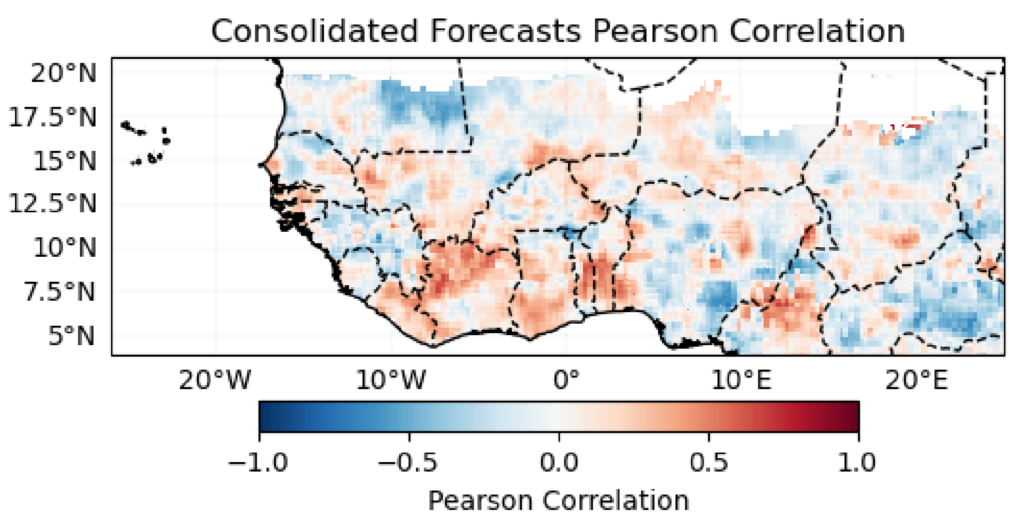
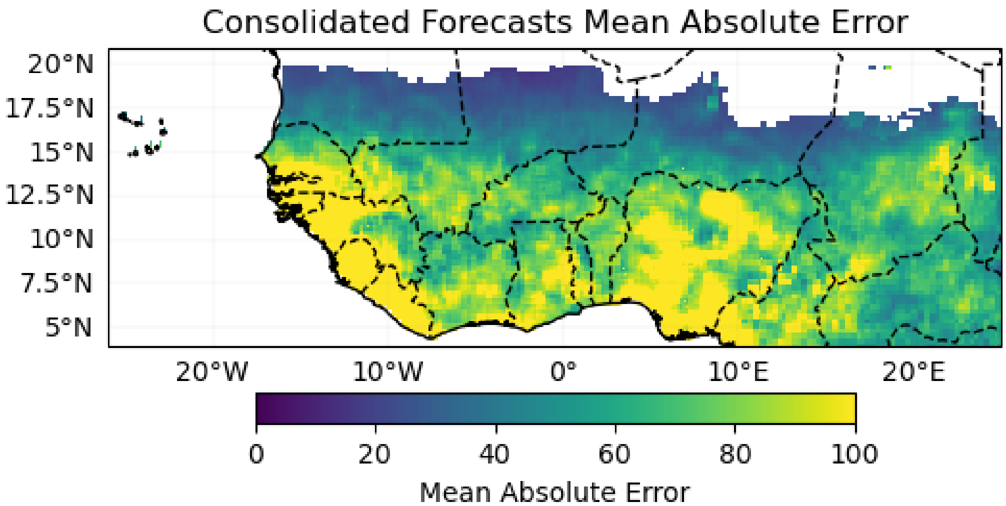


5 Consolidated Forecasts

Cross-validation

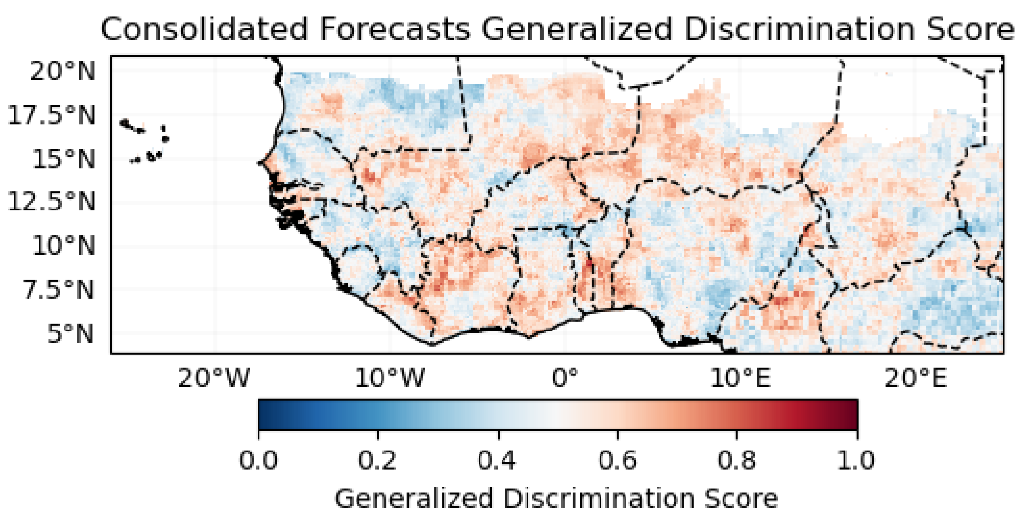
Verification Deterministic



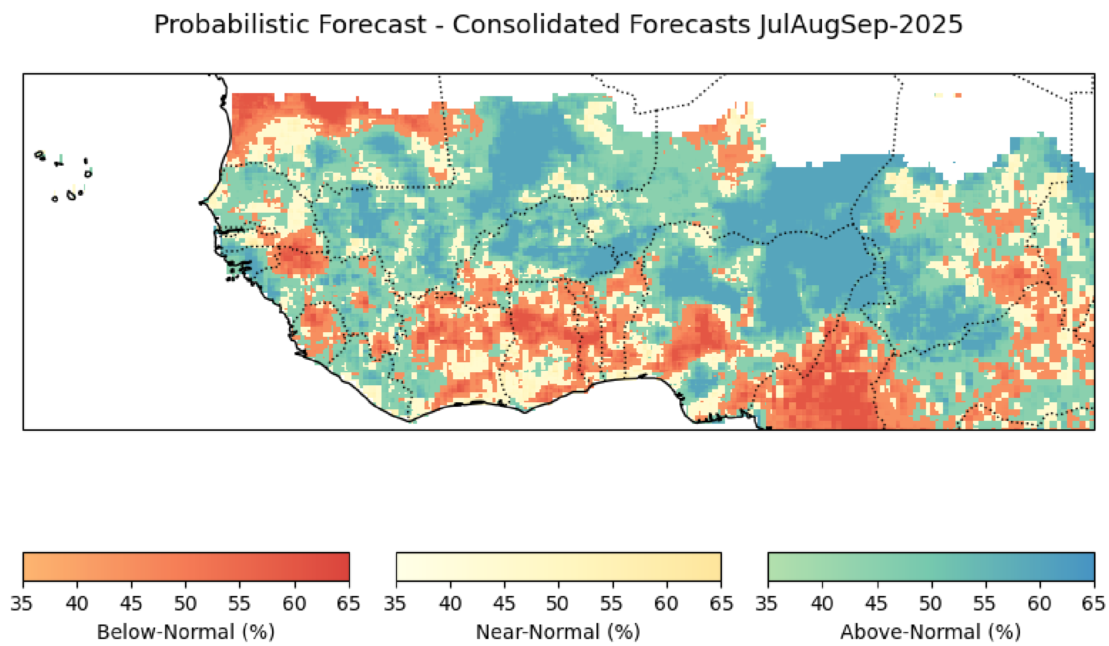


Probabilistic

`Generalized Discrimination Score`



5.0.1 Forecast



120787

Save Forecast for future evaluation

5.0.2 END ——————————-

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[NbConvertApp] Support files will be in 01_MME_JAS_2025_files/
[NbConvertApp] Making directory ./01_MME_JAS_2025_files
[NbConvertApp] Writing 74436 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
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